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Author: Avirup

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Module 1

Power Semiconductor Devices

Overview

This module introduces the principal power semiconductor devices used in renewable-energy converters, emphasizing structure, operation, characteristics, switching behavior, and device-selection context.

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Chapter 1.1: Power BJT

Chapter opening

Power electronics is often introduced through converters, inverters, and modern devices such as MOSFETs and IGBTs. That approach can obscure a more basic question: what a power switch must do in an actual circuit. The **power bipolar junction transistor**, or **power BJT**, provides a useful starting point because its strengths and limits are explicit.

A power converter requires a switch that can carry large current, block large voltage, and move between ON and OFF states under external control. A small-signal transistor is not designed for that duty at the power levels used in battery chargers, UPS systems, wind-energy interfaces, or inverter auxiliaries. The power BJT was developed for this role and became one of the earliest widely used controllable solid-state power switches in choppers and switch-mode power supplies before MOSFETs and IGBTs displaced it in many newer designs [IEEE EDS, 75th Anniversary of the Transistor], [ST AN656].

This chapter introduces the power BJT as both a device and a model for studying power-semiconductor behavior. It examines structure, quasi-saturation, safe operating area, switching behavior, and key datasheet specifications, with the aim of making the main design limits legible in a real device.

Prerequisites check

- You should know the basic idea of a PN junction diode in forward bias and reverse bias.
- You should know that a transistor can act as a switch, with **cutoff** meaning OFF and **saturation** meaning strongly ON.
- You should be comfortable with current, voltage, power, and the relation $P = VI$.
- You should know the difference between DC and AC, and you should be able to recognize that a 230 V, 50 Hz mains supply becomes a high-voltage DC bus after rectification.
- You should have a basic idea that semiconductor devices heat up when they dissipate power, and too much junction temperature can damage them.

The chapter assumes this background and treats the transistor as a physical switching device, not only as a circuit symbol.

Core content

1.1.1 History of the Power BJT

Early power-control systems relied on electromechanical relays and contactors. These devices could switch current, but they were slow, noisy, bulky, and subject to contact wear. A diode could conduct large current but could not be turned ON and OFF by command. A thyristor could be triggered ON, but it was less convenient to turn OFF. Power electronics therefore needed a solid-state device that behaved as an externally commanded switch.

The invention of the transistor in 1947 opened that path [IEEE EDS, 75th Anniversary of the Transistor]. As bipolar junction transistor technology matured, devices were developed with larger junction area, heavier current capability, higher voltage blocking ability, and packages that could remove much more heat than ordinary small-signal transistors. These became **power BJTs**. For many years, they were central to choppers, switch-mode power supplies, deflection circuits, motor controls, and early inverter stages [ST AN656].

A BJT is a three-terminal device in which base current influences a much larger collector current. That made the BJT one of the earliest practical solid-state power switches that could both turn ON and turn OFF under transistor-drive control. This was a major advantage over uncontrolled rectifiers and over devices that were harder to turn OFF.

At the same time, power BJTs brought their own difficulties. They are **current-controlled** devices. This means the drive circuit must continuously supply base current while the transistor is ON. At high power, that drive current can become large. Also, when a BJT is driven deeply into saturation, stored charge inside the device makes turn-OFF slower. These two problems became major reasons why power MOSFETs and, later, IGBTs became dominant in many new designs.

In modern high-efficiency main power stages such as kilowatt-level rooftop PV string inverters, EV traction inverters, and large battery-energy-storage converters, power BJTs are usually no longer the first choice. They remain important, however, in legacy equipment, in some low-cost or specialized switching circuits, and in engineering education. Concepts such as current gain, saturation, charge storage, secondary breakdown, and safe operating area are easier to see in the BJT than in later devices. For that reason, the power BJT remains a useful entry point to the study of MOSFETs and IGBTs.

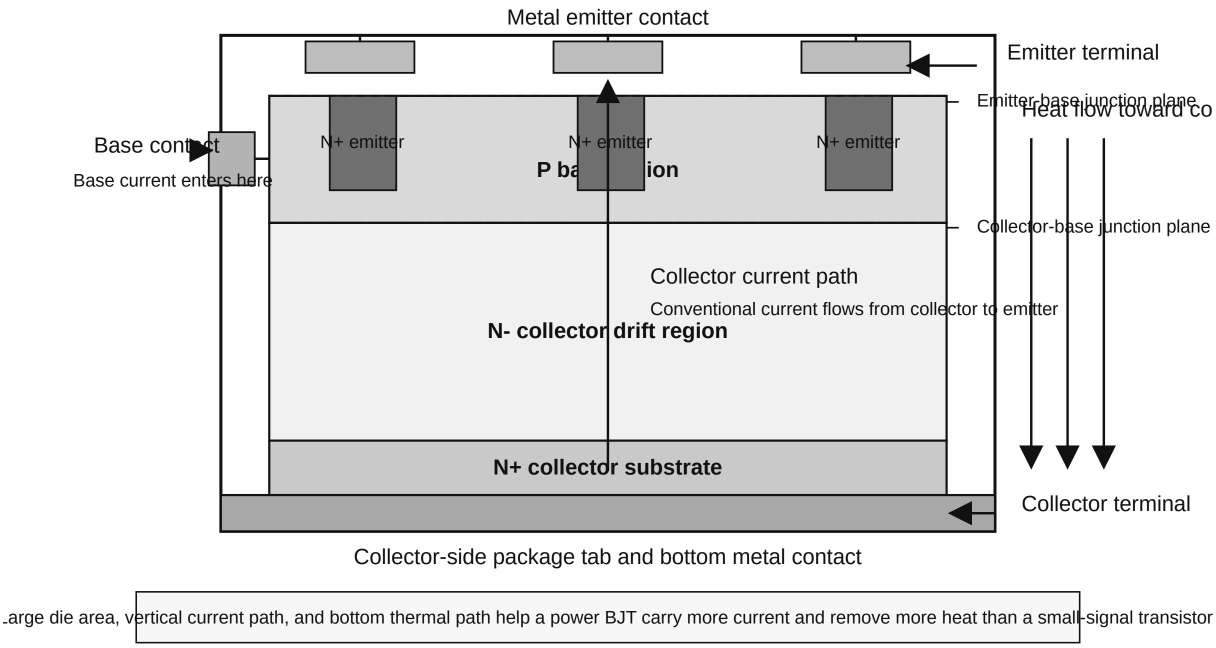
1.1.2 Structure, quasi-saturation, safe operating area (SOA)

A power BJT is not merely a larger small-signal transistor. It is designed to satisfy two competing requirements: high-voltage blocking in the OFF state and high-current conduction in the ON state. The internal structure must therefore provide both a robust voltage-supporting region and a low-resistance current path with adequate heat removal.

Physical structure of a power BJT

In power electronics, the most common discussion is around the **NPN power BJT**. The internal construction is usually vertical rather than purely lateral. That means the main current path goes from the top of the silicon die to the bottom, instead of flowing only across the surface. This helps the device use silicon area more effectively and improves current handling.

Figure 1.1 Vertical NPN Power BJT Cross-Section



Source: Author-generated academic monochrome vector diagram created for this chapter.

The **emitter** is heavily doped so that it can inject carriers effectively. The **base** is thin and lightly doped enough that a relatively small base current can control a much larger collector current. The **collector** region includes a lightly doped portion, often called a drift region, so that the device can withstand higher collector-emitter voltage in the OFF state. The die area is much larger than that of a small-signal transistor, and the package is designed to connect thermally to a heat sink.

Manufacturer documents for switching BJTs often mention internal layout choices such as **cellular emitter structure** or **multi-epitaxial planar technology**. The ST13007 datasheet, for example, states that the device uses high-voltage multi-epitaxial planar technology and a cellular emitter structure to improve switching speed while maintaining wide reverse-bias safe operating area [ST13007 Datasheet]. Such details show that switching behavior and ruggedness depend on internal geometry, not only on terminal ratings.

Basic current relations

A transistor is a three-terminal device, so current conservation must hold. The emitter current is the sum of collector current and base current:

$$\boxed{I_E = I_C + I_B} \quad (1.1)$$

Here, I_E is emitter current, I_C is collector current, and I_B is base current.

In the forward-active region, a useful first relation is the current gain:

$$\boxed{\beta = \frac{I_C}{I_B}} \quad (1.2)$$

Here, β is the DC current gain. Many datasheets write it as h_{FE}.

Equation (1.2) is most meaningful in the active region under stated test conditions. In switching circuits, the nominal gain alone is usually not trusted. Instead, a smaller **forced beta** is chosen to guarantee that the transistor enters saturation:

$$\boxed{I_B \geq \frac{I_C}{\beta_{\text{forced}}}} \quad (1.3)$$

In this expression, β_{forced} is not a fixed material constant. It is a design choice. Engineers often choose it much smaller than the datasheet h_{FE} value so that the transistor turns ON firmly under worst-case conditions.

For example, suppose a power BJT must carry $I_C = 5$ A in a switching circuit. If we choose $\beta_{\text{forced}} = 5$, then the required base current is

$$I_B = \frac{5 \text{ A}}{5} = 1 \text{ A}.$$

That is a large control current. This one number already tells us why power BJTs place a heavy burden on their drive circuits.

Operating regions as a power switch

Four operating conditions must be distinguished in power-BJT switching: **cutoff**, **active region**, **quasi-saturation**, and **hard saturation**.

Table 1.2: Practical operating regions of a power BJT

Region	External behavior	Junction picture	Power-electronics meaning
Cutoff	Very little collector current	Emitter-base not forward biased	Switch is OFF
Active region	I_C approximately controlled by base current	Emitter-base forward biased, collector-base reverse biased	Used mainly during transition, not as steady ON state in power switching
Quasi-saturation	V_{CE} has dropped significantly, but extra base drive gives diminishing reduction in V_{CE} and more stored charge	The collector side is no longer behaving like a simple reverse-biased region everywhere	Common practical region in power BJTs as they are driven hard

Region	External behavior	Junction picture	Power-electronics meaning
Hard saturation	Both junctions effectively forward biased and stored charge is high	Deeply saturated condition	Low V_{CE}, but turn-OFF becomes slower

In **cutoff**, collector current is limited to leakage and the device is OFF. In the **active region**, base current controls collector current; this region is central to transistor theory but undesirable as a steady ON state in power switching because both V_{CE} and I_C may be substantial at the same time. Between the active region and hard saturation, power BJTs often operate in **quasi-saturation**. In **hard saturation**, V_{CE} becomes relatively small, which reduces conduction loss, but stored charge rises and turn-OFF slows.

What is quasi-saturation?

Power BJTs use a thicker, more lightly doped collector structure than small-signal transistors because they must block higher voltage. As base drive increases, the transition from active mode to deep saturation is therefore not abrupt. In practice, the device often enters an intermediate region in which V_{CE} is already low, but additional base current produces only limited further reduction while increasing stored charge.

For this chapter, **quasi-saturation** means operation beyond the ordinary active region in which further base overdrive yields diminishing reduction in V_{CE} and a larger turn-OFF penalty.

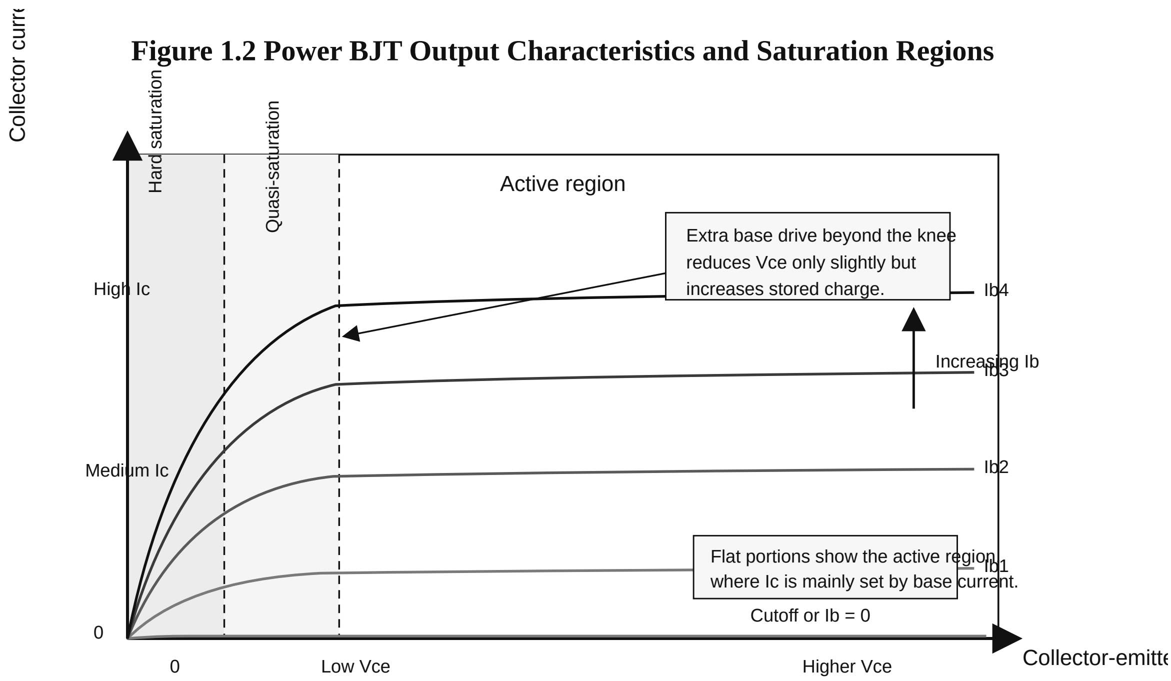


Figure 1.2 Power BJT Output Characteristics and Saturation Regions

Source: Author-generated academic monochrome vector diagram created for this chapter.

Additional base current can help ensure turn-ON, but excessive overdrive pushes the transistor deeper toward hard saturation and increases storage time. Good design requires enough drive to establish conduction without unnecessary charge storage [ST AN656].

Safe operating area (SOA)

The **safe operating area**, usually shortened to **SOA**, is the region on a datasheet graph within which the transistor can operate without destructive failure under stated conditions. Maximum collector current and maximum collector-emitter voltage cannot be checked independently, because safe operation depends on the combined stress of current, voltage, pulse duration, and temperature. SOA is therefore plotted as an I_C-V_{CE} boundary, often with separate curves for different pulse durations [onsemi AN875/D], [onsemi 2N3055 Datasheet].

The main SOA boundaries are as follows.

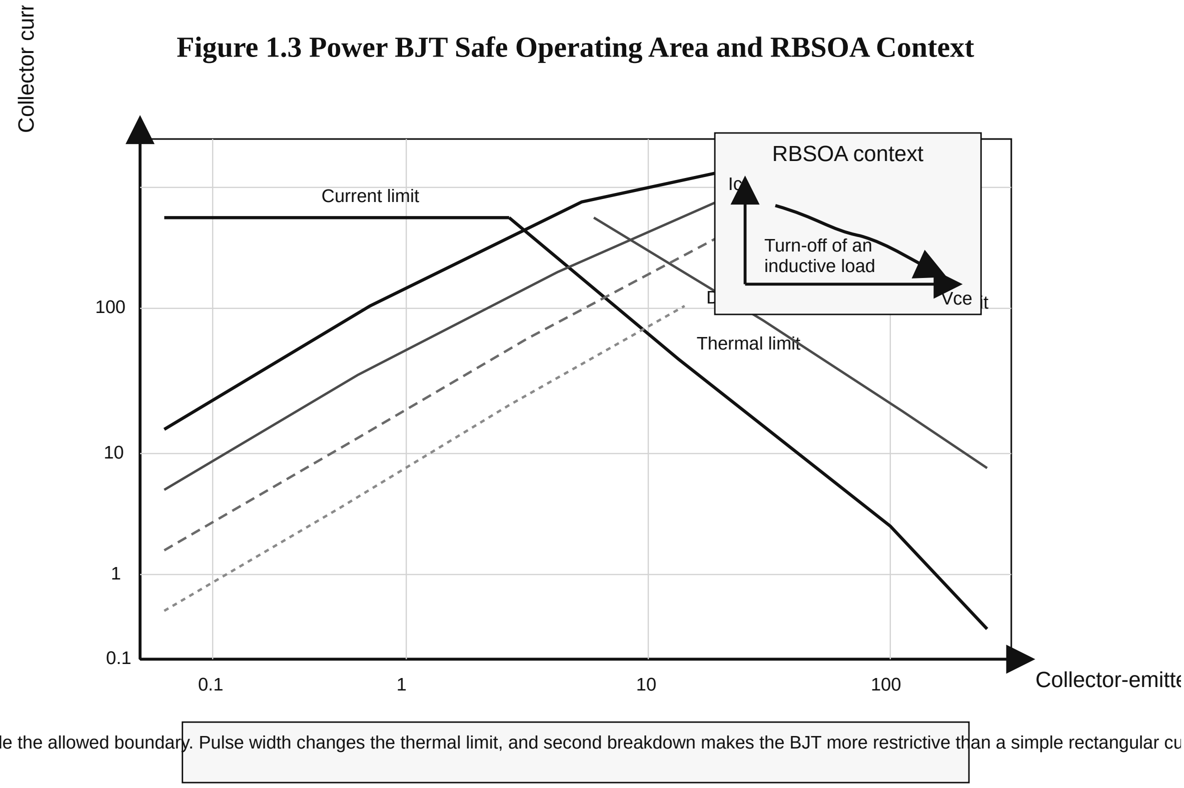
Current limit. Bond wires, silicon area, and package leads can only carry so much current.

Thermal limit. If V_{CE} and I_C are both substantial, the instantaneous power $P = V_{CE}I_C$ can become large. Over time, the junction heats. Short pulses may be tolerated better than DC, because the junction has not yet heated fully.

Second breakdown limit. This is especially important in BJTs. Localized current crowding can create a hot spot. The hot spot carries more current, heats more, and can rapidly destroy the device. This can happen even when the average power seems acceptable. BJT SOA is therefore more restrictive than a simple rectangular voltage-current box.

Reverse-bias safe operating area (RBSOA). During turn-OFF of an inductive load, the transistor may see both rising voltage and falling current while the base drive is being removed, sometimes with reverse base current extraction. Manufacturer notes treat this as a special safe-operating condition because turn-OFF can be very stressful [onsemi AN875/D], [ST AN656].

Figure 1.3 Power BJT Safe Operating Area and RBSOA Context



Source: Author-generated academic monochrome vector diagram created for this chapter.

One of the most important practical consequences is that a device may survive a high current for a very short pulse but not for DC. That is why datasheets frequently show several pulse-duration lines.

Inductive loads are particularly demanding at turn-OFF. If a transistor is carrying current in an inductor, the collector-emitter voltage can rise sharply as the current is forced to continue. The device may then see high voltage and significant current at the same moment. That combination moves the operating point across the SOA graph; if it leaves the safe region, failure can occur.

1.1.3 Switching characteristics and specifications

Switching behavior determines whether a power transistor is usable in a real converter circuit.

Why switching is not instantaneous

A power BJT does not change state instantaneously, because charge must be injected during turn-ON and removed during turn-OFF.

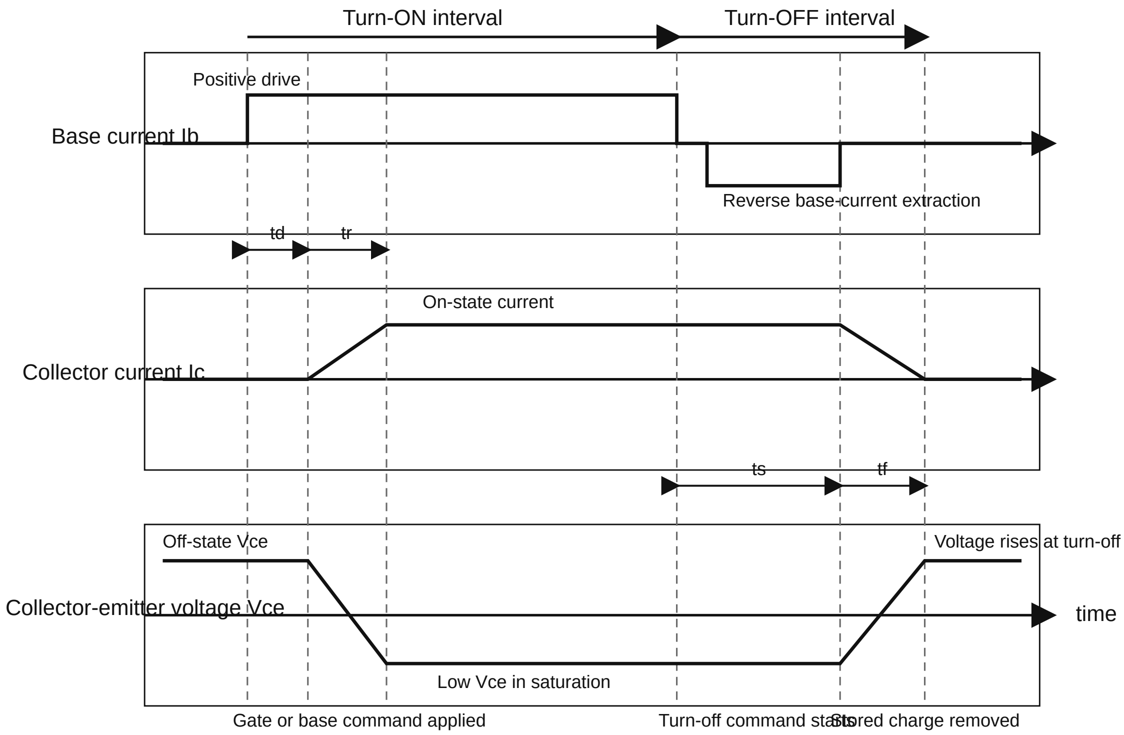
When the device turns ON, base current begins to build charge in the internal junctions and regions, so collector current does not jump instantly to its final value. When the transistor turns OFF, that stored charge does not disappear instantly either. This is the main reason that turn-OFF can be relatively slow.

The standard switching intervals used in datasheets are:

- **Delay time** t_d: the interval between application of base drive and the start of significant collector-current change.
- **Rise time** t_r: the interval during which collector current rises to its ON-state value and V_{CE} falls.
- **Storage time** t_s: after turn-OFF drive begins, the transistor may still conduct heavily because stored charge remains. This interval is storage time.
- **Fall time** t_f: the interval during which collector current actually falls and V_{CE} rises to the OFF-state value.

For power BJTs, storage time is often the most revealing interval because it captures the turn-OFF penalty created by stored charge.

Figure 1.4 Power BJT Switching Waveforms



Source: Author-generated academic monochrome vector diagram created for this chapter.

During turn-ON, the drive circuit pushes current into the base. At first, the transistor is still mostly OFF, so collector current is low and V_{CE} is high. This is the delay interval. The device then moves toward conduction: collector current rises and collector-emitter voltage falls until a low-V_{CE} ON state is reached.

During turn-OFF, base drive is removed, and many circuits apply reverse base current to extract stored charge more quickly. The transistor nevertheless continues to conduct for a storage interval before collector current falls rapidly. Deep saturation therefore improves ON-state voltage drop at the cost of slower turn-OFF.

Conduction loss and base-drive burden

When the transistor is ON in saturation, its collector-emitter voltage is not zero. It may be 1 V, 2 V, or more, depending on current and drive conditions. That causes conduction loss.

A first practical estimate is

$$\boxed{P_{\text{cond}} \approx V_{CE(\text{sat})} I_C} \quad (1.4)$$

Here, P_{cond} is conduction loss inside the transistor, $V_{CE(\text{sat})}$ is the collector-emitter saturation voltage, and I_C is collector current.

If the transistor is ON only for a fraction D of each switching cycle, where D is the duty ratio, then the average conduction loss is approximately

$$\boxed{P_{\text{cond,avg}} \approx D\, V_{CE(\text{sat})} I_C} \quad (1.5)$$

The drive circuit also consumes power because it must deliver base current. An approximate average base-drive power is

$$\boxed{P_{B,\text{avg}} \approx D\, V_{BE(\text{sat})} I_B} \quad (1.6)$$

In this equation, $V_{BE(\text{sat})}$ is the base-emitter saturation voltage.

These equations show that a power BJT dissipates power both in its collector-emitter path and in the base-drive circuit. The need for continuous base current is one reason later voltage-driven devices became attractive.

Switching-loss estimate

During switching, voltage and current overlap. That overlap creates switching loss. A simple first estimate, using a triangular overlap during rise and fall, is

$$\boxed{E_{\text{sw,approx}} \approx \frac{1}{2} V_{CC} I_C (t_r + t_f)} \quad (1.7)$$

Here, $E_{\text{sw,approx}}$ is approximate switching energy per cycle, V_{CC} is the supply or blocking voltage, I_C is the switched collector current, and t_r, t_f are rise and fall times.

Then average switching power is approximately

$$\boxed{P_{\text{sw,approx}} \approx f_s E_{\text{sw,approx}}} \quad (1.8)$$

where f_s is the switching frequency.

For BJTs, however, this approximation often underestimates turn-OFF loss when the device has been driven deeply into saturation. Storage time keeps collector current flowing before the fall interval really begins, so real turn-OFF energy may be larger than Equation (1.7) suggests.

Temperature and thermal specification

All semiconductor ratings ultimately reduce to a thermal question: how hot does the junction become?

A simple thermal relation from junction to case is

$$\boxed{T_J \approx T_C + P_D R_{\theta JC}} \quad (1.9)$$

Here, T_J is junction temperature, T_C is case temperature, P_D is device power dissipation, and $R_{\theta JC}$ is thermal resistance from junction to case.

The relation shows that, for a given case temperature, higher dissipation raises the junction temperature, while lower thermal resistance helps. Later, when heat sinks are studied in Module 2, this idea will be extended to the full path from junction to ambient.

Worked numerical example

The following estimate uses values from a real switching transistor.

Consider the ST13007, a high-voltage fast-switching NPN power transistor used in switch-mode power supplies [ST13007 Datasheet]. Its datasheet gives, among other values, a collector-emitter saturation voltage of about 1 V at collector current 2 A and base current 0.4 A, and a base-emitter saturation voltage of about 1.2 V at the same operating point. Its junction-to-case thermal resistance is 1.56 °C/W.

Imagine that this transistor is used in a low-power auxiliary SMPS inside a rooftop solar inverter control cabinet. During each cycle, the transistor is ON for duty ratio 0.35 and carries 2 A when ON. Ignore switching loss for this first estimate.

First, what forced beta is the design using?

$$\beta_{\text{forced}} = \frac{I_C}{I_B} = \frac{2}{0.4} = 5.$$

The design therefore uses forced beta 5.

Next, the average conduction loss inside the transistor is

$$P_{\text{cond,avg}} \approx D V_{CE,\text{sat}} I_C.$$

Substituting the numbers,

$$P_{\text{cond,avg}} \approx 0.35 \times 1\,\text{V} \times 2\,\text{A} = 0.70\,\text{W}.$$

Now estimate the average base-drive power:

$$P_{B,\text{avg}} \approx D\,V_{BE,\text{sat}} I_B.$$

So,

$$P_{B,\text{avg}} \approx 0.35 \times 1.2\,\text{V} \times 0.4\,\text{A} = 0.168\,\text{W}.$$

This 0.168 W is not dissipated in the collector-emitter path, but it is still power that the drive circuit must supply.

Finally, estimate the junction temperature rise above the case due only to the transistor's average conduction loss:

$$T_J - T_C \approx P_D R_{\theta\text{JC}}.$$

Taking P_D as approximately 0.70 W for this simple estimate,

$$T_J - T_C \approx 0.70 \times 1.56 = 1.092.$$

So the junction is about 1.092 °C hotter than the case.

If the case temperature is 80 °C, then the estimated junction temperature is

$$T_J \approx 80 + 1.092 \approx 81.1.$$

So the estimated junction temperature is about 81.1 °C.

This estimate is intentionally incomplete: it ignores switching loss, case-to-ambient heating, and circuit-dependent variation in saturation voltage. It nevertheless shows that collector loss can be modest in a low-power auxiliary converter while the required base-drive current remains substantial.

Reading key specifications in a datasheet

A power-transistor datasheet contains many numbers. The most important ones are grouped below.

Table 1.3: Key power-BJT specifications and what they mean

Symbol	Meaning	Why you should care
V_{CEO} or $V_{CEO(\text{sus})}$	Collector-emitter voltage rating with base open	Tells you the off-state voltage limit in a common test condition

Symbol	Meaning	Why you should care
V_{CES}	Collector-emitter voltage with base shorted to emitter	Often larger than V_{CEO}; useful in off-state stress interpretation
I_C, I_{CM}	Continuous and peak collector current	Needed for load current and transient current checks
h_{FE}	DC current gain under specified test conditions	Helps estimate base-drive needs, but does not replace forced-beta design
$V_{CE(\text{sat})}$	Collector-emitter saturation voltage	Directly affects ON-state loss
$V_{BE(\text{sat})}$	Base-emitter saturation voltage	Helps estimate base-drive burden
t_s, t_f and sometimes t_d, t_r	Storage, fall, delay, and rise times	Show switching speed and turn-OFF difficulty
SOA / RBSOA	Safe current-voltage operating boundaries	Prevents destructive operation during switching and faults
P_{TOT}	Total dissipation	Gives a package-level dissipation limit under stated conditions
$R_{\theta JC}$, T_J max	Thermal resistance and maximum junction temperature	Needed to judge temperature rise and heat-sink need

A device with higher h_{FE} is not automatically better for switching service. Voltage rating, $V_{CE(\text{sat})}$, switching times, SOA, thermal resistance, and the required base drive matter at least as much.

Worked interpretation exercise

The ST13007 datasheet provides a compact example of how a switching power-BJT datasheet is interpreted.

The title and description identify the device as a "high voltage fast-switching NPN power transistor." The description states that it uses multi-epitaxial planar technology and a cellular emitter structure for high switching speed and wide RBSOA [ST13007 Datasheet]. Even before the numeric tables are read, the intended use is clear: high-voltage switching service such as switch-mode power supplies, not low-voltage relay replacement.

The absolute maximum ratings list $V_{CES} = 700$ V and $V_{CEO} = 400$ V, along with continuous collector current $I_C = 8$ A and peak collector current $I_{CM} = 16$ A [ST13007 Datasheet]. This places the device in the high-voltage BJT family. A rectified 230 V AC supply is about 325 V DC, so a transistor with $V_{CEO} = 400$ V is in the correct general class for such service, though the design would still need margin and usually a snubber or clamp for switching spikes.

The thermal data give maximum junction temperature $T_J = 150°\text{C}$ and junction-to-case thermal resistance $R_{\theta JC} = 1.56°\text{C/W}$ [ST13007 Datasheet]. Heat-sink design therefore cannot be ignored. Even an electrically strong device can still fail thermally.

The saturation and gain data must be read together. The table gives $V_{CE(\text{sat})} = 1$ V at $I_C = 2$ A and $I_B = 0.4$ A, and $V_{BE(\text{sat})} = 1.2$ V at the same operating point [ST13007 Datasheet]. It also gives DC current gain h_{FE} values such as 16 to 40 at $I_C = 2$ A and $V_{CE} = 5$ V, depending on gain group [ST13007 Datasheet].

These entries describe two different operating conditions. The h_{FE} value is measured in the active region under specified voltage conditions, whereas the saturation voltage is measured under hard switching-drive conditions. It would therefore be incorrect to say, "The transistor has gain 20, therefore $I_B = I_C/20$ is enough for switching." The datasheet itself shows that for $I_C = 2$ A, the saturation test uses $I_B = 0.4$ A, which corresponds to forced beta 5, not 20.

The switching-time entries also depend on context. For a resistive-load test at $V_{CC} = 300$ V and $I_C = 2$ A, the datasheet gives storage time up to $4.5\ \mu$s and fall time about 350 ns [ST13007 Datasheet]. For an inductive-load test, it gives different storage and fall times under specified clamp and reverse-base-drive conditions [ST13007 Datasheet]. Switching speed therefore depends on the test circuit; there is no single universal switching time.

The datasheet includes both a safe operating area plot and a reverse-biased SOA plot [ST13007 Datasheet]. Their presence indicates intended use in real switching conditions where turn-OFF stress matters.

Table 1.4 summarizes the interpretation.

Table 1.4: Interpreting the ST13007 datasheet

Datasheet item	What it tells us	Design meaning
Title and description	High-voltage fast-switching NPN power transistor	Intended for SMPS-type switching, not general low-voltage only
`V_CEO = 400 V`, `V_CES = 700 V`	High off-state voltage class	Suitable general class for off-line DC-bus switching with proper clamp margin
`I_C = 8 A`	Current capability	Must still be checked against SOA, not used blindly
`V_CE,sat = 1 V` at `2 A` collector current and `0.4 A` base current	ON-state drop under strong base drive	Conduction loss is significant but manageable at moderate current
`h_FE` grouping	Gain varies by device group and test condition	Active-region gain is not the same as switch-design forced beta
`t_s` and `t_f` data	Turn-OFF charge storage is real and measurable	Deep saturation slows turn-OFF and increases loss

Datasheet item	What it tells us	Design meaning
SOA and RBSOA figures	Voltage and current must be checked together	Inductive switching can be destructive without protection
`R_thetaJC` and maximum `T_J`	Thermal path matters	Heat sink and temperature rise must be verified

A datasheet is best read as a coordinated set of limits and operating clues, not as a collection of isolated numbers.

How this matters in renewable-energy systems

Even when main renewable-energy converter stages use MOSFETs or IGBTs, the power BJT remains instructive. Older or lower-cost auxiliary supplies in inverters, chargers, and control cabinets often used high-voltage switching BJTs, and the same design questions appear in modern devices: drive requirement, conduction drop, switching loss, SOA, and thermal limits.

Renewable-energy systems also expose power devices to the stresses emphasized in this chapter: inductive current, high DC voltage, repeated switching, heat, and fault transients. The BJT makes these limits particularly visible, which is why it remains a useful first device in power-electronics study.

Chapter summary

- A **power BJT** is a bipolar junction transistor designed for higher current, higher voltage, and higher heat dissipation than a small-signal transistor.
- The power BJT was one of the earliest practical controllable solid-state switches used in power converters before MOSFETs and IGBTs became dominant in many new designs [IEEE EDS, 75th Anniversary of the Transistor], [ST AN656].
- In a BJT, emitter current, collector current, and base current are related by $I_E = I_C + I_B$.
- In the active region, the current gain is $\beta = I_C/I_B$, but switching design usually uses a smaller **forced beta** to ensure saturation.
- A power BJT has operating regions including cutoff, active region, quasi-saturation, and hard saturation. Quasi-saturation is important because extra base drive gives diminishing reduction in V_{CE} while increasing stored charge.
- **Safe operating area (SOA)** is a current-voltage-time-temperature boundary, not a single number. It is shaped by current limit, thermal limit, and second breakdown [onsemi AN875/D], [onsemi 2N3055 Datasheet].
- Power BJTs store charge, so switching is not instantaneous. Important switching intervals include delay time, rise time, storage time, and fall time.
- ON-state conduction loss is approximately $P_{\text{cond}} \approx V_{CE(\text{sat})} I_C$, while average base-drive power is approximately $P_{B,\text{avg}} \approx D\, V_{BE(\text{sat})} I_B$.
- A simple thermal estimate is $T_J \approx T_C + P_D R_{\theta JC}$, but real temperature rise also depends on the rest of the thermal path.

- In renewable-energy systems, power BJTs are more important today as a conceptual foundation, an auxiliary-stage device in some designs, and a bridge to understanding later devices such as MOSFETs and IGBTs.

Further reading

- ST13007 Datasheet, STMicroelectronics — A real high-voltage switching power-BJT datasheet with saturation data, switching times, SOA, and RBSOA.
- 2N3055 / MJ2955 Datasheet, onsemi — A classic power-transistor datasheet that is especially useful for learning safe operating area and second-breakdown interpretation.
- AN875/D: Power Transistor Safe Operating Area, onsemi — A concise official note on forward-bias and reverse-bias SOA and why switching stress must be checked on an I_C–V_{CE} plot.
- AN656: Power Transistors - Devices and Datasheets, STMicroelectronics — A practical manufacturer note on reading power-transistor datasheets, base-drive concerns, temperature effects, and RBSOA.
- The Transistor at 75, IEEE Electron Devices Society / IEEE Spectrum — A useful historical overview of the transistor's development, which helps place the power BJT in the larger evolution of power semiconductor devices.

Chapter 1.2: Thyristor (SCR)

Chapter opening

A **thyristor**, or **silicon controlled rectifier (SCR)**, is a four-layer power semiconductor that can be triggered into conduction by a gate signal but cannot ordinarily be turned OFF by the gate. Once the device has latched, the current through it must fall below a critical value before blocking action is restored. This behavior made the SCR one of the foundational devices of power electronics.

Commercially introduced in 1957, the SCR moved power control away from electromechanical and gas-discharge devices toward reliable solid-state switching [IEEE Milestone: Silicon Controlled Rectifier (SCR/Thyristor), 1957]. Although MOSFETs and IGBTs dominate high-frequency PWM converters, SCRs remain important in controlled rectification, phase control, crowbar protection, inrush-current limiting, soft starters, and other high-power line-frequency applications where rugged current handling and low on-state loss are more important than fast turn-OFF [ST AN4607], [Littelfuse Phase Control].

Prerequisites check

- forward and reverse bias of a PN junction
- basic transistor action from Chapter 1.1, especially current gain and switching operation
- RMS and peak values of AC quantities; for a sinusoid, $V_m = \sqrt{2}V_{rms}$
- the fact that an inductor resists sudden change of current and a capacitor resists sudden change of voltage
- the idea that semiconductor devices have electrical ratings and temperature limits

The chapter assumes familiarity with current paths, blocking states, and basic switching stress.

2.1.1 History of the Thyristor (SCR)

Before solid-state power control became practical, heaters, battery chargers, motors, and rectifier front ends were controlled by mechanical switches, rheostats, or bulky gas-filled devices. Engineers needed a semiconductor device that could block high voltage, carry high current, and be triggered by a small control signal. The SCR met that need.

IEEE's milestone document notes that General Electric introduced the silicon controlled rectifier in 1957 and that the device marked a turning point in electric-energy control [IEEE Milestone: Silicon Controlled Rectifier (SCR/Thyristor), 1957]. Its significance lay in a particular combination of properties: substantial OFF-state voltage blocking, large current capability, relatively low on-state drop, and electronic triggering by a gate signal. Its main limitation was equally important. A conventional SCR could be turned ON by the gate, but once latched it could not normally be turned OFF by the gate. Converter design therefore had to rely on natural or forced commutation.

The SCR also became the basis of a wider thyristor family. The IEEE milestone note identifies later devices such as the TRIAC and GTO as descendants of the SCR [IEEE Milestone: Silicon Controlled Rectifier (SCR/Thyristor), 1957]. The device remains important in line-frequency power control, static switches, soft starters, crowbar protection, high-power rectifiers, and applications where ruggedness and current handling matter more than very high switching frequency [ST AN4607], [Littelfuse MCR16NG Product Page], [Littelfuse Phase Control].

Table 2.1: Why the SCR became a foundational power device

Reason	Why it mattered
High-voltage blocking capability	Made direct connection to power circuits practical
High current capability	Allowed real industrial loads to be controlled
Gate triggering	Enabled electronic control without a large mechanical actuator
Low on-state drop	Reduced conduction loss compared with many older control methods
Rugged construction	Suited industrial and utility environments

SCRs are no longer the universal choice in power conversion, but they are not merely historical devices. They remain practical wherever line-frequency efficiency, surge tolerance, and latching behavior are useful.

2.1.2 Construction, two-transistor analogy, operating principle

An SCR is often described as a power device with memory. A transistor responds continuously to its control input, whereas an SCR responds to a trigger and then remains ON because of internal regenerative action until the external circuit reduces its current sufficiently.

Physical structure

An SCR is a **four-layer, three-junction, three-terminal** semiconductor device. Its terminals are:

- **Anode**
- **Cathode**
- **Gate**

The internal layers are arranged in **PNPN** order, forming three junctions labeled J_1, J_2, and J_3 [ST AN4607].

Image prompt for Figure 2.1: Create a clean textbook-style technical illustration of an SCR cross-section showing the four semiconductor layers in vertical order as P, N, P, N. Label the top terminal as anode and the bottom terminal as cathode. Show the gate connected near the cathode-side P-layer. Mark the three junctions as J_1, J_2, and J_3. Add two small side insets: one for forward blocking with anode positive, showing J_1 and J_3 forward biased and J_2 reverse biased, and one for reverse blocking with cathode positive, showing the appropriate reverse-biased junctions. Use monochrome engineering style, no decorative elements.

When the anode is positive with respect to the cathode, the outer junctions J_1 and J_3 are forward biased while the middle junction J_2 remains reverse biased. The device is then in **forward blocking**: it is forward biased overall but does not yet conduct strongly. When the cathode is positive with respect to the anode, the device is in **reverse blocking**. Only a small leakage current flows until reverse breakdown is reached, which is outside normal operating conditions [ST AN4607], [ST AN4608].

This ability to remain OFF under forward bias and then remain ON after triggering distinguishes the SCR from both a diode and an ordinary transistor.

The two-transistor analogy

The internal PNPN structure can be represented by the **two-transistor analogy**. In this model, the SCR is viewed as two tightly coupled transistor actions:

- a **PNP transistor** near the anode side
- an **NPN transistor** near the cathode side

These are not two separate packaged transistors. They are a conceptual representation of the internal structure [ST AN4607], [Teccor Thyristor Design Guide].

Image prompt for Figure 2.2: Create a textbook-style diagram of the SCR two-transistor analogy. Show one PNP transistor on the left and one NPN transistor on the right. Connect the collector of each transistor to the base of the other to illustrate regenerative feedback. Label the anode at the emitter of the PNP transistor, the cathode at the emitter of the NPN transistor, and the gate entering the base of the NPN transistor. Add current arrows and annotate "regenerative feedback" between the two collector-base cross-connections.

Let the PNP section be T_1 and the NPN section be T_2. Their coupling is the essential feature:

- the emitter of T_1 is connected to the anode
- the emitter of T_2 is connected to the cathode
- the collector of T_1 feeds the base of T_2
- the collector of T_2 feeds the base of T_1
- the gate injects current into the base region of T_2

The latching process can then be described in a short sequence:

1. With the anode positive, the SCR is forward biased but still OFF. The internal transistor actions are weak and only leakage currents exist.
2. A gate pulse supplies current to the NPN section T_2, which begins to conduct.
3. The collector current of T_2 becomes base drive for the PNP section T_1, so T_1 also begins to conduct.
4. The collector current of T_1 feeds additional base drive back into T_2. Positive feedback therefore builds rapidly between the two internal transistor actions.
5. Once the feedback is strong enough, conduction becomes self-sustaining. The SCR **latches** and remains ON until the anode current falls low enough for the regenerative loop to collapse.

A simplified current relation from the two-transistor model is [ST AN4607]

$$\boxed{I_A = \frac{I_{CBO1} + I_{CBO2} + \alpha_2 I_G}{1 - (\alpha_1 + \alpha_2)}} \quad (2.1)$$

where

- I_A is the anode current
- I_G is the gate current
- α_1 is the common-base current gain of the PNP section
- α_2 is the common-base current gain of the NPN section
- I_{CBO1} and I_{CBO2} are leakage-current terms of the two sections

The denominator shows the essential behavior. As $\alpha_1 + \alpha_2$ approaches 1, the denominator becomes small and the anode current rises sharply. The latching condition is therefore expressed conceptually as

$$\boxed{\alpha_1 + \alpha_2 \to 1} \quad (2.2)$$

Equation (2.2) is not used as a direct design formula. It explains why the SCR turns ON through regenerative feedback rather than by gradual, continuously controllable conduction.

A simple numerical intuition

For a 230 V, 50 Hz AC supply,

$$\boxed{V_m = \sqrt{2} V_{rms}} \quad (2.3)$$

so

$$V_m = \sqrt{2} \times 230 \approx 325 \text{ V}.$$

An SCR connected to this supply may remain in forward blocking during part of a positive half-cycle even while the instantaneous anode voltage is rising toward 325 V. Conduction begins only when a gate pulse arrives or forward breakover is reached. This controllable delay is the basis of phase control.

What makes the SCR attractive

The SCR combines low conduction loss in the ON state with high voltage blocking in the OFF state. That combination makes thyristors effective in high-current, line-frequency applications [ST AN4607].

The same structure also creates the device's main limitation. Charge is stored deeply in the silicon, and a conventional SCR cannot ordinarily be gate-turned-OFF. It is therefore most useful where natural commutation is available or where a separate commutation circuit can be justified.

2.1.3 Static V-I characteristics: latching current, holding current, break-over voltage, forward and reverse blocking

The static V-I characteristic identifies the main operating regions of an SCR: reverse blocking, reverse breakdown, forward blocking, and forward conduction.

Image prompt for Figure 2.3: Create a clean engineering plot of the static V-I characteristic of an SCR. Horizontal axis: anode-cathode voltage V_{AK} in volts. Vertical axis: anode current I_A in amperes. Show reverse blocking in the third quadrant, forward blocking in the first quadrant, forward breakover voltage V_{BO}, the sudden transition to ON state, and the low-voltage high-current on-state branch. Mark latching current I_L on the rising current path just after triggering and holding current I_H on the falling current path, with $I_L > I_H$. Label reverse breakdown, forward blocking, on-state, and leakage regions clearly.

The main regions

Table 2.2: Main regions of the SCR static characteristic

Region	External condition	Device behavior	Practical meaning
Reverse blocking	Cathode positive with respect to anode	Very small reverse leakage current	SCR is OFF in reverse direction
Reverse breakdown	Reverse voltage too high	Large reverse current	Abnormal and unsafe operating region
Forward blocking	Anode positive but not yet triggered	Only a small forward leakage current	SCR is forward biased but still OFF
Forward conduction	Device triggered or broken over	Large anode current with small on-state voltage	SCR is ON

In **forward blocking**, the SCR is ready to conduct but has not yet been triggered. In **forward conduction**, it behaves approximately like a closed switch with a small forward voltage drop.

Forward breakover voltage

If the forward anode-cathode voltage is increased sufficiently without a gate pulse, the SCR can enter conduction at the **forward breakover voltage** V_{BO} [ST AN4607]. This is a real turn-ON mechanism, but it is not the normal design method. Breakover triggering gives poor timing control, increases stress, and depends strongly on device condition and temperature. In practical circuits, the SCR is normally triggered by the gate at a lower and controlled voltage.

Latching current and holding current

Two current levels are particularly important:

- **Latching current**, I_L
- **Holding current**, I_H

The **latching current** is the minimum anode current that must be reached immediately after turn-ON so that the SCR remains ON when the gate signal is removed [ST AN303], [ST TYN612 Datasheet]. The **holding current** is the minimum anode current below which the SCR returns to the OFF state after conduction is already established [ST AN302], [ST TYN612 Datasheet].

In normal devices,

$$\boxed{I_L > I_H} \quad (2.4)$$

The difference arises because conduction is not yet fully established at the instant of triggering. The device needs a higher current to latch firmly than it needs to remain in conduction afterward.

Datasheet values make the distinction concrete. The ST TYN612 family lists maximum holding-current values such as 15 mA, 30 mA, or 40 mA, and latching-current values such as 30 mA, 60 mA, or 80 mA depending on the sensitivity class [ST TYN612 Datasheet]. A lightly loaded circuit may therefore trigger an SCR and still fail to keep it ON reliably.

A 50 Hz intuition

At 50 Hz, one full AC cycle lasts

$$T = \frac{1}{f} = \frac{1}{50} = 20 \text{ ms.}$$

One half-cycle therefore lasts 10 ms. In AC phase control, the load current usually decreases near the end of each half-cycle. If it falls below I_H, the SCR turns OFF and must be triggered again in the next positive half-cycle. This repeated turn-ON and natural turn-OFF makes phase-controlled rectifiers and AC voltage controllers possible.

Reverse and forward leakage

In the blocking states, the SCR still carries a small **leakage current**. The TYN612 datasheet specifies off-state leakage in the microampere range at 25 °C, with much larger values at higher junction temperature [ST TYN612 Datasheet]. The OFF state therefore means strong blocking, not zero current.

On-state branch

When the SCR is ON, the anode-cathode voltage does not fall to zero. Practical thyristors show an on-state drop on the order of 1 V to 2 V at useful current densities [ST AN4607]. The TYN612 datasheet gives a maximum on-state voltage V_{TM} of about 1.6 V at a specified pulsed-current test point [ST TYN612 Datasheet]. This relatively low drop is one reason SCRs remain efficient in high-current line-frequency systems.

2.1.4 Turn-ON methods: gate triggering, dv/dt, thermal, light, forward-voltage

An SCR may turn ON by several mechanisms, but only one of them is the normal control method. In design practice, gate triggering is intentional; the others are either special cases or effects to be avoided.

1. Gate triggering

Gate triggering is the preferred turn-ON method in most controlled circuits [ST AN4607], [Teccor Thyristor Design Guide]. A positive gate current with respect to the cathode injects carriers into the device and initiates regenerative action. In design, the gate drive is chosen with margin over the minimum trigger requirement so that the SCR turns ON reliably across temperature, manufacturing spread, and noise.

If a gate source of voltage V_s drives the gate through a resistor R_G, a first estimate of gate current is

$$\boxed{I_G \approx \frac{V_s - V_{GK}}{R_G}} \qquad (2.5)$$

where V_{GK} is the gate-cathode drop during triggering.

For example, with a 12 V gate pulse, a desired gate current of 20 mA, and $V_{GK} \approx 1.3$ V,

$$R_G \approx \frac{12 - 1.3}{0.02} = 535\ \Omega.$$

A practical design would then examine nearby standard values such as $510\ \Omega$ or $560\ \Omega$ and verify gate-current, pulse-width, and gate-power limits against the actual datasheet. The gate pulse need only last long enough for anode current to rise above the latching-current requirement.

2. Forward-voltage triggering

If the anode-cathode voltage rises to the **breakover voltage**, the SCR can turn ON without gate drive [ST AN4607]. This **forward-voltage** or **breakover** triggering is real, but it is not preferred in controlled circuits because it gives poor timing accuracy and adds electrical stress.

3. dv/dt triggering

The SCR junctions have capacitance, so a rapid change in forward voltage can create a displacement current through the internal capacitances. The capacitor relation is

$$\boxed{i = C_j \frac{dv}{dt}} \qquad (2.6)$$

where C_j is the effective junction capacitance.

If $\frac{dv}{dt}$ is large enough, the resulting displacement current can trigger the SCR unintentionally [ST AN4608], [Teccor Thyristor Design Guide]. This is one of the main reasons for using an RC snubber across the device.

4. Thermal triggering

As junction temperature rises, leakage current increases. In a four-layer structure this increase can strengthen the internal regenerative mechanism and reduce blocking stability [ST AN4607], [ST AN4608]. Thermal triggering is therefore a stress or failure mechanism, not a normal operating method.

5. Light triggering

Some thyristors are designed so that light generates the carriers needed to start conduction. These **light-activated SCRs**, or **LASCRs**, are useful where high-voltage isolation is required between control and power circuits [Teccor Thyristor Design Guide]. The underlying turn-ON mechanism is still regenerative.

Which methods are good engineering practice?

Table 2.3: SCR turn-ON methods in practice

Turn-ON method	Physical basis	Used intentionally?	Typical design view
Gate triggering	Carrier injection through gate	Yes	Normal and preferred
Forward-voltage breakover	Exceeding V_{BO}	Rarely	Avoid as routine control method
dv/dt triggering	Junction-capacitance displacement current	No	Prevent with protection
Thermal triggering	Leakage increase and loss of blocking stability	No	Prevent by thermal design
Light triggering	Optical carrier generation	Yes, in special devices	Useful where isolation is important

A practical timing picture

In AC phase control, turn-ON is usually described by the **firing angle** α, measured from the start of the positive half-cycle to the instant of gate triggering.

At 50 Hz:

- half-cycle duration = 10 ms
- $\alpha = 90^\circ$ corresponds to a delay of 5 ms
- $\alpha = 150^\circ$ corresponds to a delay of about 8.33 ms

The SCR therefore allows the start of conduction to be placed at a controlled point within each half-cycle. The mathematical consequences of firing angle are taken up in Chapter 3.1.

2.1.5 Turn-OFF and commutation methods: natural (line) commutation and forced commutation

A conventional SCR turns OFF only when the anode current falls below the holding current and the device is given enough time to recover its blocking capability. The recovery interval is described by the **turn-off time** t_q [ST AN4608], [Teccor Thyristor Design Guide], so a basic requirement is

$$\boxed{t_{\text{available off}} > t_q} \quad (2.7)$$

Turn-OFF is therefore not only a matter of reducing current. Stored charge must also be removed before the device can block forward voltage again.

Natural or line commutation

In AC circuits, turn-OFF often occurs naturally. As the line current approaches zero near the end of a half-cycle, the anode current falls below I_H. When the supply reverses, the SCR becomes reverse biased and regains its blocking state. This is called **natural commutation**, **line commutation**, or, in some classifications, **Class F** [ST AN4607].

For 50 Hz mains, each half-cycle lasts 10 ms, so the circuit repeatedly offers a turn-OFF interval that is usually much longer than the recovery time of the device. This is why SCRs are particularly well suited to controlled rectifiers and other line-commutated converters.

Forced commutation

In DC circuits, there is no natural line-current zero crossing. A conducting SCR must therefore be turned OFF by circuit action that forces its current below the holding-current level and usually applies reverse bias for a sufficient interval. This is called **forced commutation**.

The implementation varies, but the principle is the same: an inductor, capacitor, auxiliary switch, or resonant network is used to create a current or voltage condition that drives the conducting SCR current to zero or reverse long enough for recovery.

Common Class A to E naming

Many power-electronics texts classify forced-commutation methods as Classes A to E. The names and definitions vary somewhat across books, so Table 2.4 should be read as a concept-level overview rather than a universal naming standard.

Table 2.4: Concept-level view of forced commutation classes

Class	Concept	Main idea
Class A	Load or self commutation	The load itself forms an underdamped RLC path that naturally drives current to zero

Class	Concept	Main idea
Class B	Resonant-pulse commutation	An LC branch generates a resonant current pulse to oppose device current
Class C	Complementary commutation	One SCR helps turn OFF another SCR
Class D	Auxiliary commutation	An auxiliary SCR and capacitor apply reverse voltage across the main SCR
Class E	External-pulse commutation	An external pulse source forces reverse current or reverse voltage

The essential distinction is simple. In **natural commutation**, the AC source provides the turn-OFF condition. In **forced commutation**, the circuit designer must create that condition.

This distinction explains why SCRs fit line-frequency rectifiers more naturally than high-frequency DC choppers. In the former, the line itself supplies current zero. In the latter, the converter must provide it.

2.1.6 Gate-triggering circuits: R, R-C and UJT-based triggering (overview)

Once gate triggering has been identified as the preferred turn-ON method, the practical question is how to generate the gate pulse. Classical power-electronics literature presents several traditional triggering circuits, especially for line-frequency phase control [Teccor Thyristor Design Guide]. Only an overview is needed here.

R triggering

In **resistor triggering**, gate current is supplied through a resistor from a source or from the AC line through a suitable network. Changing the resistor changes the available gate current and therefore the triggering condition. The method is simple and inexpensive, but it gives poor control of firing angle, offers little isolation, and does not produce a particularly sharp or repeatable gate pulse. It is mainly useful in simple low-power circuits and instructional examples.

R-C triggering

In **R-C triggering**, a capacitor charges through a resistor during each AC half-cycle. When the capacitor voltage reaches the effective trigger threshold, gate current flows and the SCR turns ON. Changing the resistance changes the capacitor charging time and therefore the firing angle α.

At 50 Hz, a positive half-cycle lasts 10 ms. If the RC network reaches trigger level after about 5 ms, the firing angle is roughly 90°; if it reaches trigger level after about 8.33 ms, the firing angle is about 150°. This method appears in many classical phase-control circuits for dimmers, heater controls, and small fan regulators.

UJT-based triggering

A more refined classical trigger uses a **UJT**, or **unijunction transistor**, as a relaxation-oscillator pulse source [Teccor Thyristor Design Guide]. A capacitor charges gradually through a resistor. When its voltage reaches the UJT firing condition, the UJT switches and the capacitor discharges rapidly through the pulse path, producing a sharp trigger pulse for the SCR.

The value of the method is its pulse quality. A UJT trigger produces a clearer and more repeatable gate pulse than a slowly rising waveform, and it can be combined with pulse transformers where isolation is required.

High-level comparison

Table 2.5: Classical SCR triggering circuits

Trigger circuit	Main idea	Advantage	Limitation
R triggering	Gate current through a resistor	Very simple	Poor timing control, little isolation
R-C triggering	RC charging delay sets firing angle	Useful for AC phase control	Pulse quality and repeatability are limited
UJT-based triggering	Relaxation oscillator generates sharp pulses	Better timing and cleaner pulse generation	More components and extra circuit complexity

Isolation and modern practice

Traditional texts often show these trigger circuits connected directly to the line. Historically that is important, but modern practice increasingly favors isolated gate-drive arrangements using pulse transformers, optocouplers, or dedicated driver stages when power level, safety requirements, or control sophistication increase. The classical circuits remain useful because they reveal the origin of firing-angle control and the logic of pulse generation.

Gate drive must still stay within datasheet limits. Trigger pulses should exceed the minimum requirements with suitable margin, but gate current, gate power, and pulse duration must remain within the specified ratings [ST TYN612 Datasheet], [ST AN4608].

2.1.7 SCR ratings and specifications; di/dt and dv/dt protection

An SCR datasheet does not describe only one voltage rating and one current rating. A workable design must satisfy a set of limits covering blocking, conduction, triggering, surge survival, switching stress, and temperature.

The main datasheet parameters

Table 2.6: Key SCR ratings and what they mean

Symbol	Meaning	Why it matters
V_{DRM}	Repetitive peak off-state voltage	Forward blocking limit under repetitive conditions
V_{RRM}	Repetitive peak reverse voltage	Reverse blocking limit
$I_T(RMS)$	On-state RMS current	Continuous current capability for AC conduction
$I_T(AV)$	Average on-state current	Useful in rectifier service
I_{TSM}	Non-repetitive surge peak on-state current	Surge withstand, especially for faults and inrush
I^2t	Fusing or surge-energy measure	Useful for coordination with protective fuses
I_{GT}, V_{GT}	Gate trigger current and voltage	Gate-drive design
I_L, I_H	Latching current and holding current	Reliable turn-ON and turn-OFF behavior
V_{TM}	On-state voltage drop	Conduction loss
$(di/dt)_{crit}$	Critical rate of rise of on-state current	Prevents local hot-spot damage during turn-ON
$(dv/dt)_{crit}$	Critical rate of rise of off-state voltage	Prevents unwanted triggering
$R_{\theta JC}$	Junction-to-case thermal resistance	Thermal design

The TYN612 datasheet provides a representative example set: $I_T(RMS) = 12$ A, V_{DRM}/V_{RRM} variants of 600 V, 800 V, and 1000 V, gate-trigger options as low as 5 mA or 15 mA depending on version, critical di/dt around 50 A/μs under stated conditions, critical dv/dt values such as 40 V/μs or 200 V/μs depending on sensitivity class, and junction-to-case thermal resistance about 1.3°C/W [ST TYN612 Datasheet].

Choosing the voltage class

For a 230 V, 50 Hz single-phase supply,

$$V_m = \sqrt{2} \times 230 \approx 325 \text{ V}.$$

A 600 V SCR belongs to the correct general class for such a line when suitable margin and transient protection are provided.

For a 415 V three-phase system, the line-to-line peak value is

$$V_{LL,peak} = \sqrt{2} \times 415 \approx 587 \text{ V}. \quad (2.8)$$

This immediately shows why a 600 V device is often too close for comfort in three-phase line-to-line blocking situations once switching spikes and mains disturbances are considered. In such cases, 800 V or 1000 V classes are much more realistic.

Why di/dt is dangerous

When an SCR first turns ON, conduction does not spread across the full silicon area instantaneously. The process begins near the gate region and then expands. If anode current rises too quickly, a small part of the device can carry excessive current before the entire junction area is conducting, leading to local overheating and possible failure [ST AN4608], [Teccor Thyristor Design Guide].

The basic inductor relation is

$$\boxed{\frac{di}{dt} = \frac{V}{L}} \quad (2.9)$$

so a series inductance can limit current rise. If the maximum allowable di/dt is 50 A/μs and the worst-case applied voltage at turn-ON is about 325 V, the minimum inductance required is approximately

$$L \geq \frac{V}{di/dt} = \frac{325}{50 \times 10^6} = 6.5\ \mu\text{H}.$$

Even a small inductance can therefore reduce turn-ON stress significantly.

Why dv/dt is dangerous

In the OFF state, a rapidly rising voltage can create displacement current through the internal junction capacitances, as described by Equation (2.6). If that current becomes large enough, the SCR may trigger unintentionally [ST AN4608], [Littelfuse MCR16NG Product Page].

The most common protection is an **RC snubber** across the SCR. The capacitor slows the voltage rise across the device, and the resistor limits snubber current and damps oscillation. If a capacitor initially charged to voltage V discharges through the snubber resistor at turn-ON, the initial current is approximately

$$\boxed{i_C(0^+) = \frac{V}{R}} \quad (2.10)$$

The resistor is therefore essential. Without it, the snubber capacitor itself could introduce a large current pulse at turn-ON.

On-state loss and thermal design

Even though the SCR has a relatively low on-state drop, the drop is not zero. The TYN612 datasheet gives an on-state model of the form [ST TYN612 Datasheet]

$$\boxed{V_T \approx V_{T0} + I_T r_T} \quad (2.11)$$

where V_{T0} is a threshold-type on-state voltage parameter and r_T is a dynamic resistance parameter.

The on-state power loss can then be estimated as

$$\boxed{P_{on} \approx V_T I_T} \quad (2.12)$$

Using approximate values $V_{T0} = 0.85$ V, $r_T = 30$ mΩ, and $I_T = 8$ A,

$$V_T \approx 0.85 + 8 \times 0.03 = 1.09 \text{ V}$$

and therefore

$$P_{on} \approx 1.09 \times 8 = 8.72 \text{ W}.$$

With junction-to-case thermal resistance $R_{\theta JC} \approx 1.3$°C/W, the junction temperature can be estimated by

$$\boxed{T_J \approx T_C + P_D R_{\theta JC}} \quad (2.13)$$

so the simplified rise above case temperature is

$$8.72 \times 1.3 \approx 11.3\text{°C}.$$

If the case temperature is already 85°C, the junction temperature is then about 96.3°C by this first estimate. Real design must also consider waveform shape, conduction angle, heat sink performance, thermal cycling, and ambient conditions, but the example shows that thermal design cannot be separated from electrical design.

Surge and protection interpretation

SCR datasheets also specify **non-repetitive surge current** I_{TSM} and often an I^2t value. These ratings matter in faults, capacitor-charging surges, and fuse coordination. In a crowbar circuit, for example, the SCR may be required to conduct a very large current briefly while upstream protection clears the fault. In such cases, surge-current and I^2t capability may matter more than the normal RMS current rating.

Two rating errors are common in practice. One is to select the device from RMS mains voltage rather than peak voltage plus transient margin. The other is to check only average current while ignoring RMS current, surge capability, conduction angle, thermal resistance, and dynamic limits such as di/dt and dv/dt. Gate sensitivity must likewise be interpreted alongside noise immunity and trigger robustness.

Worked datasheet interpretation: TYN612

The ST TYN612 datasheet is a useful compact example because it presents the essential SCR ratings without excessive detail. Read as a design document, it immediately shows the intended role of the device: line-frequency control and protection rather than high-frequency PWM switching.

The voltage classes, gate-trigger variants, latching and holding currents, dynamic limits, and thermal data collectively describe how the device must be used. The point of the exercise is not to memorize a table of numbers, but to convert those numbers into circuit choices.

Table 2.7: Interpreting the TYN612 datasheet

Datasheet item	What it says	What it means in design
600 V, 800 V, 1000 V versions	Several blocking classes in one family	Choose based on actual line peak plus transient margin
$I_T(RMS) = 12$ A	Medium-current SCR family	Suitable for many control and protection roles, not arbitrarily for any current
$I_{GT} = 5$ mA or 15 mA versions	Different trigger sensitivities	Drive-circuit design and noise margin depend on variant
I_H and I_L given explicitly	Conduction persistence is not automatic at low current	Load current must support both turn-ON and continued conduction
Critical di/dt and dv/dt ratings	Dynamic stress limits matter	Layout and protection network are part of SCR design
$R_{\theta JC}$ provided	Thermal path is quantified	Heat sinking and junction-temperature checks are required

How this matters in renewable-energy systems

SCRs are not the main switching devices in modern high-frequency solar PV inverters, battery DC-DC converters, or EV traction inverters. Their continuing relevance in renewable-energy systems lies elsewhere.

They remain valuable in protection and line-frequency auxiliary functions. A crowbar SCR can protect a DC bus or battery-connected load by deliberately shorting the output during an overvoltage fault until a fuse or breaker clears the condition. Inrush-current limiting, soft-start functions, static bypass paths, and some controlled-rectifier interfaces likewise benefit from the SCR's rugged current handling and low line-frequency conduction loss [ST AN4607], [Littelfuse Phase Control].

The device also remains part of the wider power-conversion ecosystem, especially in large industrial power-conditioning systems and high-power line-commutated equipment. More broadly, the SCR trains the designer to ask the right questions about turn-ON, turn-OFF, transient stress, and datasheet interpretation. Those questions remain central even when later chapters move to MOSFETs and IGBTs.

Chapter summary

- A **thyristor** or **SCR** is a four-layer PNPN power semiconductor with anode, cathode, and gate terminals.
- A conventional SCR can be triggered ON by the gate, but it normally turns OFF only when anode current falls below the holding current and the device has time to recover blocking capability.
- The **two-transistor analogy** explains SCR latching as regenerative feedback between coupled PNP and NPN transistor actions.
- In the static V-I characteristic, the main regions are reverse blocking, reverse breakdown, forward blocking, and forward conduction.
- The **latching current** I_L is the current needed just after triggering to sustain conduction without the gate; the **holding current** I_H is the current below which an already conducting SCR turns OFF. Normally, $I_L > I_H$.
- Practical turn-ON mechanisms include gate triggering, forward-voltage breakover, dv/dt triggering, thermal triggering, and light triggering, but gate triggering is the normal control method.
- In AC circuits, SCRs often turn OFF by **natural or line commutation**. In DC circuits, **forced commutation** is required.
- Classical line-frequency trigger circuits include R triggering, R-C triggering, and UJT-based triggering.
- Important ratings include blocking voltage, RMS and average current, surge current, I^2t, gate-trigger requirements, latching and holding current, on-state voltage, critical di/dt, critical dv/dt, and thermal resistance.
- Safe SCR design requires voltage margin, current and thermal checks, attention to di/dt and dv/dt, and appropriate protective elements such as series inductance, snubbers, and coordinated fault protection.

Further reading

- IEEE Milestone brochure: Silicon Controlled Rectifier (SCR/Thyristor), 1957 - A concise historical source explaining why the SCR was a milestone in electric-power control.
- ST AN4607: Basics on the thyristor (SCR) structure and its application - A strong manufacturer introduction to SCR structure, operation, and application context.
- ST TYN612 Datasheet - A compact real SCR datasheet that is excellent for learning ratings, trigger parameters, I_L, I_H, di/dt, dv/dt, and thermal data.
- ST AN4608: How to select the right thyristor - Useful for understanding real selection logic, overvoltage protection use, and dynamic-stress considerations.

- Teccor Thyristor Design Guide (legacy manufacturer guide, mirrored copy) - A classic practical guide covering triggering methods, phase control, and application-oriented SCR design ideas.

Chapter 1.3: DIAC and TRIAC

Chapter opening

DIACs and TRIACs extend the thyristor-family idea of controlled switching into the domain of line-frequency AC power control. A **TRIAC** is a three-terminal bidirectional switching device used to control power from an AC source. A **DIAC** is a two-terminal bilateral trigger device commonly used to fire a TRIAC with improved symmetry.

These devices became important in practical circuits such as light dimmers, heater regulators, fan controllers, and small appliance switches. This chapter examines their operating principles, V-I characteristics, TRIAC triggering quadrants, and the basic DIAC-TRIAC phase-control circuit. It also shows why resistive loads are easier to control than inductive loads, and why TRIACs remain useful in auxiliary AC control even though modern PWM converters use other switching devices.

Prerequisites check

- You should know the basic SCR ideas from Chapter 1.2: latching, holding current, gate triggering, and natural turn-off at current zero.
- You should know the relation between RMS and peak value for a sinusoid: $V_m = \sqrt{2}V_{rms}$.
- You should be comfortable with the difference between a **resistive load** and an **inductive load**.
- You should know the basic idea of an RC charging circuit.
- You should know that mains-connected circuits require voltage margin, current margin, and transient protection.

If the SCR terms **latching current** and **holding current** are not yet clear, review Chapter 1.2 before continuing.

Core content

1.3.1 History of DIAC and TRIAC

Early AC power control often relied on either mechanical switches or resistor-based regulators. A mechanical switch provides only full ON or full OFF operation. A series resistor can reduce the power delivered to a load, but it does so by dissipating power as heat. Older fan regulators make this limitation obvious: the control element itself becomes warm because it is wasting energy.

The SCR introduced controlled switching, but it is fundamentally a unidirectional device. AC control therefore required either two SCRs connected in anti-parallel or a single device capable of controlled conduction in both half-cycles. The TRIAC emerged as that single-device solution.

Manufacturer literature describes the TRIAC as a semiconductor device used to control power from an AC source and notes its importance in domestic, building, and industrial applications [ST AN5114, Sec. 1.1].

The DIAC became important as a companion trigger device. In simple RC phase-control circuits, direct triggering of a TRIAC can lead to uneven firing in the positive and negative half-cycles. A DIAC remains in the blocking state until its breakover voltage is reached in either polarity and then switches abruptly, producing a sharp trigger pulse with reasonably good symmetry [ST DB3 Datasheet].

Table 3.1: Why DIAC and TRIAC became important

Device	Practical need it addressed	Why it mattered
DIAC	Symmetrical triggering in both half-cycles	Helped simple RC control circuits fire more evenly
TRIAC	Controlled AC power with one device instead of two anti-parallel SCRs	Reduced component count in dimmers, regulators, and small AC controllers
DIAC + TRIAC pair	Low-cost phase-angle control from mains supply	Enabled compact light dimmers, fan regulators, heater controls, and small appliance controllers

These devices are not the main switches in modern PWM inverter bridges, but their historical role remains instructive. They provide a clear introduction to bidirectional conduction, latching behavior, commutation limits, and waveform shaping at mains frequency.

1.3.2 Construction, operating principle, V-I characteristics

The DIAC

The **DIAC** is a two-terminal bilateral trigger device. It has no gate terminal and is not intended to carry the main load current continuously. Its usual function is to remain in the blocking state until the voltage across it reaches the breakover value V_{BO}, then switch abruptly and deliver a current pulse [ST AN2703, Table 2], [ST DB3 Datasheet].

At low applied voltage in either polarity, only a very small leakage current flows. As the magnitude of the applied voltage approaches V_{BO}, the device enters breakover. Once conduction begins, the DIAC voltage falls while the current rises. This negative-resistance triggered characteristic is what makes the transition sharp rather than gradual [ST AN2703, Table 2].

The ST DB3 datasheet gives representative values [ST DB3 Datasheet]:

- typical breakover voltage: 32 V
- guaranteed breakover range: 28 V to 36 V
- breakover-voltage symmetry: within 3 V
- maximum breakover current: 50 μA

In a DIAC-TRIAC phase-control circuit, these values mean that the capacitor in the trigger network can charge for much of the half-cycle and then discharge suddenly into the TRIAC gate when the DIAC reaches breakover. The DIAC therefore acts as a symmetrical trigger element rather than as the main power switch.

Image prompt for Figure 3.1: Create a clean textbook-style graph of the V-I characteristic of a DIAC. Use horizontal axis voltage V_D in volts and vertical axis current I_D in amperes. Show symmetrical positive and negative breakover behavior. Mark $+V_{BO}$ and $-V_{BO}$, the small leakage-current region near the origin, the abrupt turn-on point at I_{BO}, the negative-resistance transition where device voltage falls as current rises, and the lower on-state voltage region labeled V_F. Add brief annotations "blocking region," "breakover," and "conduction region."

Two additional parameters are commonly used. The **breakover current**, I_{BO}, is the current flowing just before switching [ST AN2703, Table 2]. The **dynamic breakover voltage**, ΔV, is the difference between the DIAC breakover voltage and the DIAC voltage at 10 mA [ST AN2703, Table 2]. Together they describe how abruptly the device enters conduction.

The TRIAC

The **TRIAC** is a three-terminal bidirectional AC switch with terminals **A1**, **A2**, and **G** for gate. In some texts A1 and A2 are written **MT1** and **MT2**. This chapter uses A1 and A2 to match common manufacturer notation [ST BTA16 Datasheet].

A useful conceptual model is two thyristor-like conduction paths connected back-to-back and controlled by a common gate region. ST AN437 uses this comparison to explain bidirectional conduction and latching behavior [ST AN437, Sec. 1.1.1]. The model is simplified, but it captures the essential operating idea: the TRIAC can be triggered in either current direction and, once triggered, remains ON until its current falls below the holding current.

Image prompt for Figure 3.2: Create a textbook-style technical illustration of a TRIAC. On the left, show the circuit symbol with terminals labeled A1, A2, and G. On the right, show a simplified internal-conduction concept: two thyristor-like conduction paths arranged back-to-back sharing a common gate region. Add a note that the figure is a simplified conceptual model, not a literal silicon cross-section. Include arrows showing possible current flow from A2 to A1 and from A1 to A2 when triggered.

TRIAC blocking and conduction behavior

When a TRIAC is connected in series with an AC load, it blocks voltage in both polarities until an adequate gate signal is applied. After triggering, the voltage across the device falls to a low on-state value and load current flows. The device then latches and remains ON until the main current falls below the **holding current**, I_H [ST AN302], [ST AN2703, Table 2].

At turn-on there is a related parameter called the **latching current**, I_L, defined as the minimum A2-to-A1 current required to maintain conduction after the gate signal is removed [ST AN303], [ST AN2703, Table 2]. As with the SCR,

$$\boxed{I_L > I_H} \quad (3.1)$$

Low-power loads can therefore be troublesome. If the load current does not rise high enough after the gate pulse, the TRIAC may fail to latch or may drop out prematurely. ST AN303 gives the practical example of a 10 W signal lamp on European 230 V mains, where the peak current can be close to the latching-current range of some devices [ST AN303, Sec. 1.1].

TRIAC V-I characteristic

The TRIAC V-I characteristic is approximately symmetrical in the first and third quadrants. In normal use the main operating regions are:

- off-state blocking for positive voltage
- off-state blocking for negative voltage
- on-state conduction for positive current
- on-state conduction for negative current

Unlike the DIAC, the TRIAC is not ordinarily used by allowing uncontrolled breakover; it is normally triggered by its gate. Once triggered, its on-state voltage is low. The ST BTA16 datasheet gives a maximum peak on-state voltage of about 1.55 V at the stated test condition, together with threshold and dynamic-resistance parameters that describe conduction [ST BTA16 Datasheet]. This low on-state drop explains why TRIAC control is usually much more efficient than resistor-based control.

Image prompt for Figure 3.3: Create a clean engineering graph of the V-I characteristic of a TRIAC. Use horizontal axis V_{A2A1} in volts and vertical axis I_T in amperes. Show first- and third-quadrant blocking regions, first- and third-quadrant conduction regions, a note that practical operation uses gate triggering rather than breakover, and mark the low on-state voltage region. Add labels for holding current I_H and indicate that current must fall below this magnitude for the TRIAC to turn OFF.

DIAC and TRIAC compared

Table 3.2: DIAC and TRIAC at a glance

Feature	DIAC	TRIAC
Number of terminals	2	3
Main role	Trigger device	AC power switch
Gate terminal	No	Yes
Direction of use	Bilateral	Bilateral
Typical use	Triggering a TRIAC or SCR	Switching or phase-controlling AC load current
Important parameter	Breakover voltage V_{BO}	Gate trigger current I_{GT}, latching current I_L, holding current I_H

Feature	DIAC	TRIAC
Current capability	Small trigger current pulses	Load current capability from small fractions of an ampere to many amperes depending on part

The DIAC and TRIAC serve different functions in the same circuit. The DIAC shapes the trigger event; the TRIAC carries the load current.

1.3.3 Modes of operation of TRIAC; DIAC-TRIAC based phase control (concept)

TRIAC phase control depends on two linked ideas: quadrant-sensitive gate triggering and delayed conduction within each half-cycle. The DIAC-TRIAC pair combines these ideas in a simple line-frequency control circuit.

TRIAC triggering quadrants

Because TRIAC operation depends on both main-terminal polarity and gate-current polarity, triggering is described by **quadrants**. The relevant quantities are the sign of the main-terminal voltage V_{A2A1} and the sign of the gate current with respect to A1 [ST AN4363, Sec. 3.1], [ST AN2703, Table 2].

Table 3.3: TRIAC triggering quadrants

Quadrant	Main-terminal voltage	Gate-current polarity	Practical note
QI	$V_{A2A1} > 0$	Gate current positive with respect to A1	Common and usually sensitive
QII	$V_{A2A1} > 0$	Gate current negative with respect to A1	Valid, but latching behavior can be less favorable
QIII	$V_{A2A1} < 0$	Gate current negative with respect to A1	Common and usually sensitive
QIV	$V_{A2A1} < 0$	Gate current positive with respect to A1	Often the least sensitive quadrant in standard TRIACs

Trigger sensitivity is not the same in all four quadrants. Standard four-quadrant devices often require the largest gate current in QIV, and many snubberless TRIACs are specified only for QI, QII, and QIII [ST BTA16 Datasheet], [ST AN4363, Table 2]. This matters because DIAC-triggered circuits naturally operate in QI and QIII, where the gate-current polarity follows the line-voltage polarity [ST AN4363, Sec. 3.2].

Image prompt for Figure 3.4: Create a clean textbook-style diagram of TRIAC triggering quadrants. Show the TRIAC symbol with A1, A2, and G. Around it, place four small labeled panels:

QI with $V_{A2A1} > 0$ and positive gate current, QII with $V_{A2A1} > 0$ and negative gate current, QIII with $V_{A2A1} < 0$ and negative gate current, and QIV with $V_{A2A1} < 0$ and positive gate current. Use clear arrows for current direction and add a note that many DIAC-triggered circuits operate in QI and QIII.

Firing angle and the DIAC trigger network

For a sinusoidal mains supply,

$$\boxed{v_s(t) = V_m \sin(\omega t)} \quad (3.2)$$

where V_m is the peak value and $\omega = 2\pi f$ is the angular frequency. For a 230 V, 50 Hz supply,

$$\boxed{V_m = \sqrt{2}V_{rms}} \quad (3.3)$$

so

$$V_m = \sqrt{2} \times 230 \approx 325 \text{ V}.$$

If the TRIAC is triggered exactly at the beginning of each half-cycle, the load receives almost the full sinusoidal waveform. If triggering is delayed, the TRIAC remains OFF during the early part of the half-cycle and turns ON only later. This delay is the **firing angle**, α.

For a resistive load:

- from the zero crossing to angle α, the TRIAC is OFF and the load voltage is zero
- from angle α to π, the TRIAC is ON and the load receives the supply voltage
- the same pattern repeats in the negative half-cycle

Phase control therefore changes the conduction interval, not the peak value of the supply. The load receives chopped sinusoidal segments rather than a reduced-amplitude sine wave.

In the classic DIAC-TRIAC control circuit, an RC network charges during each half-cycle. When the capacitor voltage magnitude reaches the DIAC breakover voltage, the DIAC switches ON and discharges the capacitor into the TRIAC gate. A small resistance gives early firing and a small value of α; a large resistance gives later firing and a larger value of α. The DIAC sharpens the gate pulse and improves half-cycle symmetry [ST DB3 Datasheet], [ST AN308, Sec. 1.3].

For 50 Hz mains, one full cycle lasts 20 ms and one half-cycle lasts 10 ms. A firing angle of 30° begins conduction early in that 10 ms interval, 90° begins near the midpoint, and 150° allows conduction only near the end. This time-domain interpretation is often useful in understanding household dimmers and fan regulators.

RMS output for a resistive load

For a purely resistive load, if the TRIAC fires at angle α in each half-cycle, the output voltage is present only from α to π in the positive half-cycle and from $\pi+\alpha$ to 2π in the negative half-cycle.

The RMS value is

$$V_{o,rms}^2 = \frac{1}{2\pi}\left[2\int_{\alpha}^{\pi} V_m^2 \sin^2\theta \, d\theta\right] \quad (3.4)$$

where $\theta = \omega t$. Using

$$\int \sin^2\theta \, d\theta = \frac{\theta}{2} - \frac{\sin 2\theta}{4},$$

we obtain

$$V_{o,rms}^2 = \frac{V_m^2}{\pi}\left[\frac{\pi-\alpha}{2} + \frac{\sin 2\alpha}{4}\right].$$

Taking the square root gives

$$\boxed{V_{o,rms} = V_m\sqrt{\frac{1}{2\pi}\left[(\pi-\alpha) + \frac{\sin 2\alpha}{2}\right]}} \quad (3.5)$$

For a resistive load R_L, the power is

$$\boxed{P_L = \frac{V_{o,rms}^2}{R_L}} \quad (3.6)$$

Equation (3.5) gives the expected limits. When $\alpha = 0$, the output is the full supply waveform. As α increases, the conduction interval shortens and the RMS output decreases. As $\alpha \to \pi$, the output power approaches zero.

Numerical example: 1 kW heater on 230 V mains

Suppose a heater is rated at 1 kW on 230 V, so its resistance is approximately

$$R_L = \frac{V^2}{P} = \frac{230^2}{1000} = 52.9\,\Omega.$$

Now suppose the TRIAC is triggered at $\alpha = 90^\circ = \pi/2$. In Equation (3.5),

- $\pi - \alpha = \pi/2$
- $\sin 2\alpha = \sin\pi = 0$

Therefore,

$$V_{o,rms} = 325\sqrt{\frac{1}{2\pi}\cdot\frac{\pi}{2}} = 325\sqrt{\frac{1}{4}} = 162.5\text{ V}.$$

The load power is then

$$P_L = \frac{162.5^2}{52.9} \approx 499\,\mathrm{W}.$$

A firing angle of 90° therefore reduces the heater power from about $1000\,\mathrm{W}$ to about $500\,\mathrm{W}$. The load still sees full mains peaks during the conducting interval, but its effective RMS voltage and power are lower.

Inductive loads, commutation, and snubbers

Inductive loads complicate TRIAC control because the current lags the voltage. For an R-L load,

$$\boxed{\tan\phi = \frac{\omega L}{R}} \quad (3.7)$$

where ϕ is the current lag angle, L is load inductance, and R is load resistance [ST AN308, Sec. 1.2].

If the current lags, the TRIAC may continue conducting after the voltage crosses zero. The next half-cycle may then begin before the previous conduction interval has fully ended. The result can be asymmetrical operation, increased commutation stress, audible noise, torque ripple, flicker, or unwanted DC components in some loads.

This is why simple DIAC-TRIAC phase-control circuits work best with resistive or only mildly inductive loads [ST AN308, Sec. 1.1-1.3]. Inductive-load applications often require an RC snubber, a snubberless TRIAC with stronger commutation capability, or a more deliberate gate-drive strategy such as pulse-train triggering [ST AN437, Sec. 1.1-1.3].

1.3.4 Applications: light dimmers, fan regulators, small AC loads

DIAC-TRIAC control is most effective where the operating frequency is the mains frequency, the load current is moderate, and waveform-quality requirements are not severe.

Light dimmers and heater control

Incandescent lamps and resistive heaters are the classic applications. Because current is nearly in phase with voltage, triggering is predictable and turn-off occurs naturally at each current zero. The relation between firing angle, RMS voltage, and delivered power is therefore straightforward. Manufacturer literature lists light dimmers and heating regulators among standard TRIAC applications [ST BTA16 Datasheet], [ST AN5114, Sec. 1.1].

Modern LED lamps are more complicated because many contain rectifiers, capacitors, and control electronics. A TRIAC dimmer designed for incandescent loads may therefore flicker or behave poorly with some LED lamps.

Fan regulators and small motor control

Electronic ceiling-fan regulators are familiar examples of TRIAC phase control. They replaced resistor-based regulators largely because they waste less power as heat. The load, however, is inductive. Motor current, torque, heating, acoustic noise, and symmetry therefore depend more strongly on commutation behavior and trigger quality than in a resistive heater circuit [ST AN302, Sec. 1.2-1.3], [ST AN308, Sec. 1.1-1.3].

TRIAC control in such circuits reduces effective voltage, not supply frequency. It is therefore not equivalent to a modern variable-frequency drive.

Small AC loads, static relays, and appliance switching

TRIACs are also used in static relays, small pumps, solenoids, appliance control boards, induction-motor starting functions, and inrush-limiting functions [ST BTA16 Datasheet]. In these applications, load type determines how simple the control circuit can be.

Table 3.4: Where DIAC-TRIAC control fits well and where it does not

Load or application	Suitability of simple DIAC-TRIAC phase control	Reason
Incandescent lamp	Very good	Load is mostly resistive
Resistive heater	Very good	Current follows voltage closely
Small ceiling fan regulator	Moderate	Works in practice, but motor is inductive and may hum or heat
Small induction motor with varying load	Limited	Phase lag and commutation issues become important
Transformer primary control	Use with caution	Magnetizing current and asymmetry can create stress
Main switch of a modern PV inverter	Poor choice	High-quality PWM control is needed instead
Grid-tied battery inverter AC bridge	Poor choice	Harmonics and control requirements are far more demanding

When simple phase-control circuits are used, the DIAC remains valuable because breakover symmetry helps reduce half-cycle imbalance. In practice this can reduce flicker, acoustic noise, and unwanted DC bias in the load current.

Worked interpretation exercise

The following condensed exercise shows how a DIAC datasheet and a TRIAC datasheet are read together for a simple 230 V mains controller.

Step 1: Check the mains-voltage class

For a 230 V, 50 Hz single-phase supply,

$$V_m = \sqrt{2} \times 230 \approx 325 \text{ V}.$$

The OFF-state device must withstand at least this peak value, plus margin for tolerance and transients. ST AN4363 indicates that 600 V TRIACs suit most single-phase mains applications, while 800 V classes are preferred for higher-voltage or higher-stress situations [ST AN4363, Sec. 2.1]. The key design habit is to compare the device blocking-voltage rating with the **peak** line voltage, not with the RMS value alone.

Step 2: Read the DIAC as a trigger device

The DB3 datasheet gives the essential DIAC data directly [ST DB3 Datasheet]:

- breakover voltage range: 28 V to 36 V
- typical breakover voltage: 32 V
- breakover symmetry: 3 V maximum
- breakover current: 50 μA maximum

These values show that the capacitor in the trigger network needs only a modest fraction of the mains peak voltage to fire the DIAC, and that positive and negative half-cycle triggering can be reasonably well balanced. The DIAC is therefore read as a symmetrical pulse-trigger element, not as the main load switch.

Step 3: Read the TRIAC against the load

The BTA16 family provides 600 V or 800 V blocking classes, an $I_{T(RMS)}$ rating of 16 A, and quadrant-dependent gate-trigger current [ST BTA16 Datasheet]. Its latching and holding currents are in the tens of milliamperes range, so ordinary lamps and heaters are usually easy loads, while very small electronic loads can be problematic. For inductive loads, the datasheet recommendation for snubberless variants or an external snubber becomes important.

In practical design, the DIAC and TRIAC must be chosen together. The DIAC determines the trigger event; the TRIAC must match the load current, voltage class, triggering quadrant, and commutation duty.

How this matters in renewable-energy systems

TRIACs are rarely the main switches in modern solar PV inverters, battery converters, wind-turbine converters, or EV traction drives. Those systems rely on self-commutated, PWM-capable devices such as MOSFETs, IGBTs, or wide-bandgap switches. The chapter nevertheless remains relevant because auxiliary AC functions inside larger systems may still use line-frequency switching, and because ideas such as firing angle, latching current, holding current, commutation, and waveform distortion carry forward to later converter topics.

Chapter summary

- A **DIAC** is a two-terminal bilateral trigger device characterized mainly by breakover behavior.
- A **TRIAC** is a three-terminal bidirectional latching AC switch with terminals A1, A2, and G.
- In simple phase-control circuits, the DIAC produces a sharp and reasonably symmetrical trigger pulse for the TRIAC.
- TRIAC triggering is described by quadrants; DIAC-triggered circuits commonly operate in QI and QIII.
- Phase-angle control delays conduction within each half-cycle, so the load receives chopped sinusoidal segments rather than a reduced-amplitude sine wave.
- For a resistive load, Equation (3.5) gives the RMS output voltage as a function of firing angle α.
- Inductive loads complicate TRIAC control because current lags voltage and commutation becomes more demanding.
- DIAC-TRIAC control is well suited to line-frequency control of resistive and mildly inductive loads, but not to the main switching function in modern PWM-based converters.

Further reading

- ST AN5114: Controlling a Triac with a phototriac - A clear manufacturer introduction to TRIAC use in mains AC control and practical triggering methods.
- ST AN4363: How to select the triac, ACS, or ACST that fits your application - Useful for learning about triggering quadrants, voltage-class selection, and the link between circuit type and device choice.
- ST DB3 / DB4 / SMDB3 Datasheet - A compact DIAC datasheet showing breakover voltage, symmetry, breakover current, and typical trigger applications.
- ST BTA16 Datasheet - A practical TRIAC datasheet for studying current rating, voltage rating, quadrant trigger current, holding current, latching current, and commutation capability.
- ST AN308: TRIAC analog control circuits for inductive loads - A valuable application note on inductive-load behavior, trigger symmetry, and reliable analog control.

Chapter 1.4: Power MOSFET

Chapter opening

Chapters 1.1 to 1.3 introduced three important power-device families: the power BJT, the SCR, and the DIAC-TRIAC pair. Together they illustrated the essential requirements of a practical power switch: it must block voltage safely in the OFF state, conduct with acceptable loss in the ON state, and change state in a controlled manner.

The **power MOSFET**, or **power metal-oxide-semiconductor field-effect transistor**, became one of the main solutions to those requirements in high-frequency converters. Its insulated gate allows voltage control with negligible steady-state gate current, its majority-carrier operation supports rapid switching, and its vertical structure makes useful voltage and current ratings possible. For that reason, power MOSFETs are common in switch-mode power supplies, battery chargers, DC-DC converters, EV auxiliary power stages, solar charge controllers, telecom rectifiers, and many low- to medium-power inverter subsystems [onsemi AN-9010/D], [Vishay AN605].

This chapter examines the historical development of the power MOSFET, the V-DMOS structure and operating principle, the meaning of output and transfer characteristics, and the switching, body-diode, thermal, and datasheet issues that govern practical use.

Prerequisites check

- You should know the basic idea of a semiconductor switch being either OFF and blocking voltage, or ON and carrying current.
- You should be comfortable with voltage, current, power, and the relation $P = VI$.
- You should know that a transistor has three terminals and that a control terminal can influence a larger power path.
- You should remember from Chapter 1.1 that power-semiconductor switching speed matters because voltage and current can overlap during transitions.
- You should know that a 230 V, 50 Hz single-phase supply has a peak value of about 325 V after rectification.

If the distinction between conduction loss and switching loss feels weak, a quick review of Chapter 1.1 will help before going further.

Core content

1.4.1 History of Power MOSFET

Before the power MOSFET became common, engineers already had usable power devices. BJTs could be turned ON and OFF, and thyristors could handle high power, but BJTs required continuous base current while conducting and conventional SCRs could not ordinarily be gate-turned-OFF. As switching frequencies increased and compact switched-mode converters became more important, industry needed a device that was faster and easier to drive.

The broader MOSFET story begins with the invention of the MOS transistor in 1959 by Mohamed Atalla and Dawon Kahng at Bell Labs [Computer History Museum, "MOS Transistor Demonstrated", 1959]. That invention was not yet the power MOSFET used in converters today, but it established the field-effect control principle that made later development possible. Instead of controlling collector current through continuous base current, as in a BJT, a MOSFET uses an electric field created by gate voltage to form a conducting channel.

For integrated circuits, the MOSFET became revolutionary because it could be scaled and fabricated densely. Power electronics, however, needed something different. A lateral MOSFET suitable for IC logic is not enough to block hundreds of volts or carry tens of amperes efficiently. A power device needs a geometry that can support high voltage in the OFF state and still offer a low-resistance current path in the ON state. That need led to **vertical power MOSFET** structures, often called **V-DMOS** or **vertical double-diffused MOS** structures [onsemi AN-9010/D], [IEEE TED, "The Trench Power MOSFET: Part I"].

The phrase "double-diffused" refers to how the channel region is formed by two diffusion steps in fabrication. The practical consequence is that current flows vertically through the silicon rather than only along the surface, which makes the device much more suitable for power conversion.

By the 1970s and 1980s, power MOSFETs became central to switch-mode power supplies, high-frequency choppers, and low-voltage motor-control systems. The onsemi application note on MOSFET basics describes the power MOSFET as a device with high input impedance, majority-carrier operation, and fast switching behavior that made it particularly attractive in power switching applications [onsemi AN-9010/D]. Later structural improvements such as trench gates further reduced ON resistance, especially in low-voltage devices [IEEE TED, "The Trench Power MOSFET: Part I"].

The comparison with other power devices clarifies where MOSFETs fit in the larger device family:

- Compared with a **power BJT**, a MOSFET is usually easier to drive because the gate ideally draws negligible steady-state current.
- Compared with an **SCR** or **TRIAC**, a MOSFET can be turned ON and OFF directly by its control signal.
- Compared with an **IGBT**, a MOSFET is usually preferred at lower voltages and higher switching frequencies because it is a majority-carrier device and therefore avoids minority-carrier storage delay [Vishay AN605], [onsemi AN-9010/D].

Because a MOSFET is a **majority-carrier device**, it does not rely on large stored minority charge of the kind that slows a saturated BJT or a thyristor. This is one of the main reasons MOSFETs switch quickly.

Table 4.1: Why the power MOSFET became important

Property	Why it mattered in practice
Voltage-controlled gate	Simplified the drive circuit compared with BJTs
Majority-carrier operation	Enabled fast switching and high-frequency use
Vertical structure	Allowed useful voltage blocking and current capability
Positive temperature coefficient of $R_{DS(\text{on})}$ in normal operation	Helped current sharing in parallel devices
Built-in body diode	Made the device naturally compatible with many converter topologies, though not without tradeoffs

In power electronics, the significance of the MOSFET lies not only in voltage control at the gate but in the combination of gate control, vertical structure, switching speed, ON-state resistance behavior, and reverse-current behavior through the body diode.

1.4.2 Vertical (V-DMOS) structure, operation, output and transfer characteristics

Why a vertical structure is needed

A 48 V to 12 V, 300 W DC-DC converter for an EV auxiliary bus requires a switch that can carry tens of amperes when ON while blocking the bus voltage when OFF. A current path confined to a thin surface region would be too resistive for efficient power conversion.

The **vertical MOSFET** solves this problem. In a V-DMOS device, current enters near the top surface and leaves through the bottom drain contact, so many parallel microscopic cells can share current across the die area while the vertical drift region supports OFF-state voltage [Vishay AN605], [onsemi AN-9010/D].

Image prompt for Figure 4.1: Create a clean textbook-style technical illustration of an N-channel vertical power MOSFET (V-DMOS) cross-section. Show source metallization at the top contacting repeated N+ source regions inside P-body regions. Show a polysilicon gate insulated by silicon dioxide over the channel area. Show the lightly doped N- drift region below the body and the N+ drain substrate at the bottom with drain metallization. Label source, gate, drain, P-body, N+ source, N- drift region, N+ drain substrate, oxide, inversion channel, and intrinsic body diode from body to drain. Use monochrome engineering style and no decorative background.

Figure 4.1 should make four structural features clear: the **gate** is insulated from the semiconductor by a thin oxide layer, the conduction channel forms near the surface under the gate when the gate-source voltage becomes sufficiently positive in an N-channel device, the main current path then continues vertically through the drift region to the drain, and the P-body with the N-drift region naturally creates the **body diode**.

Terminals and basic control idea

The main MOSFET terminals are:

- **Gate** (**G**): the control terminal
- **Drain** (**D**): one end of the main power path
- **Source** (**S**): the other end of the main power path

Most power MOSFETs also have an internal body region tied to the source terminal in the package. Because of that internal connection, the body diode appears between source and drain with a fixed polarity determined by device type.

The discussion here focuses on the **N-channel enhancement-mode power MOSFET**, because it is the most common practical device in converters.

The basic operating sequence is as follows:

1. With $V_{GS} = 0$, there is no inversion channel between source and drift region, so the device is OFF and can block drain-source voltage.
2. As the gate-source voltage V_{GS} increases, an electric field attracts carriers near the surface.
3. When V_{GS} reaches the **threshold voltage**, written $V_{GS(\mathrm{th})}$, an inversion layer begins to form.
4. Once a sufficient channel exists, drain current can flow when a drain-source voltage is applied.

Threshold voltage does not mean that the MOSFET is fully ON. It marks the onset of inversion under specified test conditions. A power MOSFET with $V_{GS(\mathrm{th})} \approx 2$ V may still require 4.5 V, 5 V, 10 V, or another specified drive level to achieve low $R_{DS(\mathrm{on})}$ in practical power operation [FQP30N06L Datasheet], [Vishay AN605].

OFF state and voltage blocking

When the MOSFET is OFF, the drain-source voltage appears mainly across the lightly doped drift region. The same tradeoff appears in other power devices: a lightly doped region helps support high voltage, but it also tends to increase ON-state resistance.

This tradeoff is one of the defining facts of power MOSFET design. As voltage rating increases, the drift region must become thicker and lighter doped, and $R_{DS(\mathrm{on})}$ rises significantly. That is why very high-voltage silicon MOSFETs are available, but their ON resistance is much higher than that of low-voltage MOSFETs of similar die size [Vishay AN605], [onsemi AN-9010/D].

This tradeoff defines the voltage classes of MOSFETs: low-voltage devices can be made very low in resistance and are well suited to high-current, high-frequency converters, whereas high-voltage devices retain good switching performance but pay a conduction-loss penalty through higher $R_{DS(\mathrm{on})}$.

ON state and channel formation

Once the gate voltage is high enough, a channel forms in the P-body under the oxide, connecting the N+ source region to the N-drift region. For small drain-source voltage, the MOSFET behaves approximately like a controlled resistance.

In that low-V_{DS} ON region, a useful practical relation is

$$\boxed{V_{DS} \approx I_D R_{DS(\text{on})}} \quad (4.1)$$

where V_{DS} is the drain-source voltage, I_D is the drain current, and $R_{DS(\text{on})}$ is the drain-source ON resistance under stated gate-drive and temperature conditions.

Unlike a BJT in saturation, a MOSFET ON-state drop is not approximately fixed. It increases roughly in proportion to current.

So if a MOSFET has $R_{DS(\text{on})} = 35$ mΩ at the chosen gate drive, and it carries 20 A, then

$$V_{DS} \approx 20 \times 0.035 = 0.70 \text{ V}.$$

The instantaneous conduction loss is then

$$P_{\text{cond}} = I_D^2 R_{DS(\text{on})}. \quad (4.2)$$

Substituting the same current,

$$P_{\text{cond}} = 20^2 \times 0.035 = 14 \text{ W}.$$

At moderate current and low voltage, the ON-state drop can be small. Because the loss varies as I^2, however, conduction loss rises quickly as current increases.

Linear region and saturation region: careful terminology

MOSFET terminology can be confusing because words used in analog electronics and words used in power electronics do not always match everyday intuition.

In many device texts, when $V_{GS} > V_{GS(\text{th})}$ and V_{DS} is small, the MOSFET is said to operate in the **ohmic region** or **linear region**. Here it behaves approximately like a voltage-controlled resistor. As V_{DS} increases beyond a certain point, the channel pinches near the drain, and the device enters what device physics texts call the **saturation region**.

For a long-channel ideal MOSFET, the boundary is commonly written as

$$\boxed{V_{DS} = V_{GS} - V_{GS(\text{th})}} \quad (4.3)$$

and the idealized saturation-region current is written as

$$\boxed{I_D \approx \frac{k}{2}\left(V_{GS} - V_{GS(\text{th})}\right)^2} \quad (4.4)$$

where k is a device-dependent constant.

For power-electronics switching, however, the MOSFET is usually not operated as an analog amplifier. The desired states are:

- the OFF state, where the device blocks voltage, or
- the strongly ON state, where $R_{DS(\text{on})}$ is low.

In converter language, "fully ON" usually means that the gate is driven well above threshold so the device enters a low-resistance state, not merely the onset of the textbook saturation region. Device-physics **saturation** is therefore not the same as the "hard saturation" idea used for BJTs.

Output characteristics

The **output characteristics** of a MOSFET plot drain current I_D against drain-source voltage V_{DS} for several fixed values of gate-source voltage V_{GS}.

Image prompt for Figure 4.2: Create a textbook-style graph of N-channel power MOSFET output characteristics. Use horizontal axis V_{DS} in volts and vertical axis I_D in amperes. Draw a family of curves for increasing V_{GS} values such as 4 V, 5 V, 6 V, 8 V, and 10 V. Show an initial near-linear ohmic region near the origin, then a bend into a current-flattening active region. Mark the boundary approximately as $V_{DS} = V_{GS} - V_{GS(th)}$. Label the low-$V_{DS}$ region as "ohmic/low-resistance region" and the higher-V_{DS} region as "current-controlled region (device-text saturation)." Use monochrome engineering style.

If V_{GS} is small, the channel is weak and only limited current can flow. As V_{GS} rises, the channel becomes stronger and the entire output curve moves upward. Near the origin, the curves are almost straight because the MOSFET behaves like a low resistance. At larger V_{DS}, the current changes less strongly with voltage.

For converter design, the leftmost part of the graph is usually the most important because it corresponds to low conduction loss. If a datasheet graph shows that the current curve at $V_{GS} =$ 5 V is much lower than at $V_{GS} = 10$ V, gate-drive voltage must be treated as a design variable rather than a formality. A MOSFET with $R_{DS(\text{on})}$ guaranteed only at $V_{GS} = 10$ V may dissipate much more power in a 5 V gate-drive circuit unless it is specifically intended for logic-level operation [FQP30N06L Datasheet].

Transfer characteristics

The **transfer characteristic** plots drain current I_D versus gate-source voltage V_{GS} at a stated drain-source voltage and temperature.

Image prompt for Figure 4.3: Create a clean textbook-style graph of N-channel power MOSFET transfer characteristics. Use horizontal axis gate-source voltage V_{GS} in volts and vertical axis drain current I_D in amperes. Show a curve that begins near zero current below threshold, then rises steeply after $V_{GS(th)}$. Mark threshold voltage, indicate that threshold is defined at a small test current, and add a second curve at higher junction temperature shifted slightly to illustrate temperature dependence. Use monochrome engineering style.

Whereas the output characteristics show how current varies with V_{DS} at fixed V_{GS}, the transfer characteristic shows how strongly drain current rises as V_{GS} increases at stated drain voltage and temperature.

The slope of the transfer curve is related to **transconductance**, written g_m, which is the change in drain current per change in gate-source voltage:

$$\boxed{g_m = \frac{\Delta I_D}{\Delta V_{GS}}} \quad (4.5)$$

High transconductance means the drain current responds strongly to gate-voltage change.

Because threshold is defined at a very small specified current, often in the milliampere range, the transfer curve should not be treated as proof of low-loss operation. Practical low-loss design still depends on the guaranteed $R_{DS(\mathrm{on})}$ values and the relevant curves at the intended gate drive [FQP30N06L Datasheet], [Vishay AN605].

Temperature effect on ON resistance

Power MOSFET datasheets usually show that $R_{DS(\mathrm{on})}$ increases with junction temperature. A device that is excellent at 25°C may therefore exhibit much higher resistance in real equipment.

If the normalized resistance at operating temperature is denoted by a factor K_T, we may write

$$\boxed{R_{DS(\mathrm{on}),T} \approx K_T\, R_{DS(\mathrm{on}),25^\circ\mathrm{C}}} \quad (4.6)$$

where K_T is greater than 1 at elevated temperature.

This temperature rise increases conduction loss, but it also gives MOSFETs a useful feature in parallel operation. If one MOSFET in a parallel group begins to carry more current, it heats more, its $R_{DS(\mathrm{on})}$ increases, and some current tends to shift to the cooler devices. This is not a complete cure for current imbalance, but it is much friendlier than the negative temperature behavior that can make BJTs hard to parallel.

1.4.3 Switching behaviour, body diode, specifications

Why the gate is easy to drive, but not effortless to drive

A MOSFET is voltage-driven in steady state, but that description is incomplete if it is interpreted too casually.

Because the gate is insulated by oxide, it ideally draws negligible DC current. The gate still presents capacitance that must be charged and discharged each switching cycle, so the driver does not supply steady current as a BJT base driver does, yet it must deliver pulse current quickly if fast switching is desired [onsemi AN-9010/D], [Vishay AN605].

Manufacturers usually describe this dynamic behavior through capacitances such as:

- C_{iss}: input capacitance

- C_{oss}: output capacitance
- C_{rss}: reverse transfer capacitance

and through total gate charge Q_g.

A practical average gate-drive relation is

$$\boxed{I_{G,\text{avg}} \approx Q_g f_s} \quad (4.7)$$

where $I_{G,\text{avg}}$ is the average current needed from the driver over time, Q_g is the total gate charge for one turn-ON event, and f_s is the switching frequency.

If the driver swings the gate by a voltage V_{GG} each cycle, the approximate gate-drive power is

$$\boxed{P_G \approx Q_g V_{GG} f_s} \quad (4.8)$$

This power is not dissipated mainly in the MOSFET channel as conduction loss. It is associated with charging and discharging the gate and with losses in the driver path.

These relations show an important tradeoff. A large MOSFET may achieve low $R_{DS(\text{on})}$ at the cost of substantial gate charge, so low conduction loss and easy high-speed switching do not automatically coincide.

Turn-ON and turn-OFF intervals

A MOSFET switching waveform is usually described through delay and transition intervals, similar in spirit to a BJT but with different physical causes:

- turn-ON delay time
- rise time
- turn-OFF delay time
- fall time

The distinctive MOSFET feature is the **Miller plateau**. During part of the switching transition, the gate voltage stops rising much even though gate current is still flowing, because that charge is being used mainly to change the drain voltage through the reverse-transfer capacitance C_{rss} [onsemi AN-9010/D].

Image prompt for Figure 4.4: Create a textbook-style set of switching waveforms for an N-channel power MOSFET. Show three aligned plots versus time: gate-source voltage V_{GS}, drain-source voltage V_{DS}, and drain current I_D. In the gate-voltage plot, clearly show initial charging, threshold crossing, a Miller plateau, then the final rise to full gate drive. In the drain-voltage plot, show V_{DS} starting high, then falling during the Miller plateau. In the drain-current plot, show I_D rising after threshold and reaching load current before V_{DS} fully falls. Label turn-on delay, rise time, turn-off delay, fall time, and Miller plateau. Use monochrome engineering style with units on axes.

During turn-ON, gate charge first raises V_{GS} to threshold, then I_D rises, then V_{DS} falls while the gate remains near the Miller plateau, and finally V_{GS} rises to the full drive value. During turn-OFF, the sequence reverses: the gate is discharged, the device passes through the Miller plateau while V_{DS} rises, and the current then falls away.

A first estimate of switching energy is

$$\boxed{E_{\text{sw,approx}} \approx \frac{1}{2} V_{DS} I_D (t_r + t_f)} \quad (4.9)$$

and the corresponding average switching loss is

$$\boxed{P_{\text{sw,approx}} \approx f_s E_{\text{sw,approx}}} \quad (4.10)$$

These are first estimates, but they capture the main trend: switching loss grows with voltage, current, transition time, and switching frequency.

The body diode

The **body diode** is an intrinsic diode created by the P-body and N-drift structure of the MOSFET. In an N-channel MOSFET used as a low-side switch, it is oriented so that it conducts when the source becomes more positive than the drain by about a diode drop. In many converter circuits this diode provides a natural path for **freewheeling current**, carries current during dead time in bridge circuits, and makes the MOSFET compatible with inductive switching arrangements.

The body diode also brings limitations. Its forward drop may be significant, and its reverse-recovery behavior may create extra loss and stress. In some high-performance converters, an external Schottky diode or a synchronous rectification strategy may still be preferred.

Datasheets therefore often specify body-diode or reverse-recovery parameters such as reverse-recovery time t_{rr} and recovered charge Q_{rr}. These become very important in hard-switched converters.

For a buck converter in a solar charge controller, for example, current in the inductor must continue when the main switch turns OFF. If the freewheel path is through the body diode of another MOSFET, then that diode's recovery behavior affects efficiency and EMI. So the body diode is not merely a "bonus diode." It is part of the switching design.

Safe operating area and avalanche ruggedness

Although MOSFETs avoid second breakdown in the same severe way as BJTs, they still have important operating limits. Datasheets commonly specify:

- maximum drain-source voltage
- continuous drain current
- pulsed drain current
- power dissipation
- thermal resistance
- safe operating area

- avalanche energy or avalanche ruggedness

If an inductive circuit forces the drain-source voltage beyond the rated blocking condition, the MOSFET may enter avalanche. Some devices are designed to survive a stated single-pulse avalanche energy under specified conditions, but this should not be treated as a normal everyday operating mode unless the application is designed for it [Vishay AN605], [FQP30N06L Datasheet].

Thermal specification

Like every power semiconductor, the MOSFET is ultimately limited by junction temperature. A simple thermal estimate is

$$\boxed{T_J \approx T_C + P_D R_{\theta JC}} \quad (4.11)$$

where T_J is junction temperature, T_C is case temperature, P_D is dissipation, and $R_{\theta JC}$ is junction-to-case thermal resistance.

If ambient-based thermal data are being used instead, a similar form applies:

$$\boxed{T_J \approx T_A + P_D R_{\theta JA}} \quad (4.12)$$

where T_A is ambient temperature and $R_{\theta JA}$ is junction-to-ambient thermal resistance under stated mounting conditions.

These equations show why a MOSFET that appears safe electrically may still fail thermally in a compact battery charger or inverter auxiliary supply.

The most important MOSFET datasheet specifications

Table 4.2: Practical meaning of major power-MOSFET specifications

Datasheet item	What it means	Why it matters
V_{DSS} or BVDSS	Maximum drain-source blocking voltage	Must exceed worst-case circuit voltage with margin
$V_{GS(\text{th})}$	Threshold voltage at small test current	Indicates turn-on onset, not full enhancement
$R_{DS(\text{on})}$	ON resistance at stated V_{GS} and temperature	Sets conduction loss
I_D	Continuous drain current	Limited by package and thermal conditions
Q_g	Total gate charge	Influences gate-drive requirement and switching speed
C_{iss}, C_{oss}, C_{rss}	Dynamic capacitances	Affect switching transitions and EMI

Datasheet item	What it means	Why it matters
t_d, t_r, t_f	Switching times	Useful for timing and first loss estimates
Body-diode and t_{rr} data	Reverse-conduction and recovery behavior	Important in inductive and bridge circuits
$R_{\theta JC}$, $R_{\theta JA}$	Thermal resistances	Needed to estimate temperature rise

Two common interpretation mistakes are:

1. assuming threshold voltage tells you the proper gate-drive voltage,
2. reading $R_{DS(\text{on})}$ without checking the gate voltage and temperature at which it is specified.

Worked interpretation exercise

The onsemi FQP30N06L Datasheet provides a useful low-voltage example. It includes blocking-voltage rating, current rating, $R_{DS(\text{on})}$ values at different gate-drive conditions, threshold data, characteristic curves, and gate-charge information.

Step 1: Read the voltage rating before anything else

The datasheet gives a drain-source voltage rating of $V_{DSS} = 60$ V [FQP30N06L Datasheet].

This places the device in the low-voltage class. It is suitable for circuits such as:

- 12 V battery systems
- 24 V battery systems
- 48 V nominal battery systems with careful surge analysis
- low-voltage DC-DC converters and battery chargers

It is not suitable as a direct switch on a rectified 230 V mains DC bus, because that bus is about 325 V under nominal conditions. Nor is it suitable for a 415 V three-phase rectified bus, which is much higher still.

Step 2: Interpret the current rating carefully

The datasheet headline gives a continuous drain current of 32 A at a specified case condition [FQP30N06L Datasheet].

This is a conditional thermal rating, not a universal statement that the device can carry 32 A in any PCB assembly. The allowable continuous current depends strongly on case temperature, package mounting, copper area, and heat removal. In a practical converter, the thermal environment may reduce the usable current substantially.

Step 3: Read $R_{DS(\mathrm{on})}$ together with gate-drive voltage

The datasheet specifies maximum $R_{DS(\mathrm{on})}$ values at different gate-drive conditions, including a lower resistance at $V_{GS} = 10$ V and a somewhat higher value at $V_{GS} = 5$ V [FQP30N06L Datasheet].

This indicates that the device can be used with logic-level drive, but it still performs better with stronger gate drive. If a converter controller can provide only 5 V at the gate, the 5 V resistance value must be used in the loss estimate, not the 10 V value.

Using the conservative value $R_{DS(\mathrm{on})} = 45$ mΩ at $V_{GS} = 5$ V for a low-voltage converter switch carrying $I_D = 12$ A, the conduction loss estimate is:

$$P_{\mathrm{cond}} = I_D^2 R_{DS(\mathrm{on})} = 12^2 \times 0.045 = 6.48\ \mathrm{W}.$$

This is already substantial for a TO-220 device without strong cooling. So even in low-voltage circuits, current must be respected.

Step 4: Read threshold voltage correctly

The datasheet gives a threshold-voltage range roughly around the few-volt level under a very small drain-current test condition [FQP30N06L Datasheet].

This marks the onset of channel formation, not low-loss switching. For power-switching design, the guaranteed $R_{DS(\mathrm{on})}$ values at 5 V and 10 V are more informative than the threshold figure.

Step 5: Use the curves, not only the table

The curves complement the summary table. The output characteristics show current capability at a given gate drive, the transfer characteristic shows how strongly conduction rises with V_{GS}, the normalized-resistance plot shows the temperature dependence of $R_{DS(\mathrm{on})}$, the gate-charge plot indicates driver demand during switching, and the body-diode plus reverse-recovery data show the cost of current commutation through the intrinsic diode [FQP30N06L Datasheet].

Step 6: Connect the part to a realistic renewable-energy task

Consider a 24 V battery-powered solar street-light controller using a synchronous buck converter to charge a battery from a PV module. A 60 V MOSFET is in the correct general voltage class for this type of low-voltage system, and the logic-level gate feature is useful if the controller IC drives 5 V gates directly. The final choice still depends on worst-case PV open-circuit voltage and surges, conduction loss at elevated temperature, body-diode behavior during dead time, and the thermal path through the PCB or heat sink.

How this matters in renewable-energy systems

Power MOSFETs are deeply embedded in renewable-energy hardware wherever voltage is moderate and switching frequency is high. In **solar PV systems**, they are common in MPPT buck, boost, and buck-boost stages, battery chargers, and low-power DC optimizers. In **battery-energy-storage systems**, they appear in bidirectional DC-DC converters, protection switches, and precharge circuits. In **EVs** and **UPS systems**, they are widely used in auxiliary converters, battery chargers, and low-voltage DC buses.

Selection in these systems depends directly on the parameters discussed in this chapter. Voltage rating determines whether the MOSFET belongs on a battery bus or a higher-voltage DC link, $R_{DS(\text{on})}$ sets conduction loss at large current, gate charge and switching times shape driver demand and switching loss, body-diode behavior affects dead time and synchronous rectification, and thermal resistance limits performance in compact enclosures.

Chapter summary

- The **power MOSFET** combines an insulated, voltage-controlled gate with majority-carrier switching and a vertical structure suited to power conversion [onsemi AN-9010/D], [Vishay AN605].
- The MOS transistor principle dates back to 1959, while power-electronics usefulness depended on later vertical structures such as V-DMOS [Computer History Museum, “MOS Transistor Demonstrated”, 1959], [IEEE TED, “The Trench Power MOSFET: Part I”].
- In a **vertical MOSFET**, current flows vertically through the die, while the drift region supports OFF-state voltage. Higher voltage rating generally increases $R_{DS(\text{on})}$.
- For an N-channel enhancement MOSFET, threshold voltage marks the start of channel formation, not the gate-drive condition for low-loss operation.
- In the ON state, the practical relations $V_{DS} \approx I_D R_{DS(\text{on})}$ and $P_{\text{cond}} = I_D^2 R_{DS(\text{on})}$ are central to conduction-loss estimation.
- Output and transfer characteristics show how strongly MOSFET behavior depends on actual gate-drive voltage and temperature.
- Gate drive is voltage-controlled in steady state but dynamic in switching, because capacitances and gate charge determine current demand and the Miller plateau.
- The intrinsic body diode, safe operating limits, and thermal resistance are part of normal device selection, not secondary details.
- In low- and medium-voltage renewable-energy converters, MOSFET choice is governed by the combined requirements of voltage rating, conduction loss, switching behavior, and heat removal.

Further reading

- B. Jayant Baliga, “The Trench Power MOSFET: Part I. History, Technology, and Prospects,” *IEEE Transactions on Electron Devices*, vol. 55, no. 12, 2008. Useful for understanding how vertical and trench MOSFET technology developed and why structure matters.
- onsemi, *AN-9010/D: MOSFET Basics*. A practical manufacturer note that explains structure, operation, capacitances, switching, and the Miller effect in power-switching terms.
- Vishay Siliconix, *AN605: Power MOSFET Basics*. A clear application-oriented reference on V-DMOS structure, ratings, ON resistance, body diode, and safe use in circuits.

- onsemi, *FQP30N06L Datasheet*. A good real datasheet for learning how to interpret threshold voltage, ON resistance, gate charge, transfer curves, and body-diode data together.
- Computer History Museum, "MOS Transistor Demonstrated, 1959." Useful for the historical origin of the MOS transistor that later enabled the power MOSFET family.

Chapter 1.5: IGBT

Chapter opening

The **insulated-gate bipolar transistor**, or **IGBT**, combines a MOS insulated gate with bipolar current conduction. That combination made it one of the standard switches in medium- and high-power converters: the gate can be driven as a voltage-controlled input, while the conduction path avoids much of the high-voltage ON-resistance penalty of a comparable silicon MOSFET [Infineon IGBT Basic Know-How], [Infineon IGBTs Overview]. For that reason, IGBTs became common in industrial inverters, UPS systems, motor drives, traction converters, battery chargers, and many solar and wind power stages. This chapter examines the origin of the device, its internal structure, its static and switching characteristics, and its practical position relative to the power BJT and the power MOSFET.

Core content

1.5.1 History of IGBT

The IGBT emerged from a specific limitation in high-voltage switching. In a three-phase inverter fed from a rectified 415 V supply, the DC link is often around 560 V to 600 V. A high-voltage BJT can operate in this range, but it needs continuous base current and suffers from stored charge during turn-OFF. A high-voltage silicon MOSFET is easier to drive, but its ON resistance rises strongly as voltage rating increases. Designers therefore needed a device that combined an insulated gate with stronger high-voltage current conduction.

The IGBT provided that combination. It was invented by B. Jayant Baliga while he was working at General Electric. NC State University notes both the importance of the invention and the 1980 date commonly associated with it [NC State, Baliga Hall of Fame], [NC State, Pack Power].

The name itself describes the device. **Insulated-gate** indicates that control is applied through a MOS-like gate separated from the semiconductor by oxide. **Bipolar transistor** indicates that the vertical current path uses bipolar carrier action rather than majority-carrier conduction alone. The result is a hybrid device that keeps the high-input-impedance gate of the MOSFET while gaining the conduction advantage of bipolar behavior [Infineon IGBT Basic Know-How].

Table 5.1: The design gap that led to the IGBT

Device	Main strength	Main limitation in power switching
Power BJT	Good conduction capability at high voltage	Needs continuous base current; turn-off slowed by stored charge

Device	Main strength	Main limitation in power switching
Power MOSFET	Easy voltage drive and fast switching	High-voltage devices suffer rising $R_{DS(\text{on})}$
IGBT	Combines insulated gate with strong high-voltage current conduction	Turn-off is slower than MOSFET because of stored charge and tail current

The IGBT did not replace every other power device. MOSFETs remain excellent at lower voltage and higher switching frequency, while thyristor-family devices remain important in very high-power or line-frequency applications. The IGBT instead became dominant in a wide middle range of industrial conversion. Infineon describes this range as extending into kilovolt classes and switching frequencies from a few kilohertz to the tens of kilohertz, which matches many motor-drive, UPS, PV-inverter, battery-charger, and wind-converter applications [Infineon IGBTs Overview]. The different conduction mechanism also explains why the ON state is described by $V_{CE(\text{sat})}$ rather than by an ON resistance, and why turn-OFF exhibits tail current.

1.5.2 Structure, operating principle, output and transfer characteristics

Why the IGBT structure is different from a MOSFET

Like a vertical MOSFET, an IGBT uses a drift region to block voltage. Unlike a MOSFET, it introduces carrier injection into that drift region during conduction. The gate forms a channel in the body region, electrons enter from the emitter side, and the P+ collector injects holes into the drift region. The additional carriers raise the conductivity of the drift region, a phenomenon called **conductivity modulation** in power-device literature [Infineon IGBT Basic Know-How]. In practical terms, a high-voltage IGBT can conduct large current with a smaller ON-state penalty than a comparable high-voltage silicon MOSFET in many applications.

Image prompt for Figure 5.1: Create a clean textbook-style technical illustration of a vertical N-channel IGBT cross-section. Show emitter metallization at the top contacting repeated N+ emitter regions inside P-body regions. Show a gate insulated by silicon dioxide over the channel area. Below the body, show an N- drift region and a P+ collector layer at the bottom with collector metallization. Label emitter, gate, collector, N+ emitter, P-body, inversion channel, N-drift region, P+ collector, oxide, and indicate electron flow from emitter toward drift region and hole injection from collector toward drift region during conduction. Use monochrome engineering style with no decorative background.

Three structural features deserve emphasis. The **gate** is insulated from the semiconductor by oxide, so the steady-state input impedance is very high. The current path is vertical, which supports high current density and high blocking voltage. The bottom **P+ collector** layer injects carriers into the drift region during conduction, and that injection gives the device its bipolar character.

Basic operating principle

The main terminals of an IGBT are:

- **Gate** (**G**)
- **Collector** (**C**)
- **Emitter** (**E**)

For the common **N-channel IGBT**, the collector is usually at the high-potential side and the emitter at the lower-potential side.

The sequence of operation is:

1. With $V_{GE} = 0$, where V_{GE} is gate-emitter voltage, no conducting channel exists in the body region, so the device is OFF and blocks collector-emitter voltage.
2. As V_{GE} rises and exceeds the threshold value, a channel forms under the gate.
3. Electrons can then flow from emitter into the drift region.
4. This channel action supports bipolar carrier injection inside the vertical structure and greatly improves conduction.

The threshold voltage is written as $V_{GE(\mathrm{th})}$. As in the MOSFET, it marks the onset of conduction rather than the recommended drive condition for low-loss switching [Infineon Discrete IGBT Datasheet Explanation]. Many discrete IGBTs are characterized at about 15 V gate drive in normal switching use [Infineon Discrete IGBT Datasheet Explanation], [Infineon IKW40N120H3 Datasheet].

ON-state behavior and saturation voltage

In power-circuit language, the IGBT ON state is usually described by its **collector-emitter saturation voltage**, written $V_{CE(\mathrm{sat})}$.

This already marks a difference from the MOSFET. A MOSFET in the ON state is usually described by $R_{DS(\mathrm{on})}$, whereas an IGBT is usually described by a voltage drop.

A first practical conduction-loss estimate is

$$\boxed{P_{\mathrm{cond}} \approx V_{CE(\mathrm{sat})} I_C} \quad (5.1)$$

where P_{cond} is conduction loss, $V_{CE(\mathrm{sat})}$ is the collector-emitter saturation voltage, and I_C is collector current.

If the device conducts only for duty ratio D, then a simple average estimate is

$$\boxed{P_{\mathrm{cond,avg}} \approx D\, V_{CE(\mathrm{sat})} I_C} \quad (5.2)$$

Consider a medium-power inverter leg in which an IGBT has $V_{CE(\mathrm{sat})} = 2.1$ V at the operating current and carries 20 A for 40% of the average switching cycle. Then

$$P_{\mathrm{cond,avg}} \approx 0.40 \times 2.1 \times 20 = 16.8 \text{ W}.$$

This estimate shows why the IGBT is attractive in medium-voltage converters: a few volts of ON-state drop may be acceptable at large current even when a comparable high-voltage MOSFET would incur a substantial resistance penalty. It also shows that an IGBT is not a zero-loss switch. A few volts at tens of amperes produces significant heat.

Output characteristics

The **output characteristics** plot collector current I_C versus collector-emitter voltage V_{CE} for several fixed values of gate-emitter voltage V_{GE}.

Image prompt for Figure 5.2: Create a textbook-style graph of N-channel IGBT output characteristics. Use horizontal axis collector-emitter voltage V_{CE} in volts and vertical axis collector current I_C in amperes. Draw a family of curves for increasing gate-emitter voltages such as 7 V, 9 V, 11 V, 13 V, and 15 V. Show that higher V_{GE} allows higher collector current. Mark a low-V_{CE} region labeled “on-state / saturation region in power-switching use” and a higher-V_{CE} region labeled “active region.” Use monochrome engineering style with clear curve labels.

The graph shows that the current capability rises strongly as V_{GE} increases. In converter service, the important distinction is practical rather than purely device-theoretic: the IGBT must block voltage in the OFF state, carry current with acceptable loss in the ON state, and move safely between those states during switching.

Transfer characteristics

The **transfer characteristic** plots collector current I_C as a function of gate-emitter voltage V_{GE} at stated test conditions.

Image prompt for Figure 5.3: Create a clean textbook-style graph of IGBT transfer characteristics. Use horizontal axis gate-emitter voltage V_{GE} in volts and vertical axis collector current I_C in amperes. Show a curve with negligible current below threshold, then rapidly increasing current after threshold. Mark $V_{GE(th)}$, and add a note that practical power switching commonly uses gate drive around 15 V, not merely threshold voltage. Use monochrome engineering style.

The transfer characteristic shows how strongly the device turns ON as the gate voltage increases. It reinforces the distinction between threshold and full gate drive: threshold voltage marks the onset of measurable conduction under specified test conditions, whereas efficient power switching requires the intended drive level from the datasheet.

Input behavior and transconductance

Because the gate is insulated, the input behaves primarily as a capacitive load rather than a steady current load. Current is therefore required mainly during transitions, while charging or discharging the gate.

A useful quantity here is **transconductance**, written g_{fs}, which relates change in collector current to change in gate-emitter voltage under stated conditions:

$$\boxed{g_{fs} = \frac{\Delta I_C}{\Delta V_{GE}}} \quad (5.3)$$

This is not usually the first parameter used for converter loss estimation, but it shows that the gate still controls the output strongly even though the output current path itself is bipolar.

Positive temperature coefficient and paralleling

Modern IGBTs often show a **positive temperature coefficient** of $V_{CE(\mathrm{sat})}$ above a certain current range. Infineon notes that with newer Trenchstop technology, a 40 A device shows a positive temperature coefficient starting from about 10 A, which helps current sharing when devices are paralleled [Infineon Discrete IGBT Datasheet Explanation].

This behavior is useful in parallel operation because a hotter device tends to develop a higher ON-state drop and therefore give up some current to a cooler device. The effect is more favorable than the current-crowding tendency that makes parallel BJTs harder to manage. It is still not a license for careless paralleling: layout, gate resistance, stray inductance, and thermal matching remain important.

1.5.3 Switching characteristics, latch-up, SOA

Why IGBT switching is different from MOSFET switching

The IGBT gate is MOS-like, but the output conduction path is bipolar. Its switching behavior therefore combines familiar MOS-gate behavior with limits set by stored charge in the drift region.

Turn-ON begins when the gate is charged, a channel forms, and collector current rises. Turn-OFF is more difficult because stored charge remains in the device after the gate drive is removed. That stored charge produces the well-known **tail current**.

The standard switching intervals are:

- **Turn-on delay time** $t_{d(\mathrm{on})}$
- **Rise time** t_r
- **Turn-off delay time** $t_{d(\mathrm{off})}$
- **Fall time** t_f

These quantities are used in datasheets and follow IEC/JEDEC-style definitions under specified test circuits [Infineon Discrete IGBT Datasheet Explanation].

The turn-off tail current

The **tail current** is the slowly decaying current that persists during turn-OFF because stored charge in the bipolar conduction path must be removed. Infineon's datasheet-explanation note explicitly states that switching-loss timing for the IGBT takes the tail-current effect into account [Infineon Discrete IGBT Datasheet Explanation].

This is a fundamental tradeoff. A MOSFET can switch very fast because it is a majority-carrier device. An IGBT benefits from conductivity modulation during conduction, but the price is stored charge and a slower turn-OFF waveform.

Image prompt for Figure 5.4: Create a textbook-style switching waveform figure for an IGBT. Show three aligned plots versus time: gate-emitter voltage V_{GE}, collector current I_C, and collector-emitter voltage V_{CE}. Mark turn-on delay time $t_{d(on)}$, rise time t_r, turn-off delay time $t_{d(off)}$, and fall time t_f. In the turn-off current waveform, show a clear current tail after the main current fall and label it "tail current due to stored charge." Use monochrome engineering style and clear timing markers.

First estimate of switching loss

Because voltage and current overlap during switching, the device dissipates energy during each transition. A simple introductory approximation is

$$\boxed{E_{\text{sw,approx}} \approx \frac{1}{2} V_{CC} I_C (t_r + t_f)} \quad (5.4)$$

where V_{CC} is the DC-link or blocking voltage, I_C is the switched current, and t_r and t_f are the rise and fall times.

The corresponding average switching-loss estimate is

$$\boxed{P_{\text{sw,approx}} \approx f_s E_{\text{sw,approx}}} \quad (5.5)$$

where f_s is switching frequency.

These equations provide a useful first estimate, but they can understate turn-OFF loss when tail current is significant. For that reason, datasheets often provide measured switching energies directly:

- **turn-on energy** E_{on}
- **turn-off energy** E_{off}
- sometimes total switching energy $E_{\text{ts}} = E_{\text{on}} + E_{\text{off}}$

Those values are measured under specified test conditions. They change with current, voltage, temperature, gate resistance, layout, and parasitics [Infineon Discrete IGBT Datasheet Explanation].

Gate charge and driver effort

Like the MOSFET, the IGBT gate must be charged and discharged. A useful first estimate for average gate-drive current is

$$\boxed{I_{G,\text{avg}} \approx Q_G f_s} \quad (5.6)$$

where Q_G is total gate charge.

If the gate swings through voltage V_{GG}, then a simple gate-drive power estimate is

$$\boxed{P_G \approx Q_G V_{GG} f_s} \quad (5.7)$$

Although the IGBT is voltage-driven, the driver is not effortless. Large devices and high switching frequencies still demand a capable gate driver.

Latch-up in the IGBT

The IGBT contains a parasitic thyristor structure because of its layered semiconductor arrangement. Under improper conditions, that parasitic structure can turn on. This unwanted condition is called **latch-up**.

Latch-up is not normal gate-controlled turn-ON. It is loss of normal control caused by triggering of the internal parasitic thyristor path. Research literature on IGBT structures analyzes this behavior explicitly [Solid-State Electronics, 1990, latch-up phenomena in IGBT structures].

Conditions that encourage latch-up include:

- excessive current density
- excessive dI/dt
- local hot spots
- insufficient latch-up immunity in the structure
- severe short-circuit or fault stress

Modern IGBTs are designed to be highly latch-up resistant in normal operation, but the concept remains important because it explains why current limits, short-circuit ratings, and safe operating area must be respected.

Safe operating area

The **safe operating area**, or **SOA**, gives the voltage-current combinations within which the device can operate safely under stated conditions.

For IGBTs, Infineon's datasheet-explanation note distinguishes two main forms [Infineon Discrete IGBT Datasheet Explanation]:

- **Forward-bias safe operating area (FBSOA)**
- **Reverse-bias safe operating area (RBSOA)**

FBSOA refers to safe conditions during forward-biased operation. RBSOA is especially relevant during turn-OFF with an inductive load, when the device experiences rising voltage and falling current at the same time.

Infineon notes that for state-of-the-art IGBTs, the RBSOA is often approximately square-shaped and bounded mainly by breakdown voltage and pulse current capability [Infineon Discrete IGBT Datasheet Explanation]. Even so, IGBT SOA remains limited by breakdown voltage, pulse current, junction temperature, stray inductance, gate-drive conditions, and switching overshoot.

Short-circuit withstand time

Many IGBTs specify a **short-circuit withstand time**, often written t_{SC}. This rating states how long the device can survive a specified short-circuit condition under stated gate drive, voltage, and temperature.

If a datasheet gives $t_{SC} = 10\,\mu$s, the meaning is narrow and practical: under the specified test conditions, the protection system must clear the fault before that survival limit is exceeded. This rating directly influences gate-driver protection and fault-detection design in inverter systems.

Thermal relation

As with all power semiconductors, switching and conduction losses appear as heat. A basic thermal relation is

$$\boxed{T_J \approx T_C + P_D R_{\theta JC}} \quad (5.8)$$

where T_J is junction temperature, T_C is case temperature, P_D is power dissipation, and $R_{\theta JC}$ is junction-to-case thermal resistance.

If ambient-based estimation is used instead, then

$$\boxed{T_J \approx T_A + P_D R_{\theta JA}} \quad (5.9)$$

where T_A is ambient temperature and $R_{\theta JA}$ is junction-to-ambient thermal resistance.

Infineon's application note also reminds us that the full thermal path includes junction-to-case, case-to-heatsink, and heatsink-to-ambient terms, so a low chip thermal resistance alone does not guarantee cool operation [Infineon Discrete IGBT Datasheet Explanation].

1.5.4 Comparison of power BJT, power MOSFET, and IGBT

Device selection in power electronics depends on voltage class, current, switching frequency, efficiency target, thermal conditions, and control burden. No single transistor is best in every regime.

Table 5.2: Practical comparison of power BJT, power MOSFET, and IGBT

Feature	Power BJT	Power MOSFET	IGBT
Control type	Current-controlled	Voltage-controlled	Voltage-controlled
Input burden in steady state	Significant base current	Very small ideal steady-state current	Very small ideal steady-state current
Main ON-state description	$V_{CE(\text{sat})}$	$R_{DS(\text{on})}$	$V_{CE(\text{sat})}$
Switching speed	Moderate to slow	Fast	Moderate; slower than MOSFET at turn-OFF

Feature	Power BJT	Power MOSFET	IGBT
Stored-charge problem	Strong	Low compared with bipolar devices	Present; causes tail current
High-voltage suitability in silicon	Reasonable, but drive is difficult	ON resistance rises strongly with voltage	Strong practical range at medium and high voltage
Typical modern role	Mostly legacy or specialized	Low- to medium-voltage, high-frequency converters	Medium- to high-power inverters and converters

Selection by voltage and frequency

As a first design rule, MOSFETs are often favored at lower voltage and higher switching frequency, whereas IGBTs are often favored at higher voltage and moderate switching frequency. BJTs are no longer the first choice in most mainstream new converter designs.

The reason follows directly from device physics. For silicon MOSFETs, high blocking voltage tends to increase drift-region resistance. The IGBT avoids the same resistance penalty by conductivity modulation, but it accepts slower turn-OFF and tail current. In practice, that makes the IGBT attractive in the 600 V, 1200 V, and higher classes when switching frequency is still in the range normally used for PWM conversion [Infineon IGBT Basic Know-How], [Infineon TRENCHSTOP IGBT6].

Application ranges

In **solar PV systems**, low-voltage MPPT buck or boost stages often favor MOSFETs, while several-kilowatt string inverters and larger central inverter stages often use IGBTs in the inverter bridge. Infineon specifically identifies 1200 V TRENCHSTOP IGBT families for solar applications [Infineon TRENCHSTOP IGBT6].

In **wind-energy converters**, the machine-side and grid-side power stages operate at substantial DC-link voltage and significant power, so IGBT modules are common in practical industrial designs. Their voltage capability, current handling, switching range, and mature module packaging suit this operating region well.

In **EV power stages**, low-voltage auxiliary DC-DC converters commonly use MOSFETs, while high-power traction inverters have long used IGBT modules in many silicon-based platforms. Silicon carbide devices now compete strongly in this area, but the IGBT remains a central reference for traction-class inverter design.

A realistic numeric selection example

Consider a three-phase inverter running from a rectified 415 V supply. The DC-link voltage may be around 600 V after margin and variation are considered.

A 60 V MOSFET is clearly unsuitable. A 600 V MOSFET may still be usable in some cases, but its conduction loss can become substantial unless the current is modest or the chosen technology is especially favorable. A 1200 V IGBT, by contrast, is a common practical class because it provides blocking-voltage margin and is designed for inverter switching in the tens-of-kilohertz range [Infineon IKW40N120H3 Datasheet], [Infineon TRENCHSTOP IGBT6].

This example does not prove that the IGBT is always the best choice. It shows why the IGBT belongs naturally in the design space of medium-voltage, medium- to high-power PWM converters.

Worked datasheet interpretation

The Infineon IKW40N120H3 datasheet provides a useful example of an inverter-class device: a 1200 V, 40 A IGBT with an anti-parallel diode in a TO-247 package [Infineon IKW40N120H3 Datasheet].

Step 1: Read the voltage class first

The datasheet gives collector-emitter voltage rating

$$V_{CE} = 1200\ \text{V}.$$

This rating places the device in the high-voltage inverter class rather than the low-voltage battery-converter class. It is suited to converter systems such as off-line industrial inverters, PV inverter stages, UPS bridges, and motor drives on 230 V or 415 V AC systems after rectification.

Step 2: Read current rating together with temperature

The datasheet lists

- $I_C = 80$ A at $T_C = 25$°C
- $I_C = 40$ A at $T_C = 100$°C

The current rating therefore depends directly on thermal condition. A headline current number is incomplete unless the associated temperature is also stated [Infineon IKW40N120H3 Datasheet].

Step 3: Interpret ON-state and gate-drive data together

The datasheet gives a typical value around

- $V_{CE(\text{sat})} = 2.05$ V at $I_C = 40$ A, $V_{GE} = 15$ V, $T_{vj} = 25$°C

with higher values at elevated junction temperature [Infineon IKW40N120H3 Datasheet].

This immediately links conduction behavior to the intended drive condition. The device is characterized for power operation at about 15 V gate drive, and conduction loss rises with both current and temperature.

If a first estimate is made at 20 A using the fixed-drop model, then

$$P_{\text{cond}} \approx V_{CE(\text{sat})} I_C \approx 2.05 \times 20 = 41 \text{ W}.$$

The estimate is crude because $V_{CE(\text{sat})}$ varies with current and temperature, but it is sufficient to show that an IGBT at significant current is a serious thermal device.

The same datasheet gives gate-emitter threshold voltage in the approximate range

$$V_{GE(\text{th})} = 5 \text{ V to } 6.5 \text{ V}$$

under a small test current [Infineon IKW40N120H3 Datasheet].

That value marks the onset of conduction, not the recommended gate-drive level for inverter operation. The saturation-voltage data are specified at 15 V, which identifies the intended drive condition.

Step 4: Read switching and protection data as test-condition data

The datasheet provides switching times and switching energies under specified conditions such as:

- $V_{CC} = 600$ V
- $I_C = 40$ A
- $V_{GE} = 0/15$ V
- gate resistances of $12\,\Omega$

It gives typical values such as

- $E_{\text{on}} = 3.2$ mJ at 25°C
- $E_{\text{off}} = 1.2$ mJ at 25°C

with higher values at elevated temperature [Infineon IKW40N120H3 Datasheet].

These are measured values under stated conditions, not fixed switching losses for every circuit.

The datasheet also lists turn-off safe operating area up to rated voltage, short-circuit withstand time $t_{SC} = 10\,\mu$s under stated conditions, and a gate-emitter voltage rating of ± 20 V with transient allowance to ± 30 V under specific conditions [Infineon IKW40N120H3 Datasheet]. Together, these ratings show that the device is intended for serious inverter duty and must be used with a proper driver, proper layout, and fast protection.

Step 5: Place the device in an application context

Consider a 5 kW to 10 kW solar string inverter or a UPS inverter operating from a several-hundred-volt DC link. A 1200 V IGBT of this class fits that environment naturally. The voltage rating provides margin, the anti-parallel diode supports bridge operation, and the switching-energy and short-circuit data are directly relevant to the design of a PWM inverter leg.

The datasheet is therefore more than a list of device limits. It identifies the voltage class, current capability, drive condition, switching behavior, protection requirements, and likely application range of the part.

Chapter summary

- The **IGBT** combines a MOS insulated gate with bipolar conduction, giving it voltage-driven input behavior and strong high-voltage current conduction [Infineon IGBT Basic Know-How].
- Its internal structure includes an N- drift region for voltage blocking and a P+ collector layer that enables conductivity modulation during conduction.
- The ON state is usually described by $V_{CE(\mathrm{sat})}$ rather than by an ON resistance, and a first conduction-loss estimate is $P_{\mathrm{cond}} \approx V_{CE(\mathrm{sat})} I_C$.
- Threshold voltage $V_{GE(\mathrm{th})}$ marks the onset of conduction, not the recommended drive level for power switching.
- The major switching limitation of the IGBT relative to the MOSFET is tail current during turn-OFF due to stored charge.
- Practical switching evaluation relies heavily on datasheet values such as E_{on}, E_{off}, SOA, and short-circuit withstand time.
- In broad selection terms, MOSFETs dominate lower-voltage high-frequency converters, while IGBTs are often preferred in medium- to high-voltage moderate-frequency inverter applications.
- The IGBT remains one of the key devices for understanding industrial drives, UPS systems, solar inverters, wind converters, and other medium- to high-power PWM systems.

Further reading

- Infineon, "IGBT-basic know-how - IGBT: how does an Insulated Gate Bipolar Transistor work?" - A practical manufacturer overview of IGBT operation, applications, and comparison with MOSFETs.
- Infineon, "Application Note: Discrete IGBT - Explanation of discrete IGBTs' datasheets" - Useful for threshold voltage, $V_{CE(\mathrm{sat})}$, switching energies, SOA, and thermal interpretation.
- Infineon, "IKW40N120H3 Datasheet" - A representative inverter-class discrete IGBT datasheet with switching energy, SOA, short-circuit, and diode information.
- Infineon, "IGBTs - Insulated gate bipolar transistors overview" - Useful for the present-day voltage and application range of industrial IGBTs.
- NC State University, "Baliga inducted into Electronic Design Engineering Hall of Fame" - A concise source for the invention significance of the IGBT and its broad real-world impact.

Module 2

Device Protection, Gate Drive and Thermal Management

Overview

This module develops the practical support circuits that make power switches reliable in real hardware, covering drive requirements, snubbers, electrical protection, and thermal design.

Contents

Chapter 2.1: Gate / Base Drive Circuits

Chapter opening

The performance of a power semiconductor depends not only on the device itself but also on the circuit that drives it. Between a low-power controller and a power switch sits the drive circuit, which must apply the correct control voltage or current, deliver adequate source and sink capability, establish the correct electrical reference, and often provide isolation or level shifting. A device with excellent ratings can still switch poorly or fail if its drive circuit is weak, noisy, or incorrectly referenced.

This chapter examines that interface. For a power BJT it is a **base-drive circuit**; for a MOSFET or IGBT it is a **gate-drive circuit**. The chapter develops the requirements of a good driver, the role of gate charge and peak current, the totem-pole output stage, dedicated driver ICs, high-side and low-side driving, and isolated-drive methods. These ideas are central to practical converters in PV systems, battery chargers, UPS hardware, motor drives, and EV auxiliary supplies, where switching quality depends as much on the driver as on the power device itself.

Prerequisites check

- forced beta and base-current requirement from Chapter 1.1
- capacitive gate behavior of the power MOSFET from Chapter 1.4
- stored-charge effects in the IGBT from Chapter 1.5
- basic relations such as $Q = CV$, $P = VI$, and the link between current, charge, and time
- the fact that converter switches may sit at potentials far above controller ground

The chapter assumes familiarity with basic switching waveforms, device terminals, and DC-bus voltage levels.

2.1.1 Requirements of a good drive circuit

A drive circuit is the interface between the low-power control stage and the power device. Its function is to force the device into the intended ON or OFF state at the intended time and with the intended electrical reference. The name changes with the device terminal: base drive for a BJT, gate drive for a MOSFET or IGBT. The system purpose is the same.

The first distinction is between current-driven and charge-driven control:

- A **power BJT** requires control current while it is ON.
- A **power MOSFET** or **IGBT** requires gate charge to be moved during switching; under steady DC gate bias, the ideal input current is very small [Infineon AN_2203 Gate Drive].

For a power BJT, the base-drive burden can be estimated from the forced-beta relation

$$\boxed{I_B \geq \frac{I_C}{\beta_{\text{forced}}}} \quad (2.1)$$

where I_B is base current, I_C is collector current, and β_{forced} is the deliberately chosen switching gain.

If a power BJT must carry 10 A and the design uses $\beta_{\text{forced}} = 5$, then

$$I_B \geq \frac{10}{5} = 2 \text{ A}.$$

This base-current requirement explains why a practical power-BJT drive stage is a substantial circuit rather than a direct logic interface.

For MOSFETs and IGBTs, the problem is different. The insulated gate does not require large steady current, but it must be charged and discharged quickly. Slow charging keeps the device in its transition region longer, increases switching loss, and can create cross-conduction in bridge circuits.

Table 2.1 summarizes the main requirements of a good drive circuit.

Table 2.1: Main requirements of a good drive circuit

Requirement	What it means in practice	Why it matters
Correct drive amplitude	Apply the correct base-emitter or gate-source / gate-emitter voltage	Too little drive raises loss; too much drive can damage the device
Adequate source and sink capability	Charge and discharge the control terminal quickly	Faster, cleaner switching and lower switching loss
Correct reference point	Drive voltage must be measured with respect to the correct source or emitter terminal	A 12 V gate signal is meaningful only if it is 12 V above the local source/emitter
Noise immunity	Resist false turn-ON from dv/dt, ringing, and parasitic inductance	Prevents shoot-through and erratic switching
Timing accuracy	Turn devices ON and OFF at the intended instants, often with dead time	Essential in half-bridges, full-bridges, and inverters
Protection support	Include or cooperate with UVLO, current sense, shutdown, and fault reporting	Prevents operation in unsafe conditions
Isolation or level shifting when needed	Transfer control safely across a large potential difference	Necessary in many high-side and mains-referenced power stages

Requirement	What it means in practice	Why it matters
Low output impedance	Hold the device firmly in OFF state and drive it strongly in transition	Reduces sensitivity to noise and Miller-coupled disturbances

Low output impedance matters because the control terminal must be held firmly at the intended voltage. A weak driver allows external disturbances to move the gate or base voltage. A strong driver resists that movement and provides active control during both turn-ON and turn-OFF.

Correct referencing is equally important. A MOSFET responds to V_{GS}, not to gate voltage measured relative to earth or controller ground. An IGBT responds to V_{GE}. If the source or emitter is moving, the driver must move with it or must be isolated from it. This is the basis of the high-side driving problem.

Image prompt for Figure 2.1: Create a clean textbook-style technical illustration showing the role of a power-device drive circuit between a low-voltage controller and a power semiconductor switch. Show, from left to right, a controller block generating a 3.3 V PWM signal, a gate/base driver block, and a power switch connected in a converter leg tied to a high-voltage DC bus. Include labels for "logic-level command," "level shifting," "source/sink current," "isolation if required," "local gate/emitter reference," and "power switch." Add a separate small comparison inset showing a BJT requiring base current continuously while a MOSFET/IGBT requires gate charge mainly during switching. Use monochrome textbook style with clear arrows and no decorative background.

The driver is therefore more than a pulse amplifier. It sets switching speed, OFF-state firmness, noise margin, and often part of the protection behavior. Insulated-gate devices do not require large steady input current, but they can still demand substantial peak drive current, and poor gate drive can create more loss in the switch than in the driver itself [Infineon AN_2203 Gate Drive].

2.1.2 MOSFET and IGBT gate driver basics: totem-pole driver and dedicated driver ICs

Gate charge and switching current

A MOSFET or IGBT gate is often described as voltage-controlled, but switching design is governed just as much by charge as by voltage. In steady state, the insulated gate draws very little DC current [Infineon AN_2203 Gate Drive]. During switching, however, the gate behaves as a capacitive load, and the driver must supply or remove charge quickly enough to produce acceptable transition times.

The fundamental relation is

$$\boxed{Q_G = \int i_G(t)\,dt} \quad (2.2)$$

where Q_G is total gate charge and $i_G(t)$ is gate current during the switching event.

Gate charge is often more useful than a single capacitance value for design estimates because the effective capacitances vary with operating condition [Infineon AN_2203 Gate Drive]. For a first estimate, if the driver current is treated as approximately constant over the critical part of the transition, then

$$\boxed{t_{\text{sw,est}} \approx \frac{Q_G}{I_G}} \qquad (2.3)$$

where $t_{\text{sw,est}}$ is estimated switching time, Q_G is required gate charge, and I_G is available gate-drive current.

The same logic gives a useful estimate of average gate-drive power:

$$\boxed{P_{\text{DRV,avg}} \approx Q_G V_{\text{DRV}} f_s} \qquad (2.4)$$

where $P_{\text{DRV,avg}}$ is average drive power, V_{DRV} is drive-voltage swing, and f_s is switching frequency [Infineon AN_2203 Gate Drive].

Equation (2.4) shows that average drive power can remain small even when peak gate current is large. That distinction is central in practice. A design may dissipate little average power in the driver while still requiring several hundred milliamperes or several amperes of peak current for fast switching.

A short numerical example

Suppose a MOSFET used in a 48 V to 12 V EV auxiliary converter has total gate charge $Q_G =$ 60 nC at the chosen drive condition, and the important part of the switching interval is limited to about 120 ns.

Using Equation (2.3),

$$I_G \approx \frac{Q_G}{t_{\text{sw,est}}} = \frac{60 \times 10^{-9}}{120 \times 10^{-9}} = 0.5 \text{ A}.$$

The driver should therefore be able to source and sink on the order of 0.5 A, and usually more once parasitics and design margin are included.

If the driver voltage swing is $V_{\text{DRV}} = 12$ V and the switching frequency is $f_s = 50$ kHz, Equation (2.4) gives

$$P_{\text{DRV,avg}} \approx 60 \times 10^{-9} \times 12 \times 50 \times 10^3 = 36 \text{ mW}.$$

The average gate-drive power is only about 36 mW, yet the peak current requirement is around 0.5 A. This is why a microcontroller pin is usually inadequate as a direct power-switch driver.

The Miller plateau

During turn-ON, gate voltage does not always rise at a uniform rate. There is often an interval in which the gate-voltage rise slows or nearly flattens while the drain-source or collector-emitter voltage is changing. This is the **Miller plateau**. During that interval, much of the gate current is diverted into the effective gate-drain or gate-collector feedback capacitance rather than into further gate-voltage rise [Infineon AN_2203 Gate Drive].

Its practical meaning is straightforward: the driver is doing critical work even while the gate voltage appears nearly stationary. That interval corresponds to the collapse of device voltage and a large share of switching loss.

Image prompt for Figure 2.2: Create a textbook-style waveform figure for MOSFET/IGBT gate driving. Show three aligned plots versus time: gate current i_G, gate-source or gate-emitter voltage, and drain-source or collector-emitter voltage. Mark initial charging, threshold crossing, Miller plateau, and final fully enhanced region. Label the plateau as the interval where device voltage is changing while gate current continues to flow. Use monochrome engineering style with axes, units, and clean annotations.

The totem-pole driver

One of the most important output structures in gate driving is the **totem-pole driver**, also called a **push-pull output stage**. One device in the driver sources current into the gate during turn-ON, and another device sinks current during turn-OFF. The result is a low-impedance path in both directions.

This is a major improvement over passive charging or discharging through large resistances. A resistor can eventually move the required charge, but a totem-pole stage does so actively and much more strongly. Discrete driver examples built from small-signal BJTs or MOSFETs are often used to raise a logic signal to the required gate-drive level and to provide the necessary source and sink current [Infineon AN_2203 Gate Drive].

In many practical designs the turn-OFF path is intentionally stronger than the turn-ON path. A strong pull-down helps prevent false turn-ON caused by $C\,dv/dt$ coupling in half-bridge and full-bridge circuits [Infineon AN_2203 Gate Drive].

A simple low-side example

If a low-side MOSFET source is tied near ground, a 3.3 V controller pulse may still be insufficient for efficient switching. The gate may require 10 V to 12 V drive, and the transition may require substantial current. A small level-shift stage followed by a totem-pole output stage can translate the logic pulse to the required amplitude and deliver strong gate current. Infineon gives a discrete BJT-based example that raises the logic pulse to a 12 V swing and provides approximately ± 0.5 A source and sink capability [Infineon AN_2203 Gate Drive]. The exact values are example-specific; the design lesson is general.

Image prompt for Figure 2.3: Create a clean textbook-style schematic-level illustration of a low-side totem-pole gate driver for a power MOSFET or IGBT. Show a 3.3 V logic input feeding a level-shift stage, then a push-pull output stage with an upper sourcing transistor and lower sinking transistor, then a gate resistor leading to the power device gate. Label source current path during turn-ON and sink current path during turn-OFF. Include the local source/emitter reference node and supply such as 12 V or 15 V. Use monochrome engineering style with clear labels and no decorative effects.

Dedicated gate-driver ICs

Discrete totem-pole stages are useful for explanation and still appear in practice, but modern converters commonly use **dedicated gate-driver ICs**. Such ICs provide a push-pull output stage together with features such as predictable peak source and sink current, UVLO, logic-level compatibility, matched timing, enable or shutdown functions, and in some families Miller clamp, fault reporting, dead-time control, or desaturation protection. Infineon notes the wide availability of low-side and dual-low-side driver ICs with different output-current classes and protection features [Infineon AN_2203 Gate Drive]. TI likewise describes modern isolated drivers with strong peak output current, UVLO, and timing features [TI UCC21520 Product Page].

Undervoltage lockout, or **UVLO**, is especially important. If the driver supply falls below a safe level, the driver inhibits normal operation rather than attempting to switch the device with inadequate drive voltage. Partial enhancement is dangerous because the current may be high while the ON-state voltage remains too large.

Drive voltage must also be taken from the specific device datasheet rather than from a generic rule. Infineon notes that standard-level power MOSFETs are commonly driven with about 10 V to 15 V gate pulses, while logic-level MOSFETs operate from lower gate voltage but may involve tradeoffs such as higher gate charge or greater susceptibility to induced turn-ON in half-bridge service [Infineon AN_2203 Gate Drive]. Threshold voltage is not the correct design target; it merely marks the onset of conduction under light test conditions. The same principle applies to IGBTs, whose recommended gate-emitter drive conditions depend on the device family and application.

2.1.3 High-side driving and isolated gate drive

Low-side and high-side driver concept

A **low-side switch** sits between the load and the negative rail. Its source or emitter is usually close to local ground, so the driver can often share the same reference. A **high-side switch** sits between the positive rail and the load. Its source or emitter moves with the switching node, so the driver output must be referenced to that moving node. A statement such as "apply 12 V to the gate" is incomplete unless it also states the reference: 12 V relative to the local source or emitter.

This is why high-side driving is more difficult. In a half-bridge, when the low-side switch turns ON, the midpoint falls near the negative rail; when the high-side switch turns ON, the midpoint rises near the positive rail. The high-side driver must therefore operate on a floating reference.

Bootstrap high-side driving

One common high-side solution is the **bootstrap circuit**. A capacitor is charged while the switching node is low. That stored charge is then used as a floating supply for the high-side driver when the switching node rises.

This approach appears in many high-side/low-side driver IC families. The Infineon IRS2101 datasheet, for example, describes a floating high-side channel intended for bootstrap operation, with up to 600 V offset capability and UVLO [Infineon IRS2101 Datasheet].

The first sizing relation is

$$\boxed{\Delta V_{\mathrm{BOOT}} \approx \frac{Q_{\mathrm{TOT}}}{C_{\mathrm{BOOT}}}} \quad (2.5)$$

where ΔV_{BOOT} is allowed bootstrap-voltage droop during the ON interval, Q_{TOT} is the total charge drawn from the capacitor during that interval, and C_{BOOT} is bootstrap capacitance.

This is a first estimate rather than a complete design equation. In practice, Q_{TOT} includes not only power-device gate charge but also driver quiescent current, leakage, and design margin.

If $Q_{\mathrm{TOT}} = 120$ nC and the allowed droop is $\Delta V_{\mathrm{BOOT}} = 0.2$ V, then

$$C_{\mathrm{BOOT}} \approx \frac{120\ \mathrm{nC}}{0.2\ \mathrm{V}} = 600\ \mathrm{nF}.$$

A practical design would choose a larger standard value to allow for tolerance and operating variation.

Bootstrap driving is compact and economical, but it is not universal. The capacitor must be refreshed, which means the switching pattern must periodically return the node to a condition that allows recharging. For that reason, bootstrap high-side drive is well suited to many PWM bridge circuits but not to every duty cycle or operating mode.

Image prompt for Figure 2.4: Create a textbook-style half-bridge driver illustration. Show a DC bus with upper high-side switch and lower low-side switch, a load connected to the midpoint, a low-side driver referenced to ground, and a floating high-side driver referenced to the switching node. Include a bootstrap diode and bootstrap capacitor between the driver supply and the high-side floating supply. Label the switching node, local emitter/source reference for each driver, charging path for the bootstrap capacitor, and high-side ON condition. Use monochrome engineering style with clear polarity and node labels.

When isolation is needed

Galvanic isolation means that no direct conductive path for steady current exists between the two sides of the interface. In gate driving, isolation may be required for safety, for noise immunity, for level shifting across large potential differences, or for some combination of these reasons. The controller may sit on a safe low-voltage domain while the power switch moves through hundreds of volts and high common-mode transient stress.

The main families are opto-coupler-based isolated drive, pulse-transformer-based isolated drive, and integrated isolated gate-driver ICs.

Opto-coupler-based isolated gate drive

An **opto-coupler** transfers the command across the isolation barrier by light. The input side drives an LED, and the output side detects that light and reconstructs the signal. In gate-drive applications, the isolated output side must still provide or control a stage capable of charging and discharging the MOSFET or IGBT gate.

The attraction of the opto-coupler method is clear galvanic isolation and good separation between the low-voltage controller domain and the noisy power domain. The main practical caution is equally clear: the isolated output side usually still needs its own local supply. The signal crosses the barrier optically, but the energy required to charge the gate on the isolated side must still be provided locally unless a specialized architecture is used.

Pulse-transformer-based isolated gate drive

A **pulse transformer** transfers the drive command magnetically. This approach has a long history in power electronics and is attractive where strong isolation and good common-mode immunity are required. ST's application note AN461 describes a pulse-transformer-based scheme for floating switches in motor drives, UPS systems, and AC switches, and shows how transformer coupling can transfer both signal information and drive energy without a floating auxiliary supply in that specific architecture [ST AN461].

Classical pulse-transformer drive, however, is constrained by volt-second balance and saturation. Infineon's isolated-gate-driving comparison note highlights these limitations and notes that simple pulse-transformer solutions are often duty-cycle-limited and suited to a bounded switching-frequency range [Infineon Isolated Gate Driving Solutions]. Specialized circuits can extend that range, but the transformer cannot be treated as an unrestricted DC-coupled interface.

Modern isolated gate-driver ICs

Many contemporary converters use **integrated isolated gate drivers** rather than a separate opto-coupler or custom pulse transformer. These devices place the isolation barrier and the output driver in a single package. TI's UCC21520, for example, is described as a dual-channel isolated gate driver with reinforced isolation, strong peak output current, UVLO, and high common-mode transient immunity [TI UCC21520 Product Page].

The integration changes packaging rather than design questions. The designer must still ask about isolation rating, channel arrangement, source and sink current, common-mode transient tolerance, and suitability for low-side, high-side, or half-bridge use.

Comparing the main approaches

Table 2.2 summarizes the main approaches at concept level.

Table 2.2: High-side and isolated driver approaches at concept level

Approach	Main strength	Main limitation or caution	Typical fit
Low-side non-isolated driver	Simple reference, low cost, easy layout	Only works when source/emitter stays near local ground	Buck converters, low-side switches, grounded stages
Bootstrap high-side driver	Compact and widely used in bridge circuits	Floating supply must be refreshed; not ideal for every duty condition	Half-bridges, inverters, synchronous legs
Opto-coupler isolated drive	Clear galvanic isolation, good noise separation	Often still needs isolated secondary supply	Mains-referenced stages, industrial inverters, battery chargers
Pulse-transformer isolated drive	Strong isolation and good dv/dt immunity	Classical circuits are constrained by volt-second balance and saturation	High-frequency isolated drive, some bridge and inverter stages
Integrated isolated gate-driver IC	Combines barrier and strong driver in one package	Cost and supply planning still matter	Modern PV, UPS, EV, and motor-drive power stages

High-side drive and isolated drive are related but not identical. A high-side driver must cope with a floating reference. Isolation is one way to achieve that, but bootstrap high-side drive is another and does not itself provide galvanic isolation.

Worked interpretation exercise

The TI UCC21520 product page is a useful example of how to read a modern gate-driver specification. TI describes the device as a dual-channel isolated gate driver with 4 A source and 6 A sink peak current, reinforced isolation of 5.7 kV RMS for one minute, minimum common-mode transient immunity greater than 125 V/ns, configuration options for two low-side drivers, two high-side drivers, or a half-bridge driver, and UVLO on all supply pins [TI UCC21520 Product Page].

Taken together, those statements identify a real power-driver component rather than a signal isolator. The peak-current ratings indicate a strong push-pull output stage. The isolation and CMTI ratings indicate suitability for fast, noisy switching environments. The channel configuration options show that the same IC can serve in several converter arrangements, and UVLO indicates that the device is designed to prevent unsafe partial-drive operation.

Table 2.3 translates the datasheet language into design language.

Table 2.3: Interpreting the UCC21520 datasheet

Datasheet statement	Plain-language meaning	Design importance
Dual-channel isolated gate driver	Two real gate-drive outputs sit behind an isolation barrier	Suitable for one half-bridge leg or two isolated switches
4 A source, 6 A sink peak	Strong push-pull output stage	Can move gate charge quickly and hold OFF state firmly
5.7 kV RMS reinforced isolation	High-grade barrier inside the package	Useful where controller and power stage must be safely separated
CMTI greater than 125 V/ns	Fast common-mode node movement will not easily corrupt the signal	Important in hard-switched inverter and converter legs
Configurable as two low-side, two high-side, or half-bridge	Flexible channel referencing and use	Fits different topologies without changing IC family
UVLO on supply pins	Driver refuses unsafe low-supply operation	Helps prevent partial enhancement and excess device loss

The same reading method applies to any driver datasheet. The essential questions are whether the part is a true gate driver or only an isolator, how strong the source and sink stages are, whether UVLO is included, what isolation and CMTI ratings are specified, and whether the device is intended for low-side, high-side, bootstrap-based, or fully isolated use.

How this matters in renewable-energy systems

Drive circuits are easy to overlook in introductory block diagrams, but they are central to the performance of practical converters. In a solar PV inverter or battery-storage converter, the controller may sit on a quiet low-voltage board while the switches operate on a noisy, high-voltage bridge node. The driver must translate those controller commands into correctly referenced, high-current switching action.

The result affects more than turn-ON and turn-OFF. Gate-drive strength, referencing, and isolation influence switching loss, EMI behavior, shoot-through immunity, fault response, and device survival. In that sense, the drive circuit is one of the places where converter theory becomes converter hardware.

Chapter summary

- A **drive circuit** interfaces the low-power control stage with the power semiconductor switch.
- A **base-drive circuit** for a BJT must supply control current while the device is ON; a first switching estimate is $I_B \geq I_C/\beta_{\text{forced}}$.
- A MOSFET or IGBT gate is insulated, so steady-state input current is small, but switching still requires charge transfer [Infineon AN_2203 Gate Drive].

- The fundamental gate-charge relation is $Q_G = \int i_G(t)\,dt$.
- A useful first switching-time estimate is $t_{\text{sw,est}} \approx Q_G/I_G$.
- A useful first average gate-drive-power estimate is $P_{\text{DRV,avg}} \approx Q_G V_{\text{DRV}} f_s$ [Infineon AN_2203 Gate Drive].
- The **Miller plateau** is the interval during which gate current continues to flow while device voltage is changing.
- A **totem-pole driver** actively sources current for turn-ON and sinks current for turn-OFF, giving low output impedance in both directions.
- **Dedicated gate-driver ICs** commonly add stronger output stages, UVLO, timing control, and sometimes additional protection features.
- A **low-side driver** is referenced to a near-ground source or emitter, while a **high-side driver** must operate on a floating reference.
- A **bootstrap high-side driver** uses stored charge on a capacitor to power a floating high-side channel; a first estimate is $\Delta V_{\text{BOOT}} \approx Q_{\text{TOT}}/C_{\text{BOOT}}$.
- **Opto-coupler-based isolated drive** transfers the command across a light-based galvanic barrier.
- **Pulse-transformer-based isolated drive** transfers gate-drive information magnetically and offers strong isolation, but classical solutions are constrained by volt-second balance and saturation [Infineon Isolated Gate Driving Solutions], [ST AN461].
- Modern **integrated isolated gate-driver ICs** combine the isolation barrier and the output driver in one package [TI UCC21520 Product Page].

Further reading

- Infineon, *Gate drive for power MOSFETs in switching applications* - A strong beginner-to-intermediate application note on gate charge, discrete drivers, driver ICs, and high-side methods.
- Infineon, *Isolated gate driving solutions* - A focused comparison of pulse-transformer and isolated-driver approaches, useful for understanding duty-cycle, CMTI, and integration tradeoffs.
- Infineon, *IRS2101(S)PbF High and Low Side Driver Datasheet* - A concise real datasheet for learning bootstrap high-side/low-side driver terminology and ratings.
- Texas Instruments, *UCC21520 Product Page and Datasheet* - A useful modern example of an isolated dual-channel gate driver with strong output current, UVLO, CMTI, and flexible channel use.
- STMicroelectronics, *AN461: An isolated gate drive for Power MOSFETs and IGBTs* - A compact note that is especially helpful for understanding pulse-transformer-based isolated gate-drive thinking.

Chapter 2.2: Snubber Circuits

Chapter opening

Correct gate or base drive is only part of reliable switching. Real power circuits also contain stray inductance, device capacitance, stored magnetic energy, and abrupt current change. These non-idealities produce voltage overshoot, current surge, ringing, false triggering, electromagnetic interference, and additional switching loss.

A **snubber circuit** is auxiliary circuitry placed around a switch, transformer, diode, or inductive path to control those transients and redirect stored energy in a controlled way [Rashid, *Power Electronics: Circuits, Devices and Applications, 4e*], [onsemi AN1048/D]. This chapter explains why snubbers are needed, distinguishes turn-ON and turn-OFF stress, and introduces the three foundational protective networks used throughout the subject: the **RC snubber**, the **RCD snubber**, and the **freewheeling diode**. These networks recur in rectifiers, choppers, inverters, battery chargers, PV converters, UPS systems, and auxiliary power supplies.

Prerequisites check

- You should remember from Chapter 1.2 that an SCR can false-trigger if the voltage across it changes too quickly.
- You should remember from Chapters 1.4 and 1.5 that MOSFETs and IGBTs do not switch instantaneously and that parasitic capacitances influence their switching behavior.
- You should be comfortable with the basic inductor relation that current through an inductor cannot jump suddenly.
- You should know that the energy stored in an inductor is $\frac{1}{2}LI^2$.
- You should remember from Chapter 2.1 that driver strength affects dv/dt and di/dt, but layout and external protection components matter too.

If the ideas of leakage inductance, Miller effect, or reverse recovery feel weak, a short review of Chapters 1.2, 1.4, 1.5, and 2.1 will make this chapter easier to absorb.

Core content

2.2.1 Need for snubbers; turn-ON and turn-OFF snubbers

Why switching stress appears even in a "simple" circuit

Consider a MOSFET switching current through an inductive load in a 48 V battery charger, or an IGBT switching a few amperes in an inverter leg fed from a rectified 230 V, 50 Hz supply. In an ideal circuit, turn-OFF would interrupt current and establish the blocking voltage. In practice, wiring, package leads, PCB traces, transformer leakage paths, and the load all contribute inductance, while the switch node and device terminals contribute capacitance. When the current path is interrupted, the stored magnetic energy in that inductance appears as overvoltage and ringing.

The magnetic energy stored in stray or leakage inductance is

$$\boxed{E_L = \frac{1}{2} L_\sigma I^2} \quad (7.1)$$

where E_L is the energy stored in the stray or leakage inductance, L_σ is that inductance, and I is the current flowing just before the switching event.

Even a small inductance can store enough energy to produce a severe transient. Suppose the effective stray or leakage inductance is only 2 μH and the current just before turn-OFF is 12 A. Then

$$E_L = \frac{1}{2} \times 2 \times 10^{-6} \times 12^2 = 144 \times 10^{-6}\,\mathrm{J} = 144\,\mu\mathrm{J}.$$

If, for a brief moment, this energy can charge only about 200 pF of effective node capacitance, the equivalent voltage associated with that energy is

$$V \approx \sqrt{\frac{2E}{C}} = \sqrt{\frac{2 \times 144 \times 10^{-6}}{200 \times 10^{-12}}} \approx 1200\,\mathrm{V}.$$

The exact spike in a practical converter depends on the available parasitic and intentional paths, so the circuit does not necessarily reach 1200 V. The estimate nevertheless shows why a converter operating from a few hundred volts can generate a much larger transient if the stored energy is left to the parasitics alone.

A **snubber circuit** is added to shape this transient by providing a controlled alternative path. Depending on topology, a snubber can

- limit the rate of rise of voltage across a device,
- limit peak overvoltage,
- limit turn-ON current surge,
- reduce ringing,
- reduce false triggering of thyristors,
- reduce stress caused by diode reverse recovery,
- trade a small controlled power loss for a large improvement in device safety [Rashid, *Power Electronics: Circuits, Devices and Applications, 4e*], [onsemi AN1048/D].

Table 7.1 summarizes the main reasons snubbers appear.

Table 7.1: Why snubber circuits are used

Problem in the power stage	What the snubber tries to do	Typical consequence if no snubber is used
Stray or leakage inductance stores energy	Provide a controlled path for that energy	Overvoltage spike at turn-OFF
Device or load capacitances resonate with inductance	Add damping or clamp voltage	Ringing, EMI, repeated stress
SCR or TRIAC sees rapid voltage rise	Reduce effective dv/dt across device	False turn-ON
Inductive load current has nowhere to go when switch opens	Provide recirculation path	Very large voltage spike or device avalanche
Diode reverse recovery interacts with switch turn-ON	Soften or absorb part of the transient	High turn-ON current stress and noise

What a snubber is

A **snubber circuit** is an auxiliary network placed around a switching element or energy-storage element to control transient voltage, transient current, or oscillation during switching.

The term covers several behaviors. Some snubbers are mainly **dissipative**, absorbing transient energy and converting it into heat in a resistor. Others are mainly **clamping** networks, steering the energy into a capacitor first and removing it later. Some are connected directly across the switch; others are associated mainly with the diode or the inductive load. The freewheeling diode is usually discussed alongside snubbers because it addresses the same switching problem from a different angle: it preserves a current path when the main switch changes state.

In every case, the essential question is the same: where does the stored energy go during the switching interval? If it is forced through a controlled auxiliary path, the stress is usually manageable. If it is left to parasitic inductance and capacitance, the resulting voltage spike and ringing determine the behavior.

Turn-OFF stress and turn-OFF snubbers

A **turn-OFF snubber** acts during the interval when the power device is being turned OFF and the current through a stray inductive path attempts to continue flowing. The associated voltage is governed by the inductor relation

$$v_L = L\frac{di}{dt}.$$

If $\frac{di}{dt}$ is large and no alternative path exists, the voltage across the switch rises sharply. A turn-OFF snubber softens that event by giving the current a temporary destination. In practice, an RC snubber lets a capacitor absorb part of the sudden voltage rise while the resistor damps the oscillation, an RCD snubber conducts selectively above a clamp condition and diverts the spike into a capacitor, and a clamp or freewheel path can keep the switch from seeing the full stress directly [onsemi AN1048/D], [onsemi AN4137].

The role of the snubber capacitor follows from

$$\boxed{i_C = C\frac{dv}{dt}} \quad (7.2)$$

or

$$\frac{dv}{dt} = \frac{i_C}{C}.$$

If a transient current must charge a capacitor, a larger capacitance produces a slower voltage rise. That is why a capacitor across a switch can reduce dv/dt. The tradeoff is equally important: increasing C lowers dv/dt, but it also increases stored energy, dissipation, and the current that may appear when the device turns ON again. Snubber design is therefore always a compromise among stress reduction, loss, cost, and switching behavior [onsemi AN1048/D].

Turn-ON stress and turn-ON snubbers

A **turn-ON snubber** acts during the interval when the device is being turned ON. The stress can come from discharge current in a previously charged snubber capacitor, reverse-recovery current in a diode that had been conducting, current surge into a capacitive node, or a large di/dt set by parasitics rather than by the load alone.

ST's application note on gate and base drive emphasizes that, in many inductive-load applications, turn-ON occurs while the freewheeling diode is conducting, so the diode's reverse recovery directly influences current rise and switching loss [ST AN509/1293]. Turn-ON snubbers therefore address current surge rather than turn-OFF overvoltage. Across practical topologies, the detail varies, but the conceptual distinction is stable:

- a turn-OFF snubber mainly restrains voltage rise and turn-OFF spike,
- a turn-ON snubber mainly restrains current surge and turn-ON stress.

Image prompt for Figure 7.1: Create a clean textbook-style technical illustration comparing unsnubbed and snubbed switch turn-OFF in an inductive circuit. Show a power switch carrying inductive current, a small stray inductance in the loop, and two aligned voltage waveforms across the switch: one with a sharp overshoot and ringing, one with a slower controlled rise and lower peak due to a snubber. Label stored energy in stray inductance, voltage overshoot, ringing, and controlled transient. Use monochrome engineering style with axes, units, and clear labels.

The remainder of the chapter is organized around three practical networks: the RC snubber, the RCD snubber, and the freewheeling diode.

2.2.2 RC and RCD snubber circuits; freewheeling diode

RC snubber circuits

The simplest named snubber is the **RC snubber**, a resistor and capacitor connected in series and placed across a device or across a problematic node. The capacitor slows the voltage change because the transient current must charge it first, while the resistor damps oscillation and limits the current that would otherwise flow too sharply into or out of the capacitor.

If only a capacitor were used, the circuit might reduce the initial dv/dt but could also create heavy current pulses and strong resonant ringing with the stray inductance. If only a resistor were used, the voltage spike would usually remain too sharp. The series RC combination is used because it addresses both the first transient and the oscillation that follows.

The RC pair introduces a time constant

$$\boxed{\tau = RC} \quad (7.3)$$

where τ is the RC time constant, R is the snubber resistance, and C is the snubber capacitance.

This equation does not by itself design the snubber, but it indicates the basic trend: larger C slows the voltage rise, larger R limits the current pulse and increases damping, and excessive values increase loss or slow the transition more than desired.

RC snubbers across thyristors One classical use of the RC snubber is across an SCR or TRIAC. onsemi's application note AN1048/D describes the simple snubber as a series resistor and capacitor placed around the thyristor, and explains that RC networks are used to control voltage transients that could falsely turn on a thyristor [onsemi AN1048/D].

This remains an important application because the SCR is sensitive to rapid dv/dt. If the anode-cathode voltage rises too quickly, the internal junction capacitances can drive enough current into the regenerative structure to trigger the device even without an intentional gate command. In this setting, the RC snubber is used primarily to prevent unwanted turn-ON and control transient behavior, not to improve efficiency.

Image prompt for Figure 7.2: Create a textbook-style figure showing an SCR with a series RC snubber connected across anode and cathode. Include a corresponding waveform panel that compares voltage across the SCR with and without the snubber during a transient. Label anode, cathode, gate, snubber resistor, snubber capacitor, false-triggering risk, and reduced dv/dt. Use monochrome engineering style with clear engineering labels.

RC snubbers across transistor switches RC snubbers are also used with MOSFETs, IGBTs, and diodes to reduce ringing on a switch node or across a device. In these applications the main objectives are limiting overshoot, damping resonance caused by stray inductance and node capacitance, reducing EMI, and lowering repetitive stress on the semiconductor.

The price is power loss. Every cycle, the snubber capacitor is charged and discharged. A useful first estimate for the average power associated with that repetitive charging is

$$\boxed{P_{RC,\text{approx}} \approx \frac{1}{2} C V^2 f_s} \tag{7.4}$$

where $P_{RC,\text{approx}}$ is the approximate average snubber-related power, C is the snubber capacitance, V is the voltage swing across it, and f_s is switching frequency.

This is a first estimate rather than a complete exact model, because the real loss distribution depends on topology, waveform, and the extent to which the capacitor charges and discharges in each cycle. It is nevertheless a useful warning sign.

Consider a converter switching at 20 kHz with an RC snubber capacitor of 1 nF that experiences about 325 V each cycle in an off-line stage. Then

$$P_{RC,\text{approx}} \approx \frac{1}{2} \times 1 \times 10^{-9} \times 325^2 \times 20 \times 10^3 \approx 1.06\,\text{W}.$$

This level of dissipation is not negligible. The resistor must be chosen and mounted so that the heat can be removed safely. Because the resistor ultimately absorbs the transient energy, RC snubbers are most suitable where the stress is moderate and the objective is waveform shaping rather than heavy energy clamping.

RCD snubber circuits

An **RCD snubber** adds a diode to the resistor-capacitor network so that the clamp acts mainly in one direction. Instead of participating equally in both voltage-rise and voltage-fall events, the network conducts primarily when the node attempts to exceed the safe level in the dangerous direction.

This makes the RCD snubber especially useful in circuits such as the flyback converter primary switch, where transformer leakage inductance produces a turn-OFF spike at the switch drain or collector. In that situation, the switch turns OFF, the leakage inductance attempts to keep current flowing, the switch-node voltage rises, the snubber diode conducts when the clamp condition is reached, the capacitor receives the diverted energy, and the resistor dissipates that energy between switching events. The RCD snubber is therefore a **clamp-plus-dissipation** network rather than a lossless solution [onsemi AN4137].

First energy estimate for an RCD snubber If most of the transformer leakage energy each cycle is dissipated through the RCD network, a useful first estimate is

$$\boxed{P_{RCD,\text{approx}} \approx \frac{1}{2} L_{lk} I_{pk}^2 f_s} \tag{7.5}$$

where L_{lk} is the leakage inductance, I_{pk} is the peak current present when the switch turns OFF, and f_s is switching frequency.

This estimate shows that the average snubber burden scales directly with leakage inductance, the square of current, and switching frequency. Consider a small isolated auxiliary supply in a wind-converter control cabinet or a PV inverter control board with

- $L_{lk} = 3\,\mu$H,
- $I_{pk} = 2.5$ A,
- $f_s = 65$ kHz.

Then

$$P_{RCD,\text{approx}} \approx \frac{1}{2} \times 3 \times 10^{-6} \times 2.5^2 \times 65 \times 10^3 \approx 0.61\ \text{W}.$$

This again is substantial enough to matter thermally, and it shows directly that reducing transformer leakage inductance reduces snubber burden.

Selective clamping in practice Compared with a simple RC network across a switch, the RCD clamp need not participate equally in the benign part of the cycle. The diode makes the response directional, reduces unnecessary current, and improves control of peak voltage in one-sided turn-OFF transients. This selectivity is why RCD networks are common around flyback primary switches, while simple RC networks are more often used for dv/dt control or damping across SCRs, TRIACs, MOSFETs, or diodes [Fairchild AN-6014].

Image prompt for Figure 7.3: Create a textbook-style schematic of a flyback converter primary showing a MOSFET switch, transformer primary, leakage inductance indicated, and an RCD snubber made of diode D_{sn}, capacitor C_{sn}, and resistor R_{sn} connected as a clamp from the switch node to the DC input rail. Add a waveform panel showing drain voltage rising to reflected output voltage plus input voltage, then a leakage-induced spike, with the RCD clamp limiting the peak. Label all parts, polarities, and current path during turn-OFF. Use monochrome engineering style.

Reading an RCD snubber in a flyback schematic In a flyback converter schematic, the group R_{sn}, C_{sn}, and D_{sn} placed near the primary switch node immediately identifies the problem being addressed: primary-side turn-OFF overvoltage rather than low-voltage output ripple. The orientation of D_{sn} shows that the network is a directional clamp, not a continuously active RC damper. During the spike, C_{sn} receives the diverted leakage energy; between switching events, R_{sn} dissipates it. The network is therefore a controlled destination for leakage-inductance energy when the primary switch turns OFF [onsemi AN4137].

This arrangement appears frequently in small isolated auxiliary supplies inside PV inverters, wind converters, EV chargers, and UPS hardware, where the main system may operate from a high-voltage bus but the control electronics still require low-voltage isolated rails.

The freewheeling diode

The **freewheeling diode**, also called a **flyback diode** or **freewheel diode**, provides a recirculating path for inductive current when the controlled switch turns OFF. Its function differs from that of an RC or RCD snubber. A snubber primarily shapes or clamps a transient; a freewheeling diode primarily preserves current continuity in an inductive path.

When the switch controlling an inductive current turns OFF, the inductor current cannot instantly become zero. If a diode is connected so that it becomes forward-biased at that moment, the current continues through the load and diode instead of forcing the switch voltage to rise destructively. This function is fundamental in DC choppers, buck converters, motor drives, relay and solenoid circuits, inverter legs during dead time, and other commutation paths [Mohan, Undeland, Robbins, *Power Electronics*], [ST STTH60AC06C Product Page].

Manufacturer literature commonly identifies ultrafast and SiC devices for freewheeling service, which reflects how standard this role is in practical converter design [ST STTH60AC06C Product Page], [ST STPSC40065C Product Page].

Current decay with a freewheeling diode When the current freewheels through a diode, the inductor sees only a small reverse voltage, roughly the diode forward drop plus any resistive drops in the loop. A useful first approximation is

$$\boxed{\frac{di}{dt} \approx -\frac{V_F}{L}} \quad (7.6)$$

where V_F is the effective forward drop of the recirculating path and L is the inductance.

Suppose a 2 mH inductor is carrying 5 A when the main switch turns OFF, and the freewheeling path imposes about 1 V across the inductor. Then

$$\frac{di}{dt} \approx -\frac{1}{2 \times 10^{-3}} = -500 \text{ A/s}.$$

The current therefore takes about

$$t \approx \frac{5}{500} = 0.01 \text{ s} = 10 \text{ ms}$$

to fall to zero in this simplified estimate.

The tradeoff is clear: a diode freewheel path gives low voltage stress, but it also causes slow current decay because the reverse voltage across the inductor is small. If a higher clamp voltage were allowed, the current would fall faster. The designer therefore balances voltage stress, current-decay rate, control requirements, efficiency, and EMI.

Selection and relation to snubbers The presence of a freewheeling diode does not automatically remove the need for an RC or RCD snubber. In many circuits, the diode carries the main inductive current after commutation, while a separate snubber still controls ringing and stray-inductance stress.

Selection of the diode itself depends on repetitive reverse-voltage rating, average and peak forward current, reverse-recovery behavior, and thermal capability. At high switching frequency, ultrafast, Schottky, and silicon-carbide diodes are often preferred because reverse recovery directly affects switching stress and ringing [ST STPSC40065C Product Page]. ST's drive note makes the same point from the transistor side: at turn-ON, the switching device often encounters the recovery behavior of the conducting freewheeling diode [ST AN509/1293].

Table 7.2 summarizes the three main protective networks in this chapter.

Table 7.2: RC snubber, RCD snubber, and freewheeling diode compared

Network	Main purpose	Main strength	Main tradeoff	Typical place
RC snubber	Reduce dv/dt and damp ringing	Simple and widely applicable	Continuous loss and extra turn-ON burden	Across SCR, TRIAC, switch, or diode
RCD snubber	Clamp overvoltage more selectively	Better suited to leakage-spike control	Dissipates leakage energy and needs tuning	Flyback primary switch, clamp node
Freewheeling diode	Provide path for inductive current after switch state change	Strong reduction of voltage stress from interrupted load current	Slow current decay if clamp voltage is low	Across inductive load or as recirculation path in converter leg

In a small isolated flyback auxiliary supply inside a grid-tied PV inverter, an RCD snubber is commonly used to keep the primary-switch drain spike within safe limits. In a buck converter charging a battery, the freewheeling diode is part of the basic current path after the main switch turns OFF. If switching frequency is high and ringing remains problematic, an additional RC snubber may still be required. The correct choice follows the transient problem being solved in that topology.

How this matters in renewable-energy systems

Snubber circuits are closely tied to renewable-energy reliability because these converters switch substantial current repeatedly, often for many hours each day and under demanding thermal conditions. In solar PV systems, snubbers appear in isolated auxiliary supplies, DC-DC stages, and inverter legs to keep switching stress within device limits and to reduce ringing that would otherwise worsen EMI. In battery chargers and battery-energy-storage converters, freewheeling paths and snubbers determine how safely inductor current commutates when switches change state. In wind-energy converters, transformer leakage inductance and converter-leg stray inductance make overvoltage control essential.

The system-level issue is not only efficiency. Robustness, service life, and electromagnetic behavior are equally important. A converter that is slightly more efficient on paper but suffers repeated overshoot, diode-recovery stress, or false triggering is not a better converter in practice. Snubbers are one of the main points at which transient control, reliability, and electromagnetic performance meet.

Chapter summary

- A **snubber circuit** is auxiliary circuitry used to control transient voltage, transient current, or oscillation during switching [Rashid, *Power Electronics: Circuits, Devices and Applications, 4e*].
- Snubbers are needed because real power circuits contain stray or leakage inductance, capacitance, stored energy, and non-instantaneous switching transitions.
- The energy stored in stray or leakage inductance is $\boxed{E_L = \frac{1}{2} L_\sigma I^2}$ from Equation (7.1).
- A capacitor can reduce voltage-rise rate because $\boxed{i_C = C\, dv/dt}$ from Equation (7.2).
- A **turn-OFF snubber** mainly reduces voltage overshoot and turn-OFF stress, whereas a **turn-ON snubber** mainly reduces current surge and turn-ON stress.
- A series **RC snubber** uses the capacitor to slow the transient and the resistor to damp it; its time constant is $\boxed{\tau = RC}$ from Equation (7.3).
- A useful first estimate of repetitive RC-snubber power is $\boxed{P_{RC,\text{approx}} \approx \frac{1}{2} C V^2 f_s}$ from Equation (7.4).
- An **RCD snubber** adds a diode so that the clamp acts more selectively, which is especially useful for leakage-induced turn-OFF spikes in flyback converters [onsemi AN4137].
- A useful first estimate of RCD-snubber dissipation is $\boxed{P_{RCD,\text{approx}} \approx \frac{1}{2} L_{lk} I_{pk}^2 f_s}$ from Equation (7.5).
- A **freewheeling diode** provides a safe recirculating path for inductive current when the main switch changes state.
- A first current-decay estimate in a diode freewheel path is $\boxed{di/dt \approx -V_F/L}$ from Equation (7.6).
- Freewheeling diodes reduce voltage stress strongly, but the low clamp voltage leads to slower current decay; diode reverse recovery therefore remains an important part of switch stress.

Further reading

- onsemi, *AN1048/D: RC Snubber Networks for Thyristor Power Control and Transient Suppression* - A very useful classic reference for understanding why snubbers are needed, especially for SCR dv/dt control and the tradeoffs involved.
- onsemi, *AN4137: Design Guidelines for Off-line Flyback Converters Using Fairchild Power Switch* - Helpful for seeing an RCD snubber in a real flyback schematic and understanding its place in practical converter design.
- Fairchild Semiconductor, *AN-6014: Green Current Mode PWM Controller FAN7602* - Useful because it distinguishes diode RC snubber design from MOSFET RCD snubber design in a real SMPS design context.
- STMicroelectronics, *Influence of gate and base drive on power switch behaviour* - Especially useful for the link between switch turn-ON stress and the reverse recovery of the conducting freewheeling diode.
- M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*, 4th ed. - A strong textbook reference for the broader role of snubbers, freewheeling diodes, and switching-stress

control across converter families.

Chapter 2.3: Overvoltage and Overcurrent Protection

Chapter opening

Previous chapters examined gate driving and transient control during switching. Those topics address normal converter operation. Protection addresses abnormal operation: line surges, regulator failure, wiring faults, stalled loads, short circuits, and device faults can impose excessive voltage, excessive current, or both.

Protection is therefore part of converter design rather than an afterthought. This chapter examines overvoltage protection using the MOV, the TVS diode, and the crowbar circuit, and overcurrent protection using fast-acting (HRC) fuses, electronic current limiting, and desaturation detection at concept level. The central design question is not which protector is "best," but which abnormal event must be survived and how the protective response should occur.

Prerequisites check

This chapter assumes familiarity with basic gate-driver function, switching transients and snubbers, RMS and peak values of sinusoidal voltage, inductor energy $\frac{1}{2}LI^2$, and ON-state voltage terms such as $V_{CE(\text{sat})}$ and V_{DS}. The discussion builds on those ideas by moving from transient control to explicit fault protection.

Core content

2.3.1 Overvoltage protection: MOV, TVS diodes, crowbar circuits

Why overvoltage protection is needed

A rooftop PV inverter may encounter several distinct overvoltage conditions. A surge can enter from the AC line during a nearby lightning event, parasitic inductance can produce a switching spike inside the converter, and a regulator failure can drive a low-voltage rail far above its intended value. All three are overvoltage events, but they differ in source, duration, and energy.

Protection method must therefore match fault type. A surge absorber, a fast local clamp, and a crowbar circuit do not serve the same purpose. For AC systems, one basic relation is immediately relevant:

$$\boxed{V_{pk} = \sqrt{2}\,V_{rms}} \quad (8.1)$$

where V_{pk} is the sinusoidal peak voltage and V_{rms} is the RMS value.

For a 230 V, 50 Hz mains supply,

$$V_{pk} = \sqrt{2} \times 230 \approx 325 \text{ V}.$$

Any protector connected across the line must first tolerate the normal operating voltage. If the normal peak value is misunderstood, a device may begin to conduct during ordinary operation rather than only during an abnormal event. Overvoltage protection also does not imply perfect regulation; many protectors allow voltage to rise to a defined clamping level above nominal, and no single protective element is suitable for every location [Bourns, *Tips on Selecting the Right MOV Surge Suppressor*], [Littelfuse 5KP Series], [onsemi AN004E/D].

The MOV

A **metal-oxide varistor**, or **MOV**, is a common surge-protection device at the input of power equipment.

Below its threshold region, the MOV presents very high impedance. During a surge, its resistance falls sharply and it diverts current away from the protected circuit. It is therefore best understood as a strongly nonlinear resistor rather than an ideal fixed-voltage ceiling. As surge current increases, the clamping voltage also increases [Bourns, *Tips on Selecting the Right MOV Surge Suppressor*].

The key terms used with an MOV are:

- **maximum continuous operating voltage (MCOV)**: the highest continuous AC or DC voltage the MOV should withstand in normal operation
- **varistor voltage**: a test-condition voltage, often specified at a current such as 1 mA DC
- **clamping voltage**: the voltage developed across the MOV at a specified surge current
- **surge current rating**: the magnitude of surge current the MOV can survive for a specified waveform

Bourns notes that MOV surge ratings are commonly specified on an **8/20 us** current waveform, meaning a standardized current pulse that rises and decays very quickly [Bourns, *Tips on Selecting the Right MOV Surge Suppressor*]. That rating immediately shows the intended service: short surge events rather than indefinite abnormal mains voltage.

Choosing the MOV from the normal voltage first To protect the AC input of a 230 V, 50 Hz UPS or PV inverter, the first selection criterion is the continuous voltage rating. It must safely exceed the actual steady operating condition, including line tolerance. Only after that should the clamping level and surge capability be checked [Bourns, *Tips on Selecting the Right MOV Surge Suppressor*].

1. Determine the highest normal operating voltage.
2. Add realistic margin for supply tolerance.
3. Then check clamping and surge ratings.

If this order is ignored, the MOV may run hot even when the system is healthy.

What the MOV does well and what it does not The MOV is simple, inexpensive, bidirectional in typical AC use, and capable of absorbing substantial surge current. It is therefore widely used at mains inputs.

Its limitations are equally important. Repeated surges age the device [Bourns, *Tips on Selecting the Right MOV Surge Suppressor*]. Sustained overvoltage can overheat the MOV and may drive it into thermal runaway, which is why fuse protection, thermal disconnects, or another upstream disconnection method is commonly required [Bourns, *Tips on Selecting the Right MOV Surge Suppressor*]. An MOV is suited to short surges; it is not by itself a remedy for long-duration overvoltage.

Image prompt for Figure 8.1: Create a clean textbook-style illustration of an MOV connected across a 230 V AC equipment input. Show normal sinusoidal mains where MOV current is nearly zero, and a second waveform panel showing a short surge on the mains with the MOV conducting large current and clamping the voltage. Label line, neutral, MOV, normal leakage region, surge current path, clamping voltage, and note that the clamp voltage rises with current. Use monochrome engineering style with clear axes and units.

A practical AC-input example In a small inverter-powered water-pump controller, an MOV across the AC input can divert surge current during a storm and reduce stress on the bridge rectifier and bulk capacitor. If the mains remains abnormally high for a prolonged period, however, the MOV cannot serve as the only protection. It must work with an upstream fuse or disconnect.

The TVS diode

A **transient voltage suppressor diode**, or **TVS diode**, serves a different role. It is used where a fast, sharply defined clamp is needed on a lower-voltage line or sensitive node. Littelfuse describes TVS diodes as devices intended to protect sensitive electronics against transient voltage events and emphasizes fast response and strong clamping capability [Littelfuse 5KP Series], [Littelfuse 3.0SMC Series].

In first approximation, the distinction from an MOV is straightforward: the MOV is often chosen for higher-energy surge handling at equipment entry points, whereas the TVS diode is often chosen for fast local protection of lower-voltage buses and interfaces.

The key TVS terms are:

- **reverse standoff voltage**, often written V_R or V_{RWM}: the maximum normal working voltage that should not trigger significant conduction
- **breakdown voltage**, V_{BR}: the region where avalanche conduction begins under the specified test current
- **clamping voltage**, V_C: the voltage across the TVS at a specified pulse current
- **peak pulse current**, I_{PP}: the surge current used in the clamping specification
- **peak pulse power**: the rated pulse power under a specified surge waveform

For a pulse event, a useful first estimate is

$$\boxed{P_{\text{pulse,approx}} \approx V_C I_{PP}} \quad (8.2)$$

where $P_{\text{pulse,approx}}$ is the approximate pulse power during the clamped event, V_C is the clamping voltage, and I_{PP} is the peak pulse current. This is an approximation rather than a full energy calculation, because real surge waveforms vary with time.

Reading the TVS ratings correctly Consider a low-voltage controller board in a battery-energy-storage system using a 48 V nominal auxiliary bus. The first selection question is the highest bus voltage present during normal operation, including charging, regulation tolerance, and ripple. The reverse standoff voltage must exceed that normal condition. Only then should the breakdown and clamping values be checked [Littelfuse 5KP Series].

The basic selection logic is:

1. The normal bus voltage must stay below the reverse standoff rating.
2. The abnormal transient must rise into the avalanche region.
3. The resulting clamping voltage must stay below what the protected electronics can survive.

If the standoff voltage is too low, the diode leaks excessively or conducts during normal operation. If the clamping voltage is too high, the protected circuit still sees excessive stress.

Where TVS diodes are useful and where they are not TVS diodes are widely used on:

- low-voltage DC buses
- gate-driver supply rails
- communication and sensing lines
- control inputs and outputs
- automotive and battery-connected electronics
- local protection points near sensitive ICs

Their value lies in fast local clamping close to the vulnerable node [Littelfuse 3.0SMC Series], [Littelfuse SLD6S Series].

They are not universal high-energy absorbers. Compared with an MOV at a mains input, a TVS diode usually protects a more local node and usually handles less total surge energy. Littelfuse also notes a typical failure mode of short circuit when a TVS is overstressed beyond its specified limits [Littelfuse 3.0SMC Series]. When surge energy or fault duration is too large, a different protector or a layered protection architecture is required.

Image prompt for Figure 8.2: Create a textbook-style voltage-current concept figure for a unidirectional TVS diode. Show a horizontal axis for voltage and vertical axis for current. Mark reverse standoff region with very small current, breakdown region, and clamping region at specified pulse current. Add labels for V_{RWM}, V_{BR}, V_C, and I_{PP}. Include a side inset showing a TVS connected across a 48 V DC auxiliary bus feeding a controller. Use monochrome engineering style with clear labels.

MOV versus TVS in plain language Table 8.1 compares the three main overvoltage methods introduced in this section.

Table 8.1: MOV, TVS diode, and crowbar compared at beginner level

Protection method	Main physical action	Strongest use case	Main limitation
MOV	Nonlinear resistance diverts surge current	Input surge suppression, especially at AC or higher-energy entry points	Degrades with repeated surges; not suited to sustained overvoltage by itself
TVS diode	Avalanche clamp limits transient voltage quickly	Fast local protection of sensitive low-voltage buses and nodes	Usually lower energy capability than a mains MOV; overstress may short the device
Crowbar circuit	Deliberately forces a low-impedance fault path when threshold is exceeded	Sustained overvoltage protection of sensitive DC supplies	Requires current limiting or fuse action elsewhere; trips hard rather than gently

The distinction is strategic: these devices absorb, clamp, and force shutdown in different ways.

Crowbar circuits

The **crowbar circuit** differs from both MOV and TVS protection. Instead of clamping the voltage while remaining a passive shunt element, it deliberately creates a low-impedance fault path when the voltage exceeds a defined threshold. In a common implementation, an SCR is triggered across the supply or output, forcing the source into current limit or causing a fuse or breaker to open [onsemi AN004E/D], [onsemi MC3423 Datasheet].

A crowbar is therefore a latching response to dangerous sustained overvoltage. For a 5 V processor board or gate-driver supply, sacrificing a fuse may be preferable to allowing the rail to rise to 12 V or 15 V for even a short time.

Threshold setting in a simple sensing scheme A common way to define the trip point is to compare a divided-down supply voltage with a reference. In a simple divider-based sense circuit, the trip voltage can be expressed as

$$\boxed{V_{\text{trip}} = V_{\text{ref}}\left(1+\frac{R_1}{R_2}\right)} \quad (8.3)$$

where V_{trip} is the supply voltage at which the crowbar should trigger, V_{ref} is the sensing reference, and R_1 and R_2 are the divider resistors.

Suppose a 5 V auxiliary rail should crowbar at about 6.25 V and the reference is 2.5 V. Then

$$6.25 = 2.5\left(1+\frac{R_1}{R_2}\right).$$

So

$$\frac{R_1}{R_2} = \frac{6.25}{2.5} - 1 = 1.5.$$

If $R_2 = 10\ \text{k}\Omega$, a nearby practical choice is $R_1 = 15\ \text{k}\Omega$.

The trip point is a design parameter that must be set deliberately.

Practical concerns in crowbar protection Crowbar design raises three practical questions. First, nuisance tripping must be avoided; some sensing schemes impose a minimum overvoltage duration before firing [onsemi MC3423 Datasheet]. Second, the SCR or other crowbar device must withstand the expected current, surge, and di/dt. Third, the fault must actually be cleared: the source must current-limit, a fuse must open, a breaker must trip, or control logic must shut the supply down. The crowbar is one element in a protection chain, not a complete standalone solution.

Image prompt for Figure 8.3: Create a clean textbook-style schematic illustration of a DC supply protected by an SCR crowbar circuit. Show the regulated supply output feeding a sensitive load, a sense network and reference that trigger an SCR when output voltage exceeds threshold, and an upstream fuse or current-limited source. Add a waveform panel showing output voltage rising abnormally, crowbar firing, voltage collapsing, and source current increasing until protection clears. Label V_{trip}, fuse, SCR, sense divider, and load. Use monochrome engineering style.

A realistic low-voltage example In a 15 V isolated gate-driver supply, if the isolated DC-DC converter fails and the output rises toward 24 V, the gate-driver IC and the power-device gate oxide may both be at risk. A brief spike could be handled by a TVS diode, but a sustained regulator failure demands a different response. A crowbar circuit is more suitable because it detects the abnormal DC voltage and forces the source into protective action.

2.3.2 Overcurrent protection: fast-acting (HRC) fuses, electronic current limiting, desaturation detection (concept)

Why overcurrent protection is needed

Overcurrent threatens semiconductors, conductors, and insulation through rapid heating and fault energy. In an inverter leg, a control failure or cable fault can drive current toward a value limited only by source impedance, DC-link impedance, and device survival. Magnitude and duration both matter, which is why the Joule integral is a useful measure:

$$\boxed{I^2t = \int i^2(t)\,dt} \qquad (8.4)$$

where I^2t is the current-squared time integral and $i(t)$ is the fault current waveform.

This relation matters because heating scales with current squared. A moderate overload for a long time and a severe short circuit for a very short time can both be destructive, but they stress the system differently. Fuses, semiconductor short-circuit ratings, and coordination discussions therefore often use I^2t as a measure of let-through thermal stress [Mersen HSJ Family].

If a fault current of 100 A flows approximately constantly for 1 ms, then

$$I^2t \approx 100^2 \times 0.001 = 10\ \mathrm{A}^2\mathrm{s}.$$

If the current is instead 300 A for the same 1 ms, then

$$I^2t \approx 300^2 \times 0.001 = 90\ \mathrm{A}^2\mathrm{s}.$$

Trip time therefore matters enormously. Three times the current produces nine times the I^2t for the same duration.

Protection is usually layered:

- one layer for catastrophic short-circuit interruption
- one layer for controlled overload limiting
- one layer for local device protection
- and, in many systems, another layer for branch or wiring protection

This chapter focuses on one representative method from each of those layers.

Fast-acting (HRC) fuses

A **fuse** remains one of the most important protective devices in power electronics.

Excess current heats a calibrated element until it melts and opens the circuit. The designation **HRC** means **high rupturing capacity**, indicating that the fuse can safely interrupt large fault current. In power-electronic service, the important distinction is often between ordinary installation fuses and semiconductor-protecting fuses selected for low let-through energy.

Mersen describes its HSJ high-speed series as combining branch-circuit protection with very low I^2t for protection of power semiconductors such as diodes, SCRs, GTOs, and solid-state relays [Mersen HSJ Family]. That description captures the essential point: in semiconductor service, the fuse matters not only because it opens, but because it limits the energy allowed through during the fault.

Why “fast” matters for semiconductors Power semiconductors can be damaged before a general-purpose fuse reacts. A fast semiconductor fuse is selected so that its let-through energy is compatible with the survival limit of the protected branch. It does not guarantee that every device will survive every fault; very fast solid-state faults may still require gate-driver shutdown or other electronic protection. The fuse remains essential as a final interruption layer.

A careful beginner distinction A general wiring fuse is selected primarily to protect cables and installation. A semiconductor fuse is coordinated with device stress, time-current behavior, and let-through energy. Some product families, including the HSJ line, address both roles [Mersen HSJ Family], but that cannot be assumed from the HRC label alone.

A realistic converter example In the DC link of a battery charger or inverter, a failed bridge device can allow the DC-link capacitor to dump very large current into the fault path. A fast semiconductor fuse can interrupt that current and limit propagation into busbars, cables, and adjacent devices. A slower overload condition, by contrast, may be better handled by controlled current limiting rather than immediate fuse operation.

What the fuse does not know A fuse responds only to current and time. It does not distinguish overload from short circuit, temporary inrush from sustained fault, or control error from wiring short. That simplicity is both its strength and its limitation, which is why converters often supplement fusing with electronic protection.

Electronic current limiting

Electronic current limiting measures current and actively alters converter operation to keep current below a chosen threshold. Unlike a fuse, it aims to preserve controlled operation or achieve a graceful shutdown before destructive heating develops.

The most basic sensing method is to place a small resistor in the current path and measure the voltage across it. From Ohm's law,

$$V_{\text{sense}} = IR_s.$$

If a control circuit compares that sense voltage with a reference V_{ref}, the current-limit threshold is approximately

$$\boxed{I_{\text{lim}} \approx \frac{V_{\text{ref}}}{R_s}} \quad (8.5)$$

where I_{lim} is the current-limit value and R_s is the sense resistance.

Suppose a battery charger uses a sense resistor of $0.05\,\Omega$ and the current-limit comparator threshold is 0.25 V. Then

$$I_{\text{lim}} \approx \frac{0.25}{0.05} = 5\text{ A}.$$

Different behaviors under current limit Once the current reaches the threshold, the controller may respond in several ways:

- **constant-current limiting**: hold the current near a set value

- **cycle-by-cycle limiting**: in a switching converter, terminate or reduce each switching pulse when the current reaches the threshold
- **foldback limiting**: reduce the allowed current further when the output voltage collapses
- **shutdown and retry**: stop switching, wait, and then attempt restart

Electronic current limiting is therefore an active control response rather than a sacrificial event.

Where electronic current limiting helps and where it does not Electronic current limiting is particularly useful in battery chargers, DC-DC converters, and auxiliary supplies where overload response should remain controlled. Its limits are equally important: it depends on sensing accuracy and response speed, it adds cost and power loss, and a severe hard short may still require a fuse or device-level shutdown path as backup. Current limiting reduces the number and severity of events that the interruption layer must clear; it does not eliminate the need for interruption.

Image prompt for Figure 8.4: Create a textbook-style figure showing electronic current limiting in a DC-DC converter. Show a power switch, load, sense resistor in the current path, a comparator or controller that monitors the sense voltage, and a control action block that reduces PWM or turns the switch off when current exceeds threshold. Add aligned waveforms for inductor current, sense voltage, and PWM command, marking the current-limit threshold. Use monochrome engineering style.

Desaturation detection (concept)

The third overcurrent-protection method in the syllabus is **desaturation detection**, usually shortened to **DESAT**.

DESAT is a gate-driver protection method widely used with IGBTs and similar devices. Instead of measuring load current directly, it infers severe overcurrent from abnormally high ON-state device voltage. Under short-circuit conditions an IGBT leaves its normal low-voltage saturation region and its collector-emitter voltage rises sharply; the driver detects that rise while the device is ON [Infineon DESAT Article].

Because the gate driver is already electrically close to the device, DESAT provides a fast local fault indication without placing a large shunt resistor in the main current path.

The basic threshold relation In a common DESAT arrangement, the detection path includes a diode from the collector side to the DESAT node. A useful simplified relation is

$$\boxed{V_{CE,\text{trip}} \approx V_{\text{DESAT,th}} - V_D} \quad (8.6)$$

where $V_{CE,\text{trip}}$ is the approximate collector-emitter voltage at which the driver interprets a fault, $V_{\text{DESAT,th}}$ is the internal DESAT threshold, and V_D is the detection-diode drop.

The exact relation varies with driver architecture, but the principle is the same [Infineon DESAT Article]. If a driver uses a DESAT threshold near 9 V and the sensing diode contributes about

1 V, the circuit may begin to interpret an IGBT collector-emitter voltage of roughly 8 V as a severe fault indicator. That is far above a normal saturated IGBT ON-state drop.

Why blanking time is needed During turn-on, the collector-emitter voltage is temporarily high even in normal operation. DESAT protection therefore includes a short **blanking time** so the device can enter its normal saturated state before fault judgment is made [Infineon DESAT Article], [TI UCC21750 Datasheet].

What happens after a DESAT event A modern driver typically does more than turn the device off instantly. TI describes the UCC21750 as providing fast DESAT protection, fault reporting, and **soft turn-off** when a fault occurs [TI UCC21750 Datasheet]. Soft turn-off matters because an abrupt shutdown during a short circuit can create even worse overvoltage through stray inductance.

At concept level, a DESAT response sequence is often:

1. The device is commanded ON.
2. A blanking interval allows normal turn-on.
3. The driver monitors the DESAT sense node.
4. If the inferred device voltage exceeds the threshold, the driver declares a fault.
5. The driver initiates controlled turn-off and often asserts a fault flag.

What DESAT is good at and what it is not DESAT is well suited to severe short-circuit protection close to the device. It does not replace branch-circuit fusing, careful layout, or snubbing, and it is not intended as general mild-overload control [Infineon DESAT Article].

Putting the three overcurrent methods together Table 8.2 compares the overcurrent methods introduced in this section.

Table 8.2: Overcurrent protection methods compared

Method	Main response style	Best at	Main tradeoff
Fast-acting / HRC fuse	Opens the circuit after overcurrent energy exceeds its design limit	High-fault-current interruption and backup protection	Sacrificial, passive, and not selective about fault cause
Electronic current limiting	Actively controls or shuts down the converter as current approaches threshold	Graceful overload handling and controlled converter behavior	Depends on sensing speed, control action, and added circuitry
DESAT detection	Uses abnormal ON-state device voltage to detect severe short circuit locally	Fast protection of IGBTs or similar devices during hard faults	Usually complements, not replaces, fuses and wider system protection

Worked case study: reading a high-speed fuse page

One useful real artifact for this chapter is the Mersen HSJ series product page, especially the HSJ200 example page. Mersen states that the HSJ line combines branch-circuit protection with very low I^2t for protection of power semiconductors such as diodes, SCRs, GTOs, and SSRs [Mersen HSJ Family].

The HSJ200 page identifies the device as a **600 VAC, 500 VDC, 200 A** high-speed fuse [Mersen HSJ Family]. These are not all the same kind of rating. The voltage ratings describe where the fuse may be used, and the ampere rating identifies its nominal current class. They do not mean that every 200 A event will be interrupted instantly.

The phrase **very low** I^2t is the most important clue for semiconductor service. It indicates that the fuse is intended not only to open the circuit eventually, but to limit let-through energy to a level more compatible with semiconductor survival. That wording distinguishes a semiconductor-oriented fuse description from a purely installation-oriented one.

When reading a fuse product page for converter use, the useful questions are:

- Is the fuse intended for semiconductor protection or only for wiring protection?
- Does it explicitly mention low I^2t or high-speed semiconductor service?
- Are the AC and DC voltage ratings compatible with the actual bus?
- Is the current class appropriate for the converter branch?

In a PV inverter, battery converter, or UPS, a fuse of this kind may appear in an AC branch, a DC-link branch, or another high-fault-current path feeding a semiconductor stage. It does not perform the same role as DESAT or active current limiting. Its purpose is physical interruption when fault energy becomes too large for the converter to manage electronically.

How this matters in renewable-energy systems

Renewable-energy systems connect long cables, switching converters, energy storage, and often the utility grid. They therefore experience both externally imposed disturbances and internally generated faults. Protection cannot be reduced to a single device type, because the events themselves are not all of the same kind.

In practical renewable-energy hardware, protection is layered. MOVs handle surge exposure at entry points. TVS diodes protect low-voltage rails and sensitive interfaces. Crowbar circuits protect supplies that must be shut down hard if regulation fails. Electronic current limiting manages overload in a controlled way. DESAT provides rapid device-level short-circuit detection in gate drivers. Fuses supply the final interruption layer when fault current must be physically cleared.

Chapter summary

- Overvoltage protection must be matched to the type of abnormal voltage: short surge, switching spike, or sustained DC overvoltage.

- For sinusoidal mains, the peak voltage is $\boxed{V_{pk} = \sqrt{2}\, V_{rms}}$ from Equation (8.1), so a 230 V RMS line has a peak of about 325 V.
- An **MOV** is a nonlinear surge suppressor selected from its continuous operating voltage first; it is effective for short surges but requires upstream disconnection for prolonged overvoltage [Bourns, *Tips on Selecting the Right MOV Surge Suppressor*].
- A **TVS diode** is a fast local clamp for sensitive electronics; reverse standoff, breakdown, and clamping ratings must be read together [Littelfuse 3.0SMC Series], [Littelfuse 5KP Series].
- A **crowbar circuit** detects excessive voltage and deliberately forces a low-impedance fault path, typically to make the source current-limit or a fuse open [onsemi AN004E/D], [onsemi MC3423 Datasheet].
- Overcurrent stress depends on both magnitude and duration, captured by $\boxed{I^2 t = \int i^2(t)\, dt}$ from Equation (8.4).
- **Fast-acting / HRC fuses** provide low let-through energy for semiconductor protection, while **electronic current limiting** actively controls overload response before interruption becomes necessary [Mersen HSJ Family].
- **DESAT** detects severe short circuit from abnormal ON-state device voltage in the gate-driver environment and complements, rather than replaces, wider system protection [Infineon DESAT Article], [TI UCC21750 Datasheet].

Further reading

- Bourns, *Tips on Selecting the Right MOV Surge Suppressor* - A practical white paper on MOV behavior, MCOV, clamping, surge waveforms, and the thermal limits that matter in real designs.
- Littelfuse 5KP Series - A good entry point for reading TVS terminology such as peak pulse power, clamping capability, and intended transient-protection role.
- onsemi, *AN004E/D: Consideration for DC Power Supply Voltage Protector Circuits* - A strong reference for understanding crowbar overvoltage protection and the practical issues of sensing, nuisance immunity, and SCR action.
- Mersen HSJ Family - Useful for learning how fuse manufacturers describe semiconductor protection, low I^2t, and branch-circuit capability in real product language.
- TI UCC21750 Datasheet - A modern example of a gate driver with DESAT, soft turn-off, UVLO, and fault reporting, very useful for connecting device-level protection concepts to practical converter hardware.

Chapter 2.4: Thermal Management

Chapter opening

Thermal management addresses the physical consequence of semiconductor loss. Every watt lost in a MOSFET, IGBT, or diode appears as heat at the device junction, and converter reliability depends on how effectively that heat is conducted to the case, transferred to a heat sink, and rejected to the surrounding air. A converter may meet its electrical specifications and still age prematurely if the junction temperature remains too high during ordinary operation.

This chapter examines the two main sources of device loss, develops the thermal-resistance model used for first-pass heat-sink selection, and concludes with an overview of natural and forced cooling [ST AN4783], [Infineon AN-1057]. The aim is practical: to connect electrical loss, datasheet thermal data, and cooling hardware into one design picture.

Prerequisites check

- You should remember from Chapters 1.4 and 1.5 that MOSFETs and IGBTs do not switch instantaneously, so voltage and current overlap during switching.
- You should remember from Chapter 2.1 that stronger or weaker gate drive changes switching speed, and therefore affects switching loss.
- You should remember from Chapter 2.2 that snubbers can reduce overshoot and ringing, but they may also dissipate power.
- You should be comfortable with the power relations $P = VI$ and, for a resistor, $P = I^2R$.
- You should know the difference between ambient temperature and device temperature, even if the thermal-resistance model is new.

If switching waveforms are unclear, Chapters 1.4, 1.5, and 2.1 should be reviewed before this chapter.

Core content

2.4.1 Power losses in semiconductor devices: conduction and switching losses (qualitative)

Why a "high-efficiency" converter still gets hot

Consider a DC-DC converter in a battery charger that processes 1 kW at 95% efficiency. The efficiency is high, but the lost power is still

$$P_{\text{loss}} = P_{\text{in}} - P_{\text{out}} = 1000 - 950 = 50\,\text{W}.$$

Fifty watts is substantial inside a compact enclosure. It is comparable to the heat dissipated by a small soldering iron, and it is generated continuously inside the converter. That heat must move from the semiconductor junctions, through the package and heat sink, and finally into the surrounding air.

The first thermal-design question is therefore direct: how much power is being turned into heat inside the semiconductor devices?

For first-pass analysis, two loss categories dominate:

- **Conduction loss**: loss while the device carries current in its ON state.
- **Switching loss**: loss during the finite interval in which the device changes state.

Different devices express these losses differently. A MOSFET usually shows conduction loss through ON-state resistance. An IGBT or diode is often approximated by an ON-state voltage drop. At high switching frequency, switching loss may dominate. At lower frequency and high current, conduction loss may dominate. Thermal analysis starts by identifying which loss mechanism is likely to be larger under the intended operating condition [Mohan, Undeland, Robbins, *Power Electronics*].

Conduction loss

When a semiconductor is ON, it is not an ideal short circuit. Current flows through a nonzero resistance or across a nonzero voltage drop, so power is dissipated.

For a MOSFET, a first conduction-loss estimate is

$$\boxed{P_{\text{cond,MOSFET}} \approx I_{\text{rms}}^2 R_{DS(\text{on})}} \quad (9.1)$$

where I_{rms} is the RMS current through the device over the interval of interest and $R_{DS(\text{on})}$ is the ON-state resistance.

This relation shows the square dependence on current. If current doubles, conduction loss becomes approximately four times larger.

Consider a low-voltage MOSFET in a 48 V to 12 V converter carrying an RMS current of 10 A. If its effective ON resistance at operating temperature is $25\ \text{m}\Omega$, then

$$P_{\text{cond,MOSFET}} \approx 10^2 \times 0.025 = 2.5\ \text{W}.$$

If the RMS current rises to 20 A, the conduction loss becomes

$$P_{\text{cond,MOSFET}} \approx 20^2 \times 0.025 = 10\ \text{W}.$$

The doubled current produces four times the loss, which is why current rating and thermal design are closely linked.

For an IGBT or diode, a first estimate often uses the ON-state voltage drop:

$$\boxed{P_{\text{cond}} \approx V_{\text{ON}} I_{\text{avg}}} \quad (9.2)$$

where V_{ON} may be $V_{CE(\text{sat})}$ for an IGBT or V_F for a diode, and I_{avg} is the average current during conduction.

If an IGBT in an inverter leg has an average ON-state drop of about 1.8 V at the relevant current and carries an average current of 6 A during its conduction interval, then

$$P_{\text{cond}} \approx 1.8 \times 6 = 10.8 \text{ W}.$$

In first-pass calculations, MOSFET conduction loss is usually treated as resistive, while IGBT and diode conduction loss is often treated as voltage-drop times current. More detailed models are possible, but these forms are adequate for thermal estimates [Mohan, Undeland, Robbins, *Power Electronics*], [Rashid, *Power Electronics: Circuits, Devices and Applications, 4e*].

Why conduction loss usually rises with temperature

Temperature affects the loss as well as the temperature rise. For many power MOSFETs, $R_{DS(\text{on})}$ increases as junction temperature increases. For many IGBTs, $V_{CE(\text{sat})}$ also changes with temperature. The STGWA40HP65FB datasheet, for example, shows $V_{CE(\text{sat})}$ increasing with junction temperature in its static characteristics and Figure 5 [STGWA40HP65FB Datasheet].

The result is a reinforcing loop:

1. Current causes loss.
2. Loss raises junction temperature.
3. Higher junction temperature often increases ON-state loss.

Thermal margin is needed partly because the loss calculated at room temperature may understate the loss after the device has heated in service.

Switching loss

Ideal switching assumes an instantaneous transition from blocking state to conducting state. Real devices require finite turn-ON and turn-OFF times, so voltage across the device and current through the device overlap during switching. That overlap produces power loss.

A simple physical estimate is

$$\boxed{P_{\text{sw,approx}} \approx \frac{1}{2} V I (t_{\text{on}} + t_{\text{off}}) f_s} \quad (9.3)$$

where V is the approximate device voltage during switching, I is the current being switched, t_{on} and t_{off} are the effective switching intervals, and f_s is the switching frequency.

The dependence is direct: higher voltage, higher current, slower switching, or higher switching frequency all increase switching loss.

Suppose an IGBT in a converter connected to a rectified 230 V, 50 Hz supply sees about 325 V during the important switching interval, switches about 8 A, has a total effective overlap time ($t_{\text{on}} + t_{\text{off}}$) of 200 ns, and operates at 20 kHz. Then

$$P_{\text{sw,approx}} \approx \frac{1}{2} \times 325 \times 8 \times 200 \times 10^{-9} \times 20 \times 10^{3} \approx 5.2\,\text{W}.$$

This is already comparable to the conduction loss of many practical devices.

Datasheets often provide switching energy directly, leading to a more practical design relation:

$$\boxed{P_{\text{sw}} \approx f_s(E_{\text{on}} + E_{\text{off}})} \quad (9.4)$$

where E_{on} and E_{off} are the turn-ON and turn-OFF energies per switching event, measured under stated test conditions.

The STGWA40HP65FB datasheet gives a typical turn-off energy of about 363 μJ at $V_{CC} = 400$ V, $I_C = 40$ A, $R_G = 5\,\Omega$, and $T_J = 25^{\circ}$C, rising to about 764 μJ at 175°C under the same nominal electrical test conditions [STGWA40HP65FB Datasheet]. Switching loss is therefore not fixed; it varies with temperature, current, voltage, and gate resistance.

The balance between conduction loss and switching loss

Table 9.1 compares the two loss mechanisms.

Table 9.1: Conduction and switching losses compared

Loss type	When it appears	Main physical cause	Usually worsens with
Conduction loss	While the device is ON	ON-state resistance or ON-state voltage drop	Higher current, higher device temperature
Switching loss	During turn-ON and turn-OFF	Voltage-current overlap during finite switching time	Higher voltage, higher current, higher frequency, slower switching

This comparison explains several earlier design tradeoffs. A stronger gate driver can reduce switching time and therefore switching loss, but excessive drive strength may increase dv/dt, di/dt, and EMI. A snubber can reduce overshoot and stress, but it may also dissipate additional power. Protection circuitry can save the device during faults, but it does not remove the need for a thermal path that continuously carries away ordinary operating loss.

A converter can therefore be thermally limited even when its efficiency is high. Switching loss can become important at moderate switching frequencies when voltage and current are large, and datasheet current ratings only have meaning under their stated thermal conditions.

Image prompt for Figure 9.1: Create a clean textbook-style technical figure showing power loss in a switching semiconductor. Use three aligned time plots versus time: device voltage, device current, and instantaneous power. Mark one conduction interval where current flows with low but nonzero ON-state voltage, and a switching interval where voltage and current overlap to create a clear power pulse. Label "conduction loss," "turn-ON switching loss," and "turn-OFF switching loss." Use monochrome engineering style with clear axes, units, and annotations.

2.4.2 Thermal resistance, heat-sink selection - simple design approach

The thermal path from junction to ambient

Heat is generated at the semiconductor junction. It then travels through the package to the case, across the mounting interface, into the heat sink, and finally to the ambient air. Any weak section in that path raises the junction temperature.

ST describes this with an electrical analogy: temperature difference corresponds to voltage difference, heat flow to current, and thermal resistance to electrical resistance [ST AN4783]. The basic relation is

$$\boxed{\Delta T = P_{\text{loss}} R_{\text{th}}} \quad (9.5)$$

where ΔT is the temperature rise across a thermal path, P_{loss} is the heat flow in watts, and R_{th} is thermal resistance in $^{\circ}$C/W.

If a section of the thermal path has $R_{\text{th}} = 2^{\circ}$C/W and carries 10 W, the temperature rise across that section is

$$\Delta T = 10 \times 2 = 20^{\circ}\text{C}.$$

This is the basis of first-pass heat-sink selection.

The main thermal-resistance terms

Table 9.2 lists the main thermal terms used in datasheets and heat-sink notes.

Table 9.2: Main thermal-management terms

Symbol	Meaning	Plain-language interpretation
T_j	Junction temperature	Temperature of the semiconductor silicon where heat is generated
T_c	Case temperature	Temperature at the package case or specified mounting surface
T_a	Ambient temperature	Temperature of the surrounding air

Symbol	Meaning	Plain-language interpretation
R_{thJC}	Junction-to-case thermal resistance	How hard it is for heat to move from silicon to package case
R_{thCS}	Case-to-sink thermal resistance	Thermal resistance of interface layer between case and heat sink
R_{thSA}	Sink-to-ambient thermal resistance	How hard it is for the heat sink to give heat to the air
R_{thJA}	Junction-to-ambient thermal resistance	Overall junction-to-air thermal path under a stated condition
TIM	Thermal interface material	Grease, pad, mica, ceramic sheet, or similar interface used between device and heat sink
$Z_{\text{th}}(t)$	Transient thermal impedance	Time-dependent thermal behavior for short or pulsed heating

For a device mounted on a heat sink, the simple series thermal path is written as [ST AN4783], [Infineon AN-1057]

$$\boxed{R_{\text{thJA}} = R_{\text{thJC}} + R_{\text{thCS}} + R_{\text{thSA}}} \quad (9.6)$$

which leads to the steady-state junction-temperature estimate

$$\boxed{T_j \approx T_a + P_{\text{tot}}(R_{\text{thJC}} + R_{\text{thCS}} + R_{\text{thSA}})} \quad (9.7)$$

where P_{tot} is the total average device loss.

Image prompt for Figure 9.2: Create a clean textbook-style thermal-path illustration for a power semiconductor mounted on a heat sink. Show, from left to right, semiconductor junction at temperature T_j, package case at T_c, thermal interface material, heat sink, and ambient air at T_a. Mark the thermal resistances R_{thJC}, R_{thCS}, and R_{thSA}. Add arrows showing heat flow outward and include the equation $T_j = T_a + P_{\text{tot}}(R_{\text{thJC}} + R_{\text{thCS}} + R_{\text{thSA}})$. Use monochrome engineering style.

A simple heat-sink selection method

First-pass heat-sink selection can be written as a short procedure:

1. Estimate the worst-case total device loss, P_{tot}.
2. Choose the highest junction temperature acceptable for the design.

3. Choose the worst-case ambient temperature around the heat sink.
4. Read the device thermal data from the datasheet, especially R_{thJC} and maximum T_j.
5. Estimate the interface thermal resistance R_{thCS}.
6. Solve for the largest allowable sink-to-ambient thermal resistance.

Rearranging Equation (9.7) gives

$$\boxed{R_{\text{thSA,max}} \approx \frac{T_{j,\text{target}} - T_a}{P_{\text{tot}}} - R_{\text{thJC}} - R_{\text{thCS}}} \tag{9.8}$$

This equation gives the largest sink-to-ambient thermal resistance acceptable for the design.

Numerical example: choosing a heat sink for a single IGBT

Suppose a 650 V IGBT in a small UPS inverter or PV auxiliary converter is expected to dissipate about 20 W at worst case. Use a design target of $T_{j,\text{target}} = 125°\text{C}$, let the ambient air near the heat sink be $45°\text{C}$, and assume the device datasheet gives $R_{\text{thJC}} = 0.53°\text{C/W}$, as the STGWA40HP65FB datasheet does [STGWA40HP65FB Datasheet]. Let $R_{\text{thCS}} = 0.5°\text{C/W}$ for the interface layer.

Using Equation (9.8),

$$R_{\text{thSA,max}} \approx \frac{125 - 45}{20} - 0.53 - 0.5.$$

First calculate the total allowable junction-to-ambient resistance:

$$\frac{80}{20} = 4°\text{C/W}.$$

Now subtract the device and interface contributions:

$$R_{\text{thSA,max}} \approx 4 - 0.53 - 0.5 = 2.97°\text{C/W}.$$

The design therefore requires a heat sink whose sink-to-ambient thermal resistance is approximately 3°C/W or lower under the actual airflow condition. A natural-convection heat sink rated at 5°C/W would be insufficient. If forced air reduces the effective sink-to-ambient thermal resistance to about 2°C/W, the design gains clear margin.

The target junction temperature is set below the absolute maximum because tolerance, aging, dust, airflow uncertainty, thermal-cycling stress, and overload reduce the real operating margin. Long-life equipment is rarely designed for continuous operation at the absolute thermal limit.

The role of the interface layer

The heat sink is only one part of the thermal path. Real device packages and real heat sinks are not perfectly flat, and air trapped between them is a poor thermal conductor. A **thermal**

interface material, or TIM, fills surface gaps and lowers thermal resistance. ST notes that grease or insulating layers such as mica, ceramic, or silicone can be used, and that contact pressure matters because the case-to-heat-sink resistance is introduced by real, nonideal contact surfaces [ST AN4783].

Even a large heat sink can perform poorly if

- the mounting surface is uneven,
- the TIM is poor or badly applied,
- the mounting pressure is inadequate,
- the assumed R_{thCS} is too optimistic for the actual interface.

Steady-state thermal resistance versus transient thermal impedance

Steady-state thermal resistance is appropriate for continuous average heating. Semiconductor heating is not always steady, however. Fault pulses, startup surges, current bursts, and intermittent overloads may be short compared with the thermal time constant of the package and heat sink.

For such cases, datasheets provide **transient thermal impedance**, usually written as $Z_{\text{th}}(t)$. ST AN4783 explains that short pulses should be evaluated with thermal-impedance curves rather than only steady thermal resistance, because the thermal capacitance of the structure delays the temperature rise [ST AN4783]. The STGWA40HP65FB datasheet makes the same point with its Figure 25, where the transient thermal response is shown using a normalized relation $Z_{\text{th}} = k \cdot R_{\text{thJC}}$ for different duty conditions [STGWA40HP65FB Datasheet].

The distinction is simple: use R_{th} for steady or average heating, and use $Z_{\text{th}}(t)$ when pulse duration matters.

Datasheet values such as R_{thJA} and total power dissipation must always be read with their stated mounting and temperature conditions. Heat-sink performance is not a fixed property either; orientation, airflow, surface area, and enclosure conditions all change the effective sink-to-ambient thermal resistance [ST AN4783], [Infineon AN-1057].

2.4.3 Forced cooling - overview

Natural and forced convection

Natural cooling is attractive because it is simple, silent, and contains no moving parts. If semiconductor losses are modest and sufficient heat-sink area is available, natural convection may be adequate.

Compact power-electronic equipment, however, is often enclosed and operated in warm surroundings. Under those conditions, the allowable heat-sink size may be limited and the sink-to-ambient thermal resistance may not become low enough without moving air. ST AN4783 distinguishes

- **natural convection**, where airflow is created by buoyancy due to temperature differences,
- **forced convection**, where airflow is created by external means such as a fan or pump [ST AN4783].

The basic convection relation is

$$\boxed{R_{\text{th}} \approx \frac{1}{h_c A_s}} \quad (9.9)$$

where h_c is the heat-transfer coefficient and A_s is the effective heat-transfer surface area [ST AN4783].

This relation shows that thermal resistance decreases when either the surface area increases or the heat-transfer coefficient increases. A fan mainly improves the second factor by moving more air past the fins.

With stronger airflow, either the same heat sink can carry more power or a smaller heat sink can carry the same power. That is why forced-air cooling is common in compact UPS systems, EV chargers, telecom rectifiers, and medium-power PV inverter cabinets.

Layout and enclosure details still matter

Airflow does not remove the need for good mechanical layout. ST gives several practical recommendations [ST AN4783]:

- fins should be vertically aligned for natural-convection cooling,
- airflow should not be blocked,
- heat-generating devices should not be placed where hot air accumulates without escape,
- forced-air flow should be arranged to follow sensible natural-convection paths.

These points matter because a good heat sink can perform poorly inside a poor airflow path. Forced cooling also introduces reliability questions that natural convection avoids: fan failure, dust accumulation, acoustic noise, blocked passages, and the possible need for airflow monitoring or thermal shutdown backup.

Comparing the main cooling approaches

At higher power density, the cooling method becomes part of the converter architecture rather than a packaging detail. Liquid cooling is not treated in detail at this level, but it should be placed beside natural and forced-air cooling in the overall design picture.

Table 9.3: Cooling approaches at beginner level

Cooling approach	Main strengths	Main limitations	Typical fit
Natural convection	Simple, quiet, no moving parts, higher mechanical reliability	Larger heat sink often needed, performance falls in cramped hot enclosures	Small chargers, low-power converters, open enclosures

Cooling approach	Main strengths	Main limitations	Typical fit
Forced-air cooling	Better heat removal for size, lower effective sink-to-ambient thermal resistance	Fan failure, dust, noise, extra power use	UPS systems, inverter cabinets, compact medium-power converters
Liquid cooling	Very high heat-removal capability in compact space	Higher cost, complexity, pump and sealing concerns	High-power EV and industrial converters

Local ambient temperature must also be interpreted correctly. In converter practice, the air around the heat sink may be far hotter than the room or outdoor temperature because the enclosure already contains other heat-generating components.

Image prompt for Figure 9.3: Create a clean textbook-style comparison figure showing the same finned heat sink under natural convection and forced-air cooling. On the left, show warm air rising naturally between vertical fins. On the right, show a fan pushing air across the fins. Label "natural convection," "forced convection," "airflow direction," "heat sink fins," and "lower effective sink-to-ambient thermal resistance with stronger airflow." Use monochrome engineering style with arrows and temperature labels, not decorative graphics.

Worked interpretation exercise

The following exercise uses the STGWA40HP65FB datasheet, which combines thermal limit, thermal resistance, conduction behavior, switching energy, and transient thermal information in one device [STGWA40HP65FB Datasheet].

Step 1: Read the thermal limit before reading the current rating

The datasheet states a maximum operating junction temperature of 175°C and gives thermal data including $R_{\text{thJC}} = 0.53^{\circ}\text{C/W}$ for the IGBT junction-to-case path [STGWA40HP65FB Datasheet].

Those two numbers immediately show that the device rating is thermal as well as electrical. The junction can rise only a limited amount above the case, and the allowable power dissipation depends on that temperature rise.

If the case is held at 90°C, the available junction rise to the absolute maximum is

$$175 - 90 = 85^{\circ}\text{C}.$$

Using Equation (9.5), the corresponding approximate allowable steady dissipation across the junction-to-case path is

$$P \approx \frac{85}{0.53} \approx 160\,\mathrm{W}.$$

This is an interpretation of the junction-to-case limit, not a recommended operating target. It simply shows how strongly the allowable dissipation depends on case temperature.

Step 2: Do not misread the power-dissipation rating

The same datasheet lists total power dissipation of 283 W at $T_C = 25°\mathrm{C}$ [STGWA40HP65FB Datasheet].

That number is conditional. It assumes a very cool case and therefore strong heat removal. It does not mean that a bare TO-247 package in still air can dissipate 283 W continuously.

Figure 1 in the datasheet, "Power dissipation vs case temperature," makes the condition explicit [STGWA40HP65FB Datasheet]. As case temperature rises, the allowable dissipation falls. Thermal rating is therefore conditional, not absolute.

Step 3: Connect electrical loss to thermal behavior

The datasheet gives static and switching data that can be used in loss estimation:

- $V_{CE(\mathrm{sat})}$ is around 1.6 V typ. at one stated operating point,
- typical turn-off energy is given for stated current, voltage, gate resistance, and temperature,
- Figure 17 shows switching energy increasing with junction temperature [STGWA40HP65FB Datasheet].

Electrical loss and thermal behavior are therefore coupled. If junction temperature rises, switching energy may rise as well, which increases dissipation and further raises device temperature.

Step 4: Use the thermal-impedance curve correctly

Figure 25 in the datasheet presents thermal impedance in normalized form, showing that transient thermal response depends on pulse duration and duty ratio [STGWA40HP65FB Datasheet]. One static resistance value is not sufficient for every operating condition.

If the device sees short overload pulses, the immediate junction temperature rise is governed by transient thermal impedance, not only by the steady-state heat-sink resistance.

Step 5: Translate the datasheet into design questions

Table 9.4: Reading the STGWA40HP65FB for thermal meaning

Datasheet item	What it means in plain language	Why it matters thermally
$T_J = 175°\mathrm{C}$ max	The silicon cannot safely exceed this limit	Sets the absolute ceiling for thermal design

Datasheet item	What it means in plain language	Why it matters thermally
$R_{\text{thJC}} = 0.53°\text{C/W}$	Heat still sees resistance even before leaving the package	Affects junction rise for every watt dissipated
$P_{\text{TOT}} = 283$ W at $T_C = 25°\text{C}$	Large dissipation is possible only with a cool case	Prevents misreading the package as self-cooling
Figure 1: power dissipation vs case temperature	Hotter case means less allowable dissipation	Shows why heat sink and airflow are decisive
Figure 17: switching energy vs temperature	Switching loss changes with junction temperature	Electrical and thermal design interact
Figure 25: thermal impedance	Short pulses heat the die differently from steady power	Needed for pulsed-stress judgment

The same questions should be asked of any power-device datasheet, not only this one.

How this matters in renewable-energy systems

Thermal management is closely tied to renewable-energy hardware because these converters often operate for long hours, at significant current, and in warm outdoor or semi-enclosed environments. In solar PV systems, the MPPT stage and inverter stage both generate semiconductor loss continuously while rooftop temperature may already be high. In battery chargers and battery-energy-storage converters, thermal design affects efficiency, current capability, and charging reliability. Wind-energy interfaces, EV auxiliary converters, on-board chargers, and UPS systems all place similar pressure on junction temperature because power density, operating duration, and enclosure heating act together.

The larger design lesson is that efficiency and reliability meet in the thermal design. Junction temperature influences semiconductor lifetime, loss, allowable current, and safe operating margin. For many practical converters, the thermal design determines whether an electrically correct circuit remains reliable under real climate and enclosure conditions.

Chapter summary

Thermal management begins with loss estimation. Conduction loss depends on ON-state resistance or ON-state voltage drop, switching loss depends on voltage-current overlap and switching energy, and both often increase as junction temperature rises. Electrical behavior and thermal behavior are therefore coupled rather than separate design problems.

Thermal design then converts loss into temperature rise through the path from junction to ambient. Equations (9.5) to (9.8) provide a first-pass method for steady-state junction-temperature estimation and heat-sink selection, while transient thermal impedance extends the analysis to pulsed operation. Natural convection, forced-air cooling, and liquid cooling differ mainly in how effectively they reduce the effective device-to-environment thermal resistance.

Further reading

- M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*, 4th ed. - A strong textbook source for introductory treatment of device losses, power conversion efficiency, and practical semiconductor behavior.
- Ned Mohan, Tore M. Undeland, and William P. Robbins, *Power Electronics: Converters, Applications and Design* - Especially useful for connecting device-loss concepts to converter operation and design tradeoffs.
- STMicroelectronics, *AN4783: Thermal effects and junction temperature evaluation of Power MOSFETs* - A very helpful application note on thermal resistance, thermal impedance, convection, and practical heat-sink thinking.
- Infineon, *AN-1057: Heatsink Characteristics* - A concise practical reference for heat-sink terminology, interface resistance, and first-pass heat-sink selection.
- STMicroelectronics, *STGWA40HP65FB Datasheet* - A real device datasheet that is especially useful for learning how thermal ratings, power dissipation, switching energy, and transient thermal impedance are presented in practice.

Module 3

AC to DC Converters (Controlled Rectifiers)

Overview

This module explains controlled rectifier circuits that convert AC to adjustable DC, with attention to firing angle control, waveforms, average output relations, and renewable-energy applications.

Contents

Chapter 3.1: Single-phase Controlled Rectifiers

Chapter opening

Previous chapters treated the SCR primarily as a device: its latching behavior, triggering, protection, and thermal limits. This chapter turns that device into a converter. A controlled rectifier uses SCRs to convert AC to DC while controlling the instant at which conduction begins in each cycle [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6], [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

Control of the conduction interval is useful whenever the DC side must be adjustable rather than fixed. Battery chargers, DC drives, soft-start circuits, and line-frequency front ends all use delayed firing to vary average output. This chapter develops that idea through the single-phase half-wave controlled rectifier, the fully controlled bridge, and the semi-controlled bridge, and then collects the average- and RMS-voltage expressions used later in the module [Littelfuse AN1003].

Prerequisites check

- You should remember from Chapter 1.2 that an SCR can be triggered by a gate pulse but does not turn OFF by gate control in ordinary operation.
- You should be comfortable with sinusoidal AC voltage, including the relation between RMS value and peak value.
- You should know that a resistor makes current follow voltage, while an inductor resists sudden change of current.
- You should remember from Chapter 2.1 that firing pulses must be synchronized correctly with the AC waveform.
- You should remember from Chapter 2.2 that an inductive load often needs a freewheeling path when the main source is removed.

If SCR latching and natural turn-off are unclear, review Chapter 1.2. If average value and RMS value are not yet comfortable, review those now before continuing.

3.1.1 Single-phase half-wave controlled rectifier with R and R-L loads; effect of freewheeling diode

Phase control in the half-wave rectifier

In the single-phase half-wave controlled rectifier, an SCR is connected in series with the load and the AC source. During each positive half-cycle the SCR is forward biased, but forward bias alone does not produce conduction. The device conducts only after a gate pulse is applied. Delaying the gate pulse by an angle α delays conduction and therefore controls the average output voltage [Littelfuse AN1003], [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6].

The supply is written as

$$\boxed{V_m = \sqrt{2}\,V_{s,rms}} \quad (10.1)$$

where $V_{s,rms}$ is the RMS value of the sinusoidal source and V_m is its peak value.

For a 230 V, 50 Hz single-phase supply,

$$V_m = \sqrt{2} \times 230 \approx 325 \text{ V}.$$

The 230 V RMS mains value therefore corresponds to a peak of about 325 V.

The firing instant is described by the **firing angle** α, measured from the zero crossing of the positive half-cycle. If the SCR is triggered at $\omega t = \alpha$, conduction begins at that angle provided the load current exceeds the holding current [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6], [ST AN4607].

Image prompt for Figure 10.1: Create a clean textbook-style technical illustration of a single-phase half-wave controlled rectifier using one SCR and a load. Include three vertically aligned waveform plots versus electrical angle ωt: source voltage v_s, gate pulse train showing a pulse at firing angle α in each positive half-cycle, and output voltage v_o for a resistive load that is zero from 0 to α and follows the positive sine wave from α to π. Clearly mark 0, α, π, and 2π. Use monochrome engineering style with axes and units.

Half-wave controlled rectifier with R load

With a purely resistive load, current and voltage are in phase. After the SCR is fired at angle α, the output voltage follows the source until the current falls to zero at $\omega t = \pi$. The SCR then turns OFF by line commutation, because the AC supply itself drives the current to zero [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6].

For an R load:

- from 0 to α, the SCR blocks and $v_o = 0$,
- from α to π, the SCR conducts and $v_o = V_m \sin \omega t$,
- from π to 2π, the source is negative, the SCR is reverse biased, and $v_o = 0$.

The average output voltage over one full cycle is therefore

$$\boxed{V_{o,avg} = \frac{V_m}{2\pi}(1 + \cos\alpha)} \quad (10.2)$$

[Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6], [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

For a 230 V RMS source, $V_m \approx 325$ V. If the firing angle is $\alpha = 60^\circ$, then $\cos 60^\circ = 0.5$, so

$$V_{o,avg} = \frac{325}{2\pi}(1 + 0.5) \approx 77.6 \text{ V}.$$

Phase control does not reduce the peak of the source. It changes the portion of the positive half-cycle that is applied to the load.

Half-wave controlled rectifier with R-L load

With a series R-L load, the current cannot change instantaneously. When the source voltage passes through zero, energy stored in the inductor can keep the current flowing, so conduction may continue into the negative half-cycle.

During SCR conduction, the load current satisfies

$$\boxed{L\frac{di_o}{dt} + Ri_o = V_m \sin \omega t} \quad (10.3)$$

where L is the inductance, R is the resistance, and i_o is the load current.

Instead of ending at π, conduction may continue to an **extinction angle** β, where $\beta > \pi$ [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6]. Two consequences follow:

- the output voltage becomes negative between π and β while the SCR continues conducting,
- the average output voltage becomes smaller than in the resistive-load case for the same firing angle.

If device drops are neglected, the average output voltage over one cycle is

$$\boxed{V_{o,avg} = \frac{V_m}{2\pi}(\cos \alpha - \cos \beta)} \quad (10.4)$$

where β depends on R, L, the source frequency, and the firing angle [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

The reduction in average output is the direct result of the negative-voltage interval introduced by continued inductive current.

Effect of a freewheeling diode

A freewheeling diode connected across the load provides an alternate current path when the source voltage becomes negative. The inductive current then circulates through the load and diode instead of through the source and SCR [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6], [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

The output-voltage waveform changes as follows:

- from 0 to α, $v_o = 0$,

- from α to π, the SCR conducts and $v_o = V_m \sin \omega t$,
- from π onward, the freewheeling diode conducts, current continues through the load, and the load voltage is approximately zero.

The average output voltage becomes

$$\boxed{V_{o,avg} = \frac{V_m}{2\pi}(1 + \cos \alpha)} \quad (10.5)$$

The expression matches the half-wave resistive case because the diode removes the negative load-voltage interval, even though the current waveform remains inductive.

Image prompt for Figure 10.2: Create a clean textbook-style technical illustration comparing a single-phase half-wave controlled rectifier with an R-L load, first without and then with a freewheeling diode across the load. Show two output-voltage waveforms versus ωt: in the first, the output follows the source from α to β and becomes negative between π and β; in the second, the output follows the source from α to π and becomes approximately zero during the freewheeling interval. Also show corresponding load-current waveforms that continue after π. Label α, π, β, and the freewheeling interval clearly. Use monochrome engineering style.

The freewheeling diode provides three practical benefits:

- it improves the average output voltage for the same firing angle,
- it eliminates the negative load-voltage interval,
- it smooths the load current and reduces the reactive burden on the source.

3.1.2 Single-phase full-wave fully controlled bridge rectifier with R and R-L loads

A half-wave circuit uses only one half-cycle of the supply. In practice, full-wave conversion is preferred because it uses the source more effectively, raises the ripple frequency at the output, and offers broader control flexibility. The **single-phase fully controlled bridge rectifier**, often called the **single-phase full converter**, achieves this using four SCRs [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6].

Fully controlled bridge with R load

Let the SCR pairs be (T_1, T_2) and (T_3, T_4). With a resistive load, each pair conducts from the firing instant to the natural current zero of that half-cycle:

- T_1 and T_2 conduct from α to π,
- T_3 and T_4 conduct from $\pi + \alpha$ to 2π.

The output therefore contains two positive pulses per source cycle. The average output voltage is

$$\boxed{V_{o,avg} = \frac{V_m}{\pi}(1 + \cos \alpha)} \quad (10.6)$$

which is twice the half-wave resistive-load result [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

For the same 230 V supply and $\alpha = 60^\circ$,

$$V_{o,avg} = \frac{325}{\pi}(1 + 0.5) \approx 155.2\,\text{V}.$$

This value is twice the half-wave result because both half-cycles are used.

Fully controlled bridge with R-L load and continuous current

With an R-L load and sufficiently large inductance, load current may remain continuous throughout the cycle. Each SCR pair then conducts for 180° electrical, and commutation occurs when the next pair is fired [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6].

The average output voltage becomes

$$\boxed{V_{o,avg} = \frac{2V_m}{\pi}\cos\alpha} \qquad (10.7)$$

Three standard cases follow directly:

- if $\alpha = 0^\circ$, then $V_{o,avg} = \dfrac{2V_m}{\pi}$,
- if $\alpha = 90^\circ$, then $V_{o,avg} = 0$,
- if $\alpha > 90^\circ$, then $V_{o,avg}$ becomes negative.

A negative average output becomes possible because a continuous-current DC side can drive power back toward the AC source under line commutation. That operating region is the basis of **line-commutated inverter** behavior [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

For a 230 V source:

- at $\alpha = 30^\circ$, $V_{o,avg} \approx \dfrac{2 \times 325}{\pi} \times 0.866 \approx 179.3\,\text{V}$,
- at $\alpha = 75^\circ$, $V_{o,avg} \approx \dfrac{2 \times 325}{\pi} \times 0.259 \approx 53.5\,\text{V}$,
- at $\alpha = 105^\circ$, $V_{o,avg} \approx \dfrac{2 \times 325}{\pi} \times (-0.259) \approx -53.5\,\text{V}$.

By crossing $\alpha = 90^\circ$, the converter crosses from positive to negative average output.

Image prompt for Figure 10.3: Create a clean textbook-style technical illustration of a single-phase fully controlled bridge rectifier using four SCRs. Include two waveform sets. In the first set, show the R-load case with gate pulses for the two SCR pairs and output voltage pulses from α to π and from $\pi + \alpha$ to 2π. In the second set, show the R-L continuous-current case with nearly constant positive load current and output voltage that alternates between positive and negative segments according to the firing angle. Clearly label SCR pairs, α, π, and the conduction intervals. Use monochrome textbook engineering style.

The fully controlled bridge therefore has a wider operating range than the half-wave circuit or the semi-converter.

3.1.3 Single-phase semi-controlled (half-controlled) bridge rectifier - concept

A **semi-controlled bridge rectifier**, also called a **half-controlled bridge** or **semi-converter**, uses two SCRs and two diodes. It sits between the diode bridge and the full SCR bridge: the output is adjustable, but only two devices require gate control [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6], [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6]. In each half-cycle one SCR and one diode conduct. With an inductive load, the diode arrangement also provides a natural freewheeling path.

Average output voltage of the semi-converter

For the standard inductive-load case with continuous current and ideal devices, the average output voltage is

$$\boxed{V_{o,avg} = \frac{V_m}{\pi}(1 + \cos\alpha)} \quad (10.8)$$

For $0^\circ \leq \alpha \leq 180^\circ$, the term $(1 + \cos\alpha)$ is never negative. The semi-converter therefore cannot produce negative average output voltage in normal operation and is a **one-quadrant converter** in the usual introductory sense.

Example: charger-type supply

Suppose an isolated secondary provides 18 V RMS to a single-phase semi-converter used in a small battery charger. Then

$$V_m = \sqrt{2} \times 18 \approx 25.5\,\mathrm{V}.$$

If the firing angle is $\alpha = 60^\circ$,

$$V_{o,avg} = \frac{25.5}{\pi}(1 + 0.5) \approx 12.2\,\mathrm{V}.$$

Changing α therefore changes the average DC output without requiring four controlled devices.

Operating features of the semi-converter

Because freewheeling occurs naturally, the load voltage does not go negative in the way it can for the full-controlled bridge under continuous-current conditions. The current is correspondingly smoother for many one-directional loads. The tradeoff is reduced control authority: negative average output voltage is not available.

Table 10.1 compares the three main single-phase topologies discussed in this chapter.

Table 10.1: Comparing single-phase controlled-rectifier topologies

Topology	Controlled devices used	Uses both half-cycles?	Natural freewheeling path with inductive load?	Can average output become negative?	Typical application picture
Half-wave controlled rectifier	1 SCR	No	No, unless a freewheeling diode is added	No practical inversion mode	Foundational teaching circuit, low-cost power control
Fully controlled bridge	4 SCRs	Yes	No inherent freewheeling path	Yes, when current is continuous and $\alpha > 90^\circ$	Controlled DC supplies, classical drives, line-commutated converters
Semi-controlled bridge	2 SCRs + 2 diodes	Yes	Yes	No	Battery chargers, one-way variable DC front ends

Image prompt for Figure 10.4: Create a clean textbook-style technical illustration of a single-phase semi-controlled bridge rectifier with two SCRs and two diodes feeding an R-L load. Show gate pulses only for the SCRs. Include output-voltage and load-current waveforms for continuous-current operation. Mark source-connected intervals where output follows the rectified sine wave, and freewheeling intervals where load voltage is approximately zero while current continues. Use monochrome engineering style with clear device labels and conduction intervals.

3.1.4 Expressions for average and RMS output voltage; effect of firing angle

Average output voltage measures the DC component delivered by the rectifier. RMS output voltage determines heating and effective power in resistive elements. Two rectifier waveforms can have the same average value and different RMS values, so both quantities are required in analysis [Littelfuse AN1003], [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

RMS output voltage for half-wave controlled rectifier with R load

For the half-wave controlled rectifier with a resistive load,

$$V_{o,rms} = \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\pi} V_m^2 \sin^2\theta \, d\theta}.$$

Carrying out the integration gives

$$\boxed{V_{o,rms} = \frac{V_m}{2}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}} \quad (10.9)$$

The limiting cases are consistent. At $\alpha = 0^\circ$, $V_{o,rms} = V_m/2$, which is the known RMS value of an uncontrolled half-wave rectified sine wave. As α approaches 180°, the term inside the square root approaches zero, so the RMS output also approaches zero.

For the earlier 230 V example with $\alpha = 60^\circ$,

$$V_{o,rms} = \frac{325}{2}\sqrt{1 - \frac{60^\circ}{180^\circ} + \frac{\sin 120^\circ}{2\pi}} \approx 145.8 \text{ V}.$$

In that case the average output is about 77.6 V, while the RMS output is about 145.8 V. The two quantities must not be confused.

RMS output voltage for full-wave R-load and semi-converter waveforms

For the single-phase fully controlled bridge with an R load, or for the semi-converter output-voltage waveform during source-connected intervals with ideal freewheeling, the RMS output becomes

$$\boxed{V_{o,rms} = \frac{V_m}{\sqrt{2}}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}} \quad (10.10)$$

This expression is larger than the half-wave RMS value because both half-cycles contribute. The limiting cases are again consistent:

- at $\alpha = 0^\circ$, $V_{o,rms} = V_m/\sqrt{2} = V_{s,rms}$,
- at $\alpha \to 180^\circ$, $V_{o,rms} \to 0$.

RMS output voltage for full-controlled bridge with continuous current

For the ideal single-phase fully controlled bridge with continuous current and negligible overlap, the output voltage at every instant is either $+v_s$ or $-v_s$. When it is squared for RMS calculation, the sign disappears. The RMS output voltage therefore equals the source RMS voltage:

$$\boxed{V_{o,rms} = V_{s,rms}} \quad (10.11)$$

The RMS value remains fixed because the waveform changes sign and timing, not its squared magnitude over the cycle.

Effect of firing angle

The firing angle affects more than the average output voltage. It changes:

- the average DC output,
- the RMS output in pulsed-output cases,
- the timing and distortion of the source current, and therefore the power factor,
- the boundary between discontinuous and continuous current in inductive loads [Rashid, *Power Electronics: Circuits, Devices and Applications*, 4e, Ch. 6], [Mohan, Undeland, Robbins, *Power Electronics: Converters, Applications and Design*, 3e, Ch. 6].

Table 10.2 collects the principal expressions from this chapter.

Table 10.2: Summary of important average and RMS expressions

Circuit and load condition	Average output voltage	RMS output voltage	Main note
Half-wave controlled rectifier, R load	$\frac{V_m}{2\pi}(1+\cos\alpha)$	$\frac{V_m}{2}\sqrt{1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}}$	Conduction from α to π
Half-wave controlled rectifier, R-L load without freewheeling diode	$\frac{V_m}{2\pi}(\cos\alpha-\cos\beta)$	Depends on β and exact current continuity	Negative output interval can appear
Half-wave controlled rectifier, R-L load with freewheeling diode	$\frac{V_m}{2\pi}(1+\cos\alpha)$	Depends on current waveform	Freewheeling prevents negative output voltage
Fully controlled bridge, R load	$\frac{V_m}{\pi}(1+\cos\alpha)$	$\frac{V_m}{\sqrt{2}}\sqrt{1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}}$	Two positive pulses per cycle
Fully controlled bridge, R-L load, continuous current	$\frac{2V_m}{\pi}\cos\alpha$	$V_{s,rms}$	Average becomes negative for $\alpha > 90^\circ$
Semi-controlled bridge, continuous current	$\frac{V_m}{\pi}(1+\cos\alpha)$	Same waveform trend as full-wave pulsed case when freewheeling intervals are ideal	One-quadrant operation

How this matters in renewable-energy systems

Single-phase controlled rectifiers are not the dominant form of modern low- and medium-power AC-DC conversion, but they remain foundational. Firing angle, natural commutation, current continuity, and freewheeling reappear throughout the study of controlled chargers, line-frequency converters, and higher-power front ends.

Thyristor-based AC-DC conversion also remains relevant where line-frequency operation and high power matter. Hitachi Energy notes the continued use of high-power thyristors in applications ranging from soft starters to HVDC converter stations [Hitachi Energy Thyristors], [Hitachi Energy HVDC Converter Stations]. At the same time, newer charger and converter designs often move toward PWM rectifiers for improved size and efficiency [NPTEL Line Commutated and PWM Rectifiers]. Controlled rectifiers therefore remain best understood as the classical basis from which modern AC-DC conversion is developed.

Chapter summary

- A controlled rectifier controls the start of conduction by delaying SCR firing through the angle α.
- In the half-wave rectifier, an R load gives Equation (10.2), while an R-L load can extend conduction to an extinction angle β and reduce average output through a negative-voltage interval.
- Adding a freewheeling diode to the half-wave R-L circuit removes that negative-voltage interval and gives Equation (10.5).
- The fully controlled bridge uses both half-cycles. With an R load it gives Equation (10.6); with continuous current it gives Equation (10.7) and can produce negative average output when $\alpha > 90^\circ$.
- The semi-converter gives Equation (10.8), includes natural freewheeling, and remains a one-quadrant AC-to-DC converter.
- RMS expressions complement average values: Equations (10.9) to (10.11) show that firing angle changes heating and waveform effectiveness as well as average DC output.

Further reading

- M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*, 4th ed. - A strong textbook source for first-principles treatment of single-phase controlled rectifiers, waveforms, and average-value derivations.
- Ned Mohan, Tore M. Undeland, and William P. Robbins, *Power Electronics: Converters, Applications and Design*, 3rd ed. - Especially useful for continuous current, semi-converters, and the interpretation of α and β.
- Littelfuse, *AN1003: Phase Control Using Thyristors* - A practical note on delay angle, conduction angle, and the change in output voltage with phase control.
- STMicroelectronics, *AN4607: Basics on the thyristor (SCR) structure and its application* - A concise refresher on SCR behavior and on applications in UPS, photovoltaic, and related power circuits.
- Hitachi Energy, *HVDC converter stations* - A high-level link showing how controlled AC-DC conversion scales from textbook rectifiers to grid-scale converter stations.